Thesis for the degree of *Doctor of Philosophy*

Liquid Crystalline Polymer Composites-Based 3D Shape-Programmable Platform of Locomotive Electronics

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Abstract

Liquid Crystalline Polymer Composites-Based 3D Shape-Programmable Platform of Locomotive Electronics

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Conventional electronic circuits are typically constructed on rigid substrates with low strain tolerance and limited integration onto curved surfaces. To overcome this, researchers have actively introduced flexible or stretchable polymers including hydrogels and elastomers to electronics. These electronics can conform more easily to curved structures through high strain tolerance, making them suitable for integrating with the human body with multi-curvilinear structures. However, these flexible materials must accompany physical contact and external tension or compression for shape deformation, limiting precise control in small-scale devices even at the centimeter scale. In this dissertation, I demonstrate strategies to achieve remotely controllable shape-reconfiguration of flexible polymers which have a high potential to be introduced in a multi-functional form factor of next-generation electronics. To remotely reconfigure the initial shape of the polymer into a programmed design, we should induce stress gradients inside the structure. This was achieved through frontal photopolymerization (FPP) of photocurable resins containing photo-absorbers and the fabrication of liquid crystalline (LC) polymer composites with

programmed molecular alignment. Firstly, by introducing the FPP of photocurable resin including a photo-absorber, I achieved a crosslinking density gradient within the cured polymer structure, inducing an internal stress gradient. After FPP, differences in stress relaxation along the thickness direction resulted in 3D shape-reconfiguration. Dimensions of final structure were controlled by adjusting spatiotemporal parameters such as light pattern design and exposure time. Each cured structure served as a unit module and was assembled into scaled-up structures with more complex geometries, which were then utilized as conductive frameworks through metal coating. Subsequently, we fabricated LC polymer composites which are a great candidate for the 3D shape-programmable platform of electronics. Shape-reconfiguration of LC polymer composites is not limited to the immediate photopolymerization but can be controlled by spatiotemporally applying external photonic stimuli. LC polymers possess anisotropic thermal contraction and expansion due to their anisotropic molecular structure as well as internal molecular alignment, allowing programmed shape-reconfiguration in response to temperature changes. By forming bilayer composites with 2D materials exhibiting excellent electrical conductivity such as reduced graphene oxide (rGO) and MXene, I successfully integrated the high electrical conductivity of these materials into the LC polymer without disrupting its molecular alignment based on high filler loading for electrical percolation. These 2D materials generate a stable interface with the LC polymer through secondary bonding, stably maintaining its shape-reconfigurability and electrical conductivity during repeated shape-reconfigurations. Moreover, rGO and MXene exhibit excellent photothermal conversion efficiency, enabling remote temperature control of the LC polymer composites through infrared irradiation. Specifically, by introducing photo-responsive azobenzene functional groups into the LC polymer composites, remote shape control was achieved via trans-cis photoisomerization of azobenzene under UV irradiation. When the crosslinking density of LC polymer was reduced to engineer glass transition temperature closer to room temperature, reversible shape-recovery was achieved immediately after removal of photothermal stimuli. Collectively assembled LC polymer composites allowed various types of shape-reconfigurations and multi-functional locomotion such as linear movement, rotational movement, and jumping, by controlling the symmetricity of the assembled structure and introducing snap-through instability. The introduction of composite materials that can program shape reconfiguration opens possibilities for expansion into a platform of next-generation electronic devices.

Chapter 1 Introduction

1.1 Motivation

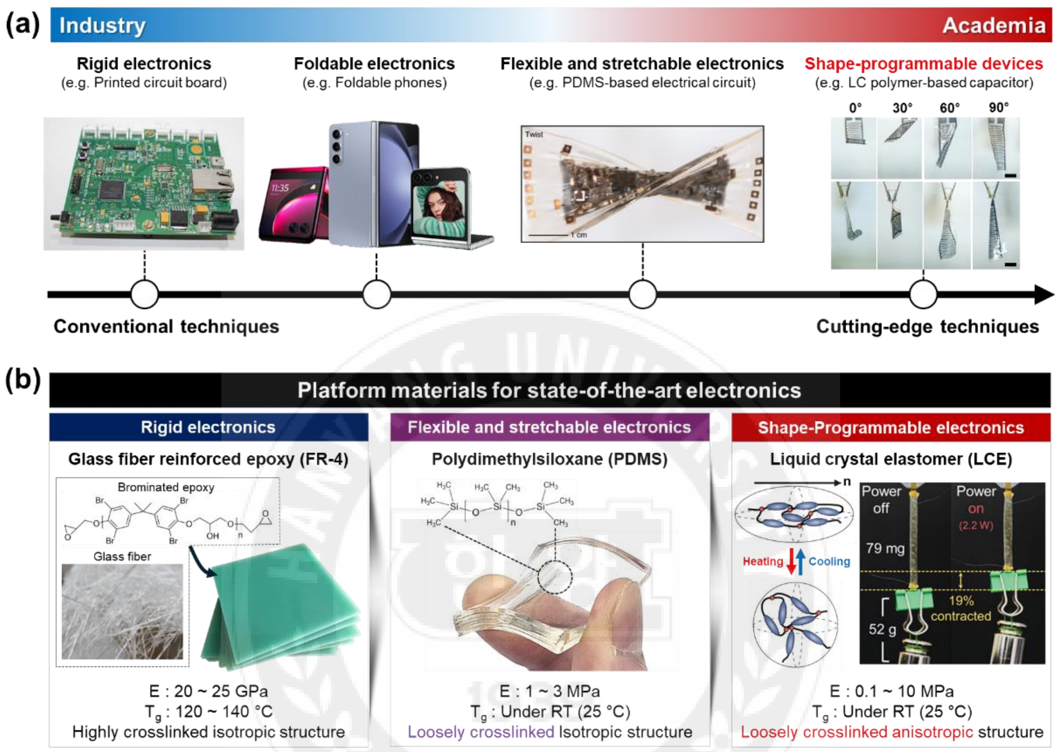

Figure 1.1 (a) Technical roadmap of state-of-the-art electronics from rigid electronics to shape-programmable electronics[1–3]. Adapted from ref. 2. Copyright 2018, Springer Nature, CCBY. Adapted from ref. 3. Copyright 2022, Elsevier. (b) Platform materials for electronics[4–6]. Adapted from ref. 4. Copyright 2024, Springer Nature. Adapted from ref. 5. Copyright 2022, Wiley-VCH. Adapted from ref. 6. Copyright 2022, Springer Nature.

Beyond rigid conventional electronics, flexible and stretchable electronics have tremendously received attention as next-generation electronics that can transform their original shape to the desired complex shape under on-demand mechanical strain (**Figure 1.1a**)[7]. For example, in the industry, major companies, such as Samsung Electronics and LG Electronics, are releasing foldable cell phones and rollable TVs as commercial products. These devices can be folded or rolled up for convenient storage, offering spatial advantages. When unfolded, they provide a larger display, enhancing the user experience with more versatility and comfort. While the industry focuses on practical applications like foldable phones and rollable TVs, academia delves into more cutting-edge and challenging research for next-generation electronics by introducing advanced platform materials beyond rigid glass fiber-reinforced epoxy-based printed circuit boards (PCBs) (**Figure 1.1b**). For example, many researchers have developed flexible and stretchable electronics using soft, flexible, stretchable polymers such as poly(dimethylsiloxane) (PDMS) as a platform material[8,9]. Due to the intrinsic flexibility of these materials, flexible and stretchable devices can be applied on complex curvilinear surfaces like the human body[10,11]. However, these devices require mechanical stretching and compressing forces accompanying physical contact to achieve shape-reconfiguration for the desired shape. Furthermore, continuous mechanical force must be applied to maintain the reconfigured shape. These passive characteristics for shape-reconfiguration cause limitations in precisely controlling the shapes of small devices manually, typically ranging from a few millimeters to centimeters[12]. To push innovation of active type shape-reconfiguration, introduction of smart material is essential for the platform materials of next-generation electronics which actively sense external stimuli and then change their shape remotely into a programmed shape.

For this, I introduce stimuli-responsive and shape-programmable polymers as a promising platform material for next-generation electronics. In particular, liquid crystalline (LC) polymer is an excellent candidate for the platform material that can change its original shape by responding to untethered external stimuli such as thermal conduction and convection, and photonic irradiation[13]. Due to their unique shape programmability, LC polymers have already attracted significant attention in various fields, including soft actuators[14], soft robotics[15], antennas[16], and locomotive electronics[3,17,18]. Unlike conventional PDMS with isotropic network structure, LC polymers exhibit anisotropic thermal contraction and expansion due to their anisotropic molecular structure and molecular alignment. To trigger remotely controlled shape-reconfiguration, we should integrate LC polymer with photochemically active moieties (e.g., azobenzene) and photothermally active fillers (e.g., graphene and MXene). Furthermore, the introduction of graphene and MXene is an effective strategy to achieve electrical conductivity simultaneously with remotely controlled shape-reconfiguration, as graphene and MXene have both high electrical conductivity and photo-thermal conversion efficiency. This dissertation starts with describing strategies to achieve 3D complex polymer framework by inducing internal crosslinking density gradient inside photocurable polymer via frontal photopolymerization. Subsequently, the integration strategy of LC polymer with functional moieties with the aforementioned fillers is introduced to achieve a shape-reconfigurable platform for locomotive electronics exhibiting high electrical conductivity and various types of shape-reconfiguration and actuation under untethered external photonic stimuli. I believe that these approaches expand the potential of shape-reconfigurable materials, which can be applied to platforms for next-generation electronics, going beyond flexible and stretchable electronics.

1.2 Frontal Photopolymerization

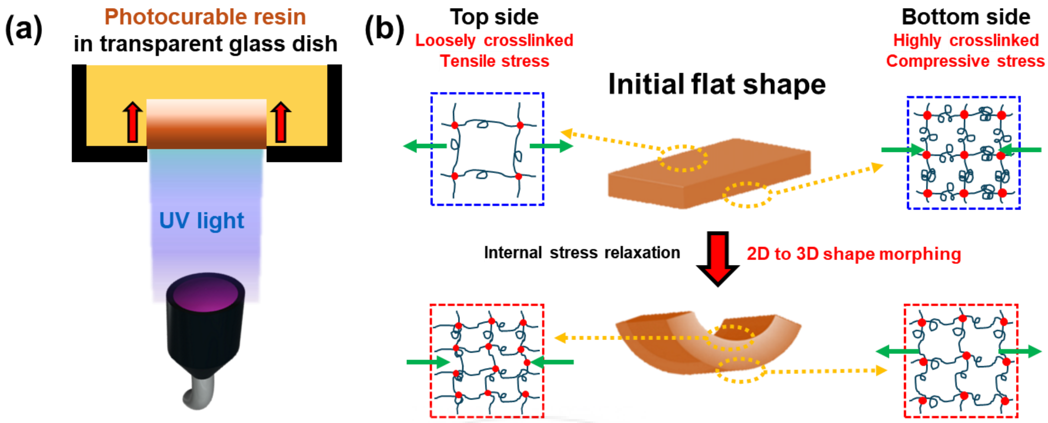

Figure 1.2 (a) Schematic illustration of frontal photopolymerization (FPP). (b) Schematic illustration indicates crosslinking gradient inside cured polymer and shape morphing mechanism via stress relaxation.

Front photopolymerization (FPP) is a type of advanced photocurable 3D printing technology in which the photopolymerization is initiated from a specific front side and is generally propagated on one side along the direction of light irradiation[19]. FPP is started with vertical irradiation of UV light to the bottom of a transparent dish containing photocurable resin (**Figure 1.2a**). Photopolymerization is initiated at the front where it contacts the surface of the dish, and the propagation proceeds with a gradually decreasing light intensity compared to the initial level as the resin simultaneously absorbs light. A crosslinking density gradient is generated in the cured structure between the bottom and top, due to the gradual decrease in light intensity according to the penetrating depth (**Figure 1.2b**). The cured structure adheres to the surface of the dish, inducing compressive stress, as the adhesion restricts the intrinsic shrinkage caused by photopolymerization. As the light intensity gradually decreases along the propagation direction, the crosslinking density also decreases. Consequently, the shrinkage by photopolymerization diminishes, leading to a reduction in compressive stress along the propagation direction. Eventually, tensile stress is generated toward the opposite direction to compressive stress. When FPP is terminated, internal stress is released as the cured structure is detached from the dish, and then the structure is bent toward UV light irradiation direction as the accumulated internal stress is released. If the gradient of light intensity is not significant, the overall deviation in the crosslinking density is small which causes a minimum degree of shape-reconfiguration. On the other hand, by maximizing the light intensity gradient by using a photo-absorber in the resin, we can induce dramatic shape-reconfiguration of the cured polymer. The degree of shape-reconfiguration can be controlled by adjusting light intensity and light irradiation time. Furthermore, patterned light irradiation with various designs can induce free-standing, complex 3D reconfigured shapes, which can be applied to platforms of electronics. In this dissertation, the strategy for achieving 3D complex-structured polymers through spatiotemporally controlled FPP will be described in Chapter 2.

1.3 Thermotropic Liquid Crystal

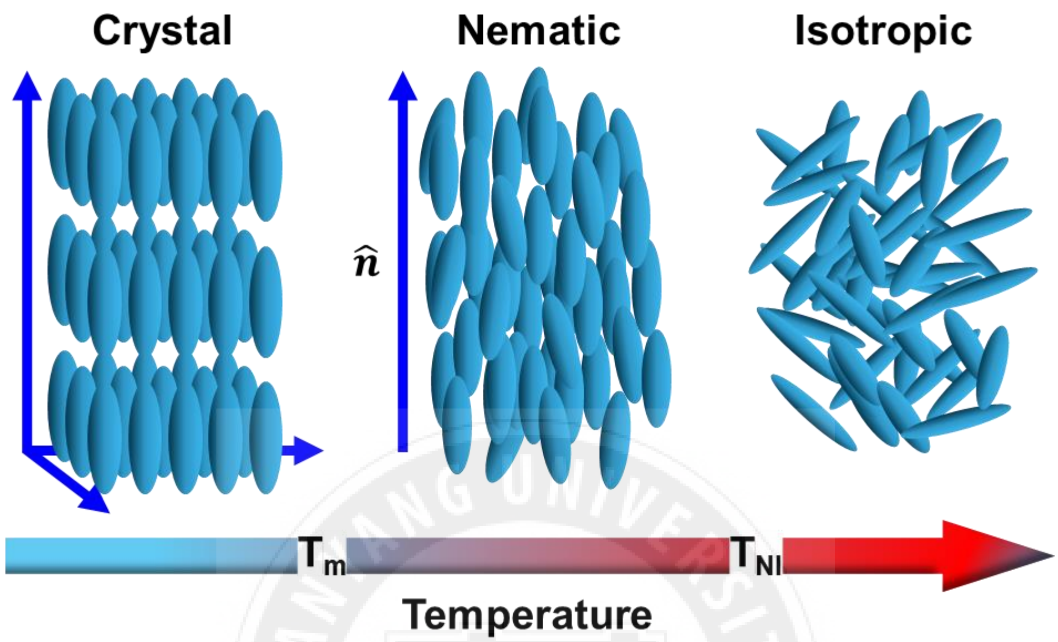

Figure 1.3 Schematic illustration indicates temperature-dependent phase transition of thermotropic LC molecules. The blue rod indicates the mesogen unit. T_m and T_{NI} mean melting temperature and nematic to isotropic temperature, respectively.

LC polymers are a class of polymers primarily composed of thermotropic LC molecules that exhibit LC phases depending on temperature (**Figure 1.3**)[20]. These thermotropic LC molecules typically possess rod-like or disc-like shapes, which confer structural anisotropy, and benzene-rich chemical structure allows for intermolecular secondary interactions such as π-π stacking. At temperatures below their melting point, these LC molecules form a crystalline phase showing both positional and orientational order. As the temperature exceeds the melting point, LC molecules become a nematic phase, where only orientational order remains while positional order is lost. As the temperature rises, LC molecules experience increased Brownian motion, which disrupts their orientation. However, the anisotropic shape of the LC molecules introduces spatial constraints, and favorable secondary interactions between the molecules provide enthalpic stabilization, compensating for the entropic penalty associated with maintaining orientational order. This is the origin of the nematic phase in thermotropic LC molecules, a behavior not typically observed in typical isotropic molecules. When the temperature is increased beyond the nematic-to-isotropic transition temperature (T_{NI}), these LC molecules lose all orientational order and become isotropic liquid phases. In other words, we can control the molecular order of thermotropic LC molecules by adjusting temperature.

1.4 Liquid Crystalline Polymers

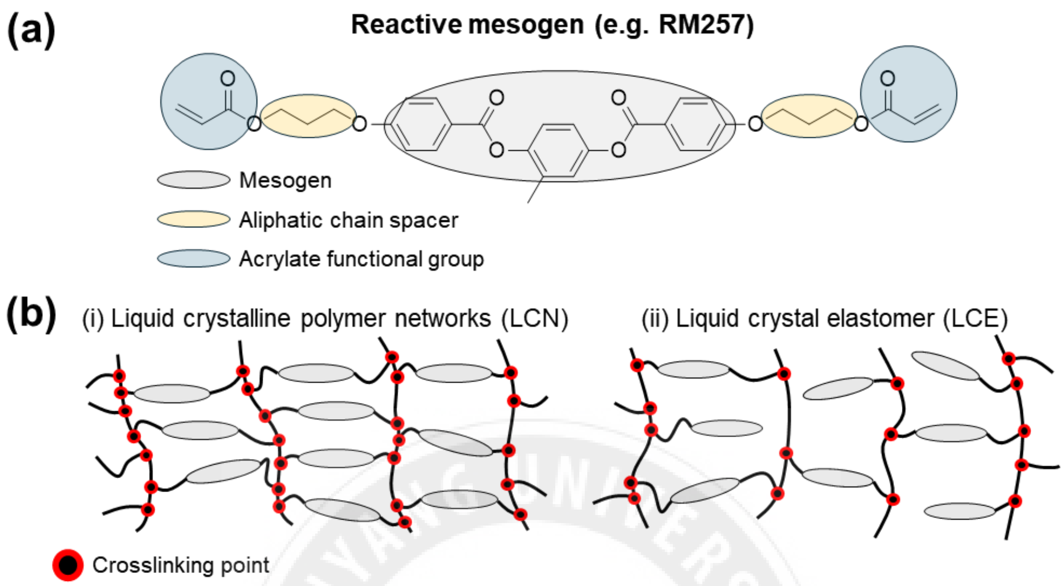

Figure 1.4 (a) Chemical structure of RM257, one of the well-known reactive mesogens. (b) Schematic illustrations of (i) liquid crystalline polymer networks (LCN) and (ii) liquid crystal elastomer (LCE) which show different crosslinking densities.

For LC polymers, it is essential to physically connect LC molecules through covalent bonding. To achieve this, acrylate-functionalized thermotropic LC molecules are commonly used, typically incorporating an aliphatic chain spacer. These molecules are referred to as reactive mesogens (**Figure 1.4a**). In LC polymers, the aliphatic chain spacer plays a crucial role in providing flexibility and mobility to the LC molecules, even when they are constrained by the covalent bonds formed through the acrylate group. This flexibility allows the LC molecules to retain their ability to change their alignment order depending on applied temperature, despite being part of a cross-linked polymer network. As a result, the alignment of LC molecules in the polymer can be controlled by temperature, leading to anisotropic thermal expansion and contraction. To achieve LC polymers, reactive mesogens must be photopolymerized in the nematic phase to maintain orientational order and low viscosity. The orientational order should be maintained to achieve anisotropic thermal contraction and expansion. Furthermore, low viscosity is essential for high conversion of photopolymerization by inducing active motion of molecules with premature gelation.

The crosslinking density of LC polymers can be easily controlled by varying the length of the aliphatic spacer chain in the reactive mesogen or the number of acrylate functional groups. LC polymers are typically classified based on their mechanical stiffness into liquid crystalline polymer networks (LCN) and liquid crystal elastomer (LCE) (**Figure 1.4b**)[13]. Traditionally, LCE is an LC polymer with lower crosslinking density showing elastomeric character compared to glassy LCN showing higher crosslinking density. In detail, LCEs have a glass transition temperature (Tg) below room temperature, a distinguishing characteristic of elastomers, an elastic modulus of 0.1-5 MPa, and a strain responsivity of greater than 90%. In contrast, LCNs have a glass transition temperature in the range of 40-120°C, an elastic modulus of 1-2 GPa at room temperature, and a strain responsivity of approximately 5%. In other words, depending on target applications, the mechanical properties of LC polymers can be freely controlled by adjusting the crosslinking density by introducing different types and ratios of reactive mesogens. This allows for fine-tuning of shape-reconfigurability, which is heavily influenced by mechanical stiffness.

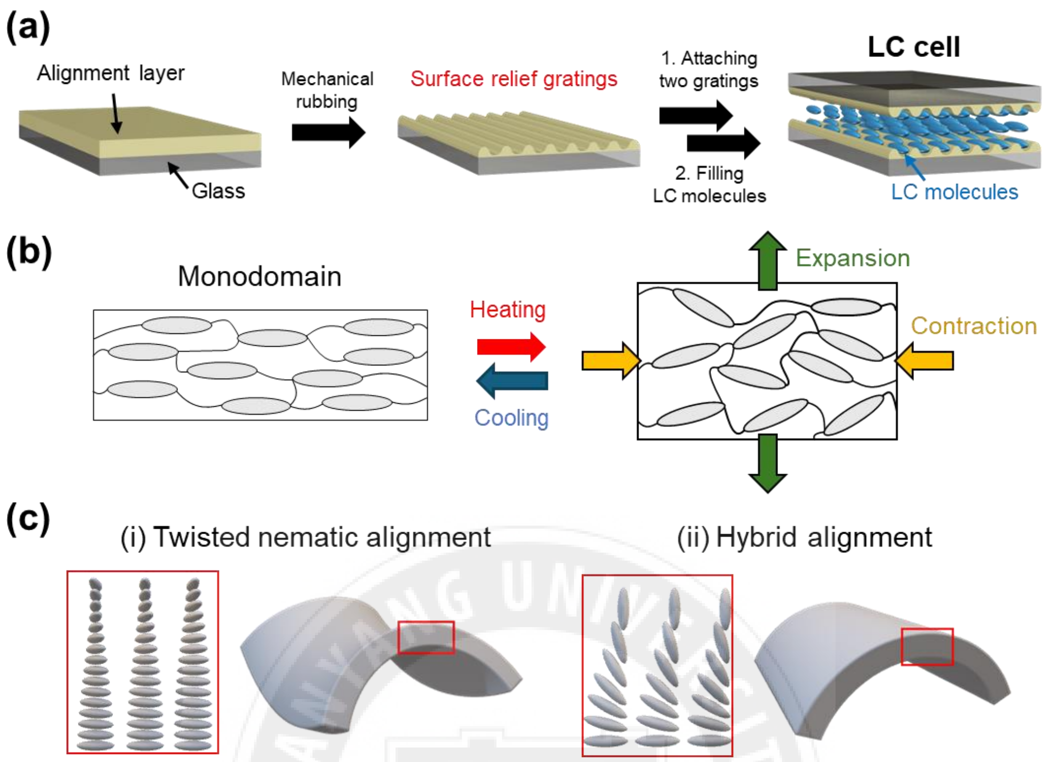

Figure 1.5 (a) Schematic illustration of preparing LC cell with surface relief gratings. (b) Monodomain LC polymer and anisotropic thermal expansion and contraction. (c) (i) Twisted nematic and (ii) hybrid aligned LC polymer and their temperature-dependent reconfigured shape.

During the photopolymerization of LC molecules, alignment can be controlled by various techniques including magneto-, photo-, and surface-induced methods. Typically, magneto-induced alignment requires expensive, high-strength magnets (often 1-5 T), making it less practical. On the other hand, photo-induced alignment is limited by small processing areas and the use of costly materials. In contrast, surface-induced alignment offers a more practical and cost-effective solution, making it ideal for large-scale applications. Hence, in this dissertation, surface-induced alignment was utilized due to its scalability, simplicity, and efficiency. In this surface-induced method, alignment layer-coated glass LC cell should be prepared in advance. The LC cell is typically composed of an alignment layer-coated glass pair. Polyimide and polyamide are typically used for the alignment layer (**Figure 1.5a**). These polymers are coated onto a glass slide, and mechanical rubbing is applied to create surface relief grating on the alignment layer. Mechanically rubbed glass pairs are assembled as alignment layers of each glass face each other, with a gap between them of several micrometers to tens of micrometers. This assembled glass pair is called an LC cell, where LC molecules are filled inside, aligning with the direction of the surface relief grating. By filling LC molecules into LC cells with controlled surface relief grating orientation, controlled alignment of LC molecules can be achieved, and then LC polymers with controlled alignment can be obtained through photopolymerization. LC polymers reconfigure into various shapes depending on the applied alignment state in response to temperature. First, Monodomain alignment refers to LC molecules being uniformly aligned in the same direction across the entire sample (**Figure 1.5b**). As the temperature increases, the LC molecules become disordered, resulting in anisotropic thermal expansion and contraction. As the temperature increases, LC polymer expands perpendicular to the alignment direction and contracts parallel to the alignment direction. Beyond 2D expansion and contraction, LC polymers with complex twisted nematic (TN) and hybrid alignments exhibit 3D shape-reconfiguration such as bending and coiling (**Figure 1.5c**)[21]. For TN-aligned LC polymer, LC molecules are aligned parallel to the surfaces of the cell, with a 90-degree twist between the substrates. In hybrid aligned LC polymer, one substrate induces planar alignment (parallel to the surface), while the other induces homeotropic alignment (perpendicular to the surface) before photopolymerization. These types of LC polymers exhibit orthogonal direction of thermal contraction and expansion between the bottom and top regions of the LC polymer. This difference induces directional 3D shape-reconfiguration due to the strain gradient generated across the polymer.

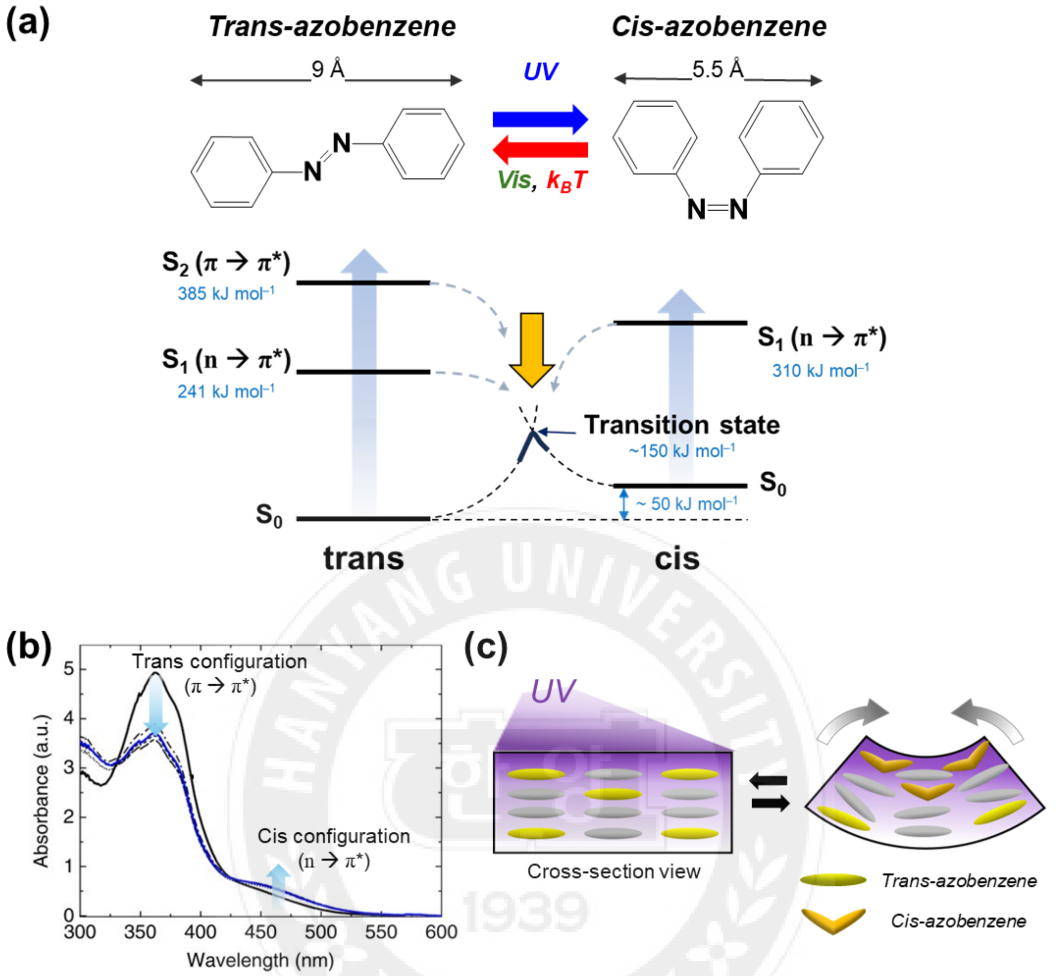

Figure 1.6 (a) Photoisomerization of azobenzene and schematic illustration of energy state diagrams for photoisomerization[22]. (b) UV-vis spectra of azo-LCN films with light (320–500 nm) irradiation at 200 mW cm^{-2} intensity[23]. Adapted from ref. 22. Copyright 2024, Springer Nature. (c) Skin-bulk effect of azobenzene functionalized LC polymer under UV irradiation.

The introduction of azobenzene, recognized as a molecular machine, enables LC polymers to perform photochemically driven actuation through photoisomerization[23,24]. As the azobenzene is covalently bonded with LC molecules inside LC polymers, the azobenzene can mechanically disorder the alignment of LC molecules via trans-cis photoisomerization induced by UV light absorption. During photoisomerization, a microscopic contraction occurs in the molecular length of azobenzene, shrinking from 9 Å to 5.5 Å (**Figure 1.6a**). In terms of energy level, azobenzene has two primary isomeric states: the thermodynamically stable trans-configuration and the less stable cis-configuration with a relative destabilization of approximately 50 kJ mol^{-1} compared to the trans configuration. For trans-to-cis photoisomerization, the excitation and subsequent relaxation of the electronic states associated with the azo bond are fundamental. Trans-azobenzene exhibits two distinct excited states, each corresponding to different absorption bands: one in the visible region (S_1) and the other in the UV region (S_2). Electronic excitation through $\pi \rightarrow \pi^*$ transition promotes bonding electrons to the S_2, typically absorbing light within the 300-400 nm range (UV spectrum). Conversely, n $\rightarrow$ π^* transitions lead to excitation to the S_1, which absorbs light in the 400-500 nm range (Visible spectrum). These excitations facilitate structural reconfiguration in azobenzene from trans to cis configuration via diverse pathways including rotation, inversion, concerted inversion, and inversion-assisted rotation[25]. Cis-azobenzene primarily undergoes excitation via n $\rightarrow$ π^* transitions, absorbing light within the 400-500 nm range. Additionally, the energy gap between S_0 of cis-azobenzene and the transition state is relatively small, around 100 kJ mol^{-1}, enabling cis-trans back isomerization under the influence of thermal energy k_BT (Boltzmann thermal energy). These photoisomerization and back-isomerization processes are reversible, allowing azobenzene to transition dynamically between forms. When exposed to continuous UV light, this reversibility leads azobenzene to an equilibrium state, known as the photostationary state, where trans- and cis-azobenzene coexist in a stable with fixed ratio.

Figure 1.6b indicates a UV-Vis absorption spectrum, demonstrating the photostationary state concentration of an azobenzene-functionalized liquid crystalline polymer before and after UV light irradiation. Initially, the spectrum exhibits a strong $\pi \rightarrow$

π^* absorption peak at 365 nm, characteristic of the trans-azobenzene configuration. Upon UV light exposure, this peak diminishes markedly as azobenzene undergoes photoisomerization to the cis configuration, leading to an increase in the n $\rightarrow$ π^* absorption peak in the visible region. These spectral changes highlight that the photoisomerization process is indeed achieved even when the azobenzene is restricted by covalent bonding to the LC molecule. As I mentioned before, the azobenzene connected inside LC polymer pulls adjacent LC molecules inducing disordered alignment under UV light. This microscopic contraction causes macroscopic deformation in the LC polymer through the skin-bulk effect (**Figure 1.6c**). Due to the short penetration depth of UV light, only azobenzene in the surface region is activated, generating a stress gradient through the thickness, leading to directional bending actuation. As a result, achieving a stress gradient is essential for directional 3D shape-reconfiguration of LC polymer. This can be accomplished through alignment programming of LC polymers and integration of azobenzene.

1.5 Overview and Objectives

The final goal of the dissertation is the development of a shape-reconfigurable platform for next-generation electronics that can perform multi-functional locomotion. Therefore, it is essential to introduce electrical conductivity into nonconductive polymers. In this dissertation, I first emphasize that the generation of an internal stress gradient is essential for directional shape-reconfiguration, which can be achieved by inducing a crosslinking density gradient or a molecular alignment gradient. In Chapter 2, I explain strategies to achieve 3D structured polymers through internal stress engineering by using FPP and collectively assembled designs for scaled-up complex 3D frameworks of electronics. As I mentioned in Section 1.2, the UV light intensity gradient during FPP induces a crosslinking density gradient of cured polymer structure, and after FPP, this structure bends toward the direction of UV exposure via internal stress relaxation. The 3D free-standing building units were prepared by spatiotemporally controlled FPP, and then a scaled-up 3D architecture was achieved by collective assembly of these building units. Finally, the assembled structures demonstrated enhanced load-resistance properties and were applied to electronics by coating them with silver paste.

Beyond shape-reconfiguration occurring only at the moment of irreversible internal stress relaxation, in Chapter 3, I introduce an azobenzene-functionalized liquid crystalline polymer network/reduced graphene oxide (azo-LCN/rGO) bilayer for stimuli-responsive, shape-reconfigurable electronics. There are several approaches to impart electrical conductivity to LC polymers by integrating electrically conductive fillers. Besides the type of conductive filler used, the method of incorporating it into the LC polymer matrix plays a critical role in determining the overall conductivity and shape-reconfigurability. The previous methods for preparing LC polymer composites are initiated by injecting an LC mixture with dispersed fillers into an LC cell. However, this approach restricts the filler content to below 1 wt% due to the significantly increased viscosity of the LC monomer and filler mixture, which hinders effective injection into the LC cell. Such a low concentration of conductive fillers fails to establish an electrical percolation for high electrical conductivity, limiting this method for highly conductive LC polymer composites. Furthermore, fillers inside LC polymers reduce shape-reconfigurability compared to neat LC polymers by disordering the alignment state. In the case of metallic filler, the high density and modulus of the filler also reduce shape-reconfigurability by increasing bending stiffness. To address this limitation, a bilayer structure was introduced that does not hinder the molecular alignment of LC molecules.

As a synergistic effect, the coefficient of thermal expansion (CTE) mismatch between the LC polymer and the filler improves shape-reconfigurability, inspired by bimetallic strips. In terms of electrical conductivity, fillers are located in the open spaces of the bilayer structure, which does not require a high filler content to achieve electrical percolation. As a filler, reduced graphene oxide (rGO) was integrated into the azo-LCN layer using a customized rGO-coated LC cell, achieving a thickness of approximately 10 times thinner (~2 μm) than the azo-LCN layer. The rGO has many advantages, including low density, facile solution-based processability, and high electrical conductivity (~400 S cm^{-1} in this case). Due to π-π interactions between the mesogen and rGO, the azo-LCN/rGO bilayer exhibited a stable interface. As a result, the azo-LCN/rGO bilayer was applied to stretchable and shape-reconfigurable electronics by kirigami engineering.

In Chapter 4, I demonstrate enhanced electrical conductivity and reversible shape-reconfiguration of LC polymer composites by developing a liquid crystal elastomer/MXene (LCE/MXene) bilayer. MXene is a new type of 2D material exhibiting metallic conductivity above 5,000 S cm^{-1} without any reduction process, unlike rGO. Due to the high photothermal conversion efficiency of MXene, the LCE/MXene bilayer performs reversible shape-reconfiguration under toggling near-infrared light via the photothermal effect. Furthermore, a stable interface between the LCE and MXene was generated via hydrogen bonding between mesogen and oxide/hydroxide surface terminal groups of MXene, enabling 1,000 cycles of repeated actuation without any deterioration of actuation or electrical performance. Importantly, asymmetric assembled designs and snap-through designs were applied to achieve multifunctional locomotion such as straightforward movement, directional rotation, and rapid jumping behavior. Lastly, I will describe the conclusions and summary in Chapter 5.

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Chapter 2 Programmable Building Blocks via Internal Stress Engineering for 3D Collective Assembly

2.1 Introduction

Numerous natural objects accomplish various scaled-up 3D geometries through collective assembly in response to environmental conditions. On a molecular scale, surfactants dispersed in the solvent act as building units to be assembled into micelles in forms of a sphere, rod, and lamellar. This assembly is accomplished by concentration changes of the surfactant, one of an environmental condition.[1,2] In DNA systems, the collective assembly ensures a diversity of genetic information depending on the assembled structures, by altering the arrangement of four bases (guanine, adenine, cytosine, and thymine) corresponding to four types of building units.[3-5] On the macroscopic scale, a honeycomb structure is an example of a collectively assembled structure based on hexagonal hollow building blocks. This honeycomb structure achieves a high specific mechanical strength while consuming minimal materials.[6,7] Hence, collective assembly of basic building units is a reasonable way to construct scaled-up 3D structures with enhanced functionality. Furthermore, the concept of collective assembly has previously been reported in many fields including metamaterials,[8-10] electronics,[11, 12] and actuators.[13-16]

2D to 3D shape-reconfiguration via control of internal stress has inherent advantages to achieve rapid fabrication and high programmability.[17-20] For this, frontal photopolymerization (FPP) is an advanced light-based 3D printing technology for preparing programmed 3D polymeric building units via internal stress engineering.[26-28] During FPP, UV light passes through the resin with gradually reduced light intensity according to the Beer-Lambert law. This induces a crosslinking density gradient in the cured polymer monolith along the thickness direction, finally inducing a stress gradient between the top and bottom surfaces of the cured polymer. In terms of materials, diacrylate-functionalized monomers have been widely applied in 3D printing techniques due to their high reactivity, high resolution, and straightforward handling.[22,23] In particular, diacrylate-functionalized poly(ethylene glycol) (PEGDA) is widely employed for numerous bio-applications owing to high biocompatibility and nontoxicity.[24,25] Many research groups have studied achieving programmed 3D shape of the polymer via FPP employing PEGDA-based photocurable resin in many fields including fabrication of origami structures,[27,29,30] microfluidics,[31] and microparticles.[32]

In this study, we described the relationship between various 2D patterns and 3D morphed structures by systematically investigating FPP conditions and constructing a collectively assembled structure using diversely designed basic building blocks. Nontoxic PEGDA-based resin was employed as a photocurable resin. Spatially controlled FPP was conducted by using 2D patterned photomasks with various designs that selectively block the irradiated UV light. Subsequently, photocured 3D building units with various curvilinear shapes were prepared by controlling the curing time of FPP which is a temporal parameter. We hierarchically designed shape-reconfiguration based on FPP by systematically investigating dimensions of single rectangular building elements, expanding them into 3D freestanding structures by the primary assembly of the rectangular element, and ultimately demonstrating complex scaled-up structures by the secondary assembly of the 3D freestanding structures. First, we investigated the effect of the rectangular photomask pattern dimensions and UV exposure conditions on the resulting dimensions of the 3D structure, including curvature, thickness, and internal stress difference between the top and bottom surfaces. In primary assembly, 3D free-standing structures were fabricated using newly assembled 2D patterns via radially combining and rearranging the rectangular patterns. Furthermore, we introduced spiral 2D patterns to demonstrate various helical structures and twisted-radial-leg freestanding structures with controlled handedness. Finally, we successfully demonstrated scaled-up hierarchical 3D structures by collectively assembling multileg building blocks inspired by objects and famous landmarks, such as a three-leaf clover and the Great Pyramid in Giza. Importantly, the scaled-up 3D structure withstood approximately 150 times its weight and demonstrates the potential for applications in complex, curved, and hollow frameworks of electronics through a simple silver coating on its surface.

2.2 Results and Discussion

2.2.1 Residual Stress-Induced Shape-Reconfiguration

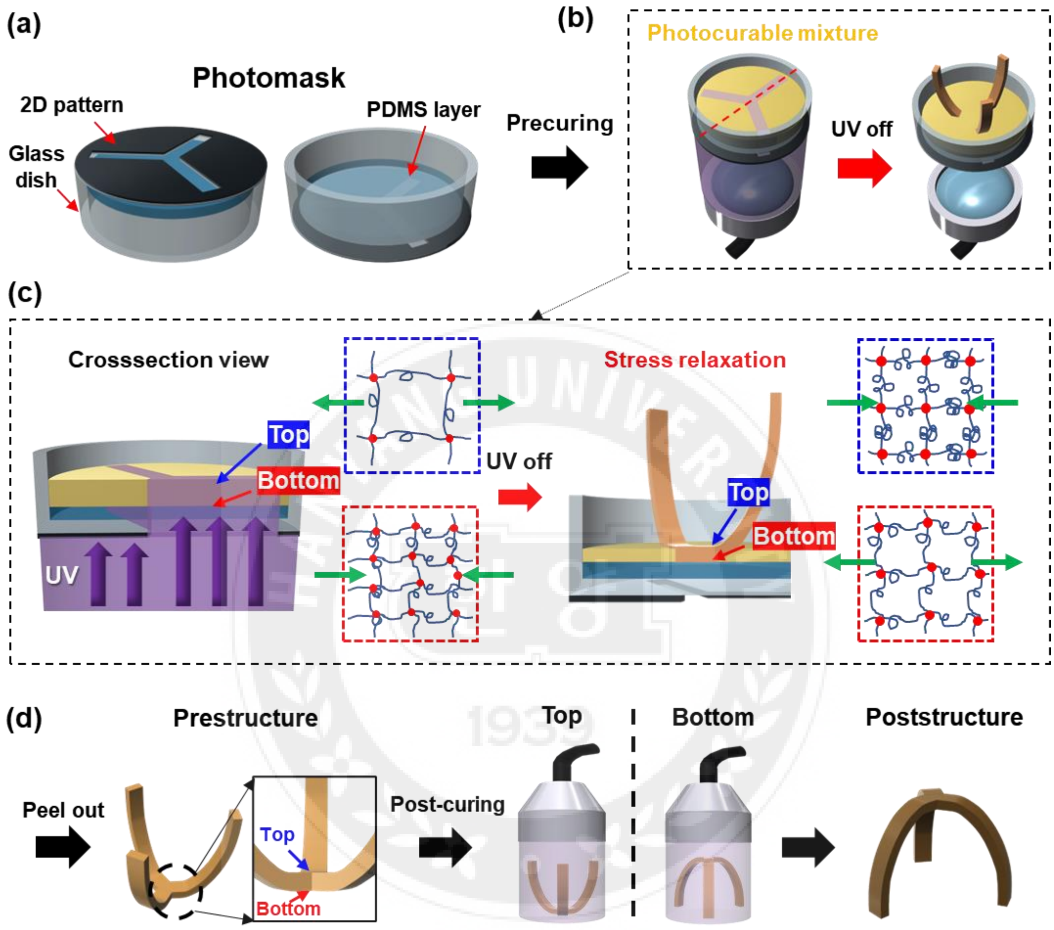

Figure 2.1 Schematic diagram for 2D to 3D shape-reconfiguration. (a) The photomask of 2D patterned layer, PDMS layer, and glass dish. (b) Precuring photocurable resin using 365 nm UV. (c) Shape-reconfiguration mechanism of cured polymer monolith after precuring, where green arrows indicated the direction of internal stresses developed in the top (Blue) and bottom (Red) regions. (d) 3D prestructure after precuring. Subsequently, postcuring was performed by selecting one of the bottom and top regions of the prestructure with UV light of the same wavelength (3 min). The illustration represented a bottom direction postcuring result.

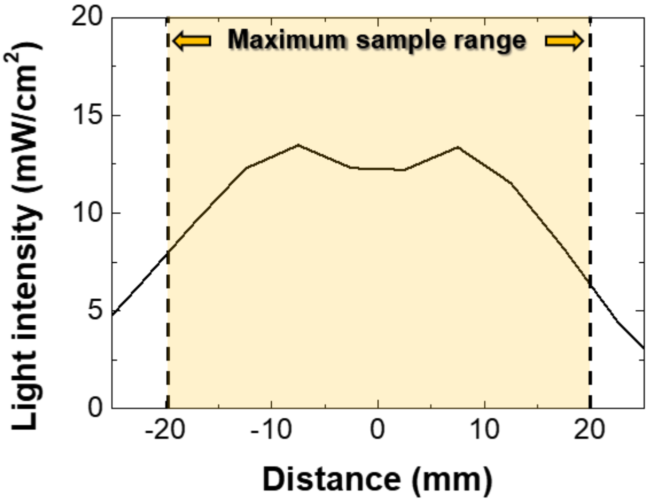

Figure 2.2 Light intensity profile of collimated UV light (365 nm).

Figure 2.3 Fabrication of 3D building block by FPP. (a) 5 s precuring process. A 3D structure was created because the internal stress was released over time after UV irradiation. (b) 3-leg structures peeled off from the photomask. The morphed structure went through the cleaning process and formed a 3D structure within 1 min.

As illustrated in **Figure 2.1**, FPP was employed to achieve 2D to 3D shape-reconfiguration. The introduction of a photomask with a 2D pattern allowed for spatially controlled FPP (**Figure 2.1a**). To create a curved 3D prestructure, the precuring step was initiated by irradiating collimated UV light (365 nm) to the photomask containing a photocurable mixture (**Figure 2.1b**). The intensity of the collimated UV light utilized in the precuring process was 12.3 mW cm^{-2} (16.5 mW cm^{-2} for postcuring). The light intensity profile is demonstrated in **Figure 2.2.**

$$I\ (x) = I_0 e^{-\alpha x} \quad (2.1)$$

As UV light penetrated inside the photocurable mixture, the initial light intensity (I_0) decreased exponentially with the optical path length (x) (**Equation (2.1)**). The absorption coefficient of the medium (α) depends on the wavelength of light and intrinsic absorbency of the medium. Reduced light intensity (I) is predominantly determined by the optical path length (x) under a constant composition ratio of photocurable mixtures and a uniform wavelength and intensity of light source. Photopolymerization can be described as the transition from a monomer to a polymer initiated by the photo-initiator activated through UV light absorption. A stronger intensity of UV light activates a larger number of photo-initiators that create denser crosslinking per unit volume and larger volumetric shrinkage.[30, 31] According to the Beer-Lambert law, the exponential decay of the light intensity generates a crosslinking density gradient along the thickness of the cured polymer monolith, which results in a strain gradient.

The acrylate chemistry was subjected to high volumetric shrinkage of about ~5% upon crosslinking and the tightly crosslinked bottom region, where light first hit, created high compressive forces due to the volumetric contractions.[33] According to the previous study on the FPP, the generated compressive force was dismissed as it approached the newly cured top region due to gradually decreased light intensity and a tensile force was generated at the top region.[27, 30] After termination of the precuring process, accumulated internal stresses were released in the opposite direction to generate shape-reconfiguration. The precured 2D polymer film transformed into a 3D curvilinear concave structure toward the top region as the internal stress was released (**Figure 2.1c**). After the 3D prestructure was removed from the photomask, the 3D curved structure was generated simultaneously within 1 min (**Figure 2.3**). The prestructure became a fixed stable shape with high elastic modulus as we additionally applied the postcuring step (**Figure 2.1d**). The detailed effect of the temporal parameter on the dimension of the reconfigured shape will be discussed in Section 2.2.2. The final dimension of the poststructure depends on the light irradiation direction during the postcuring process and details will be discussed in Section 2.2.3.

2.2.2 Rectangular Curvilinear 3D Structures

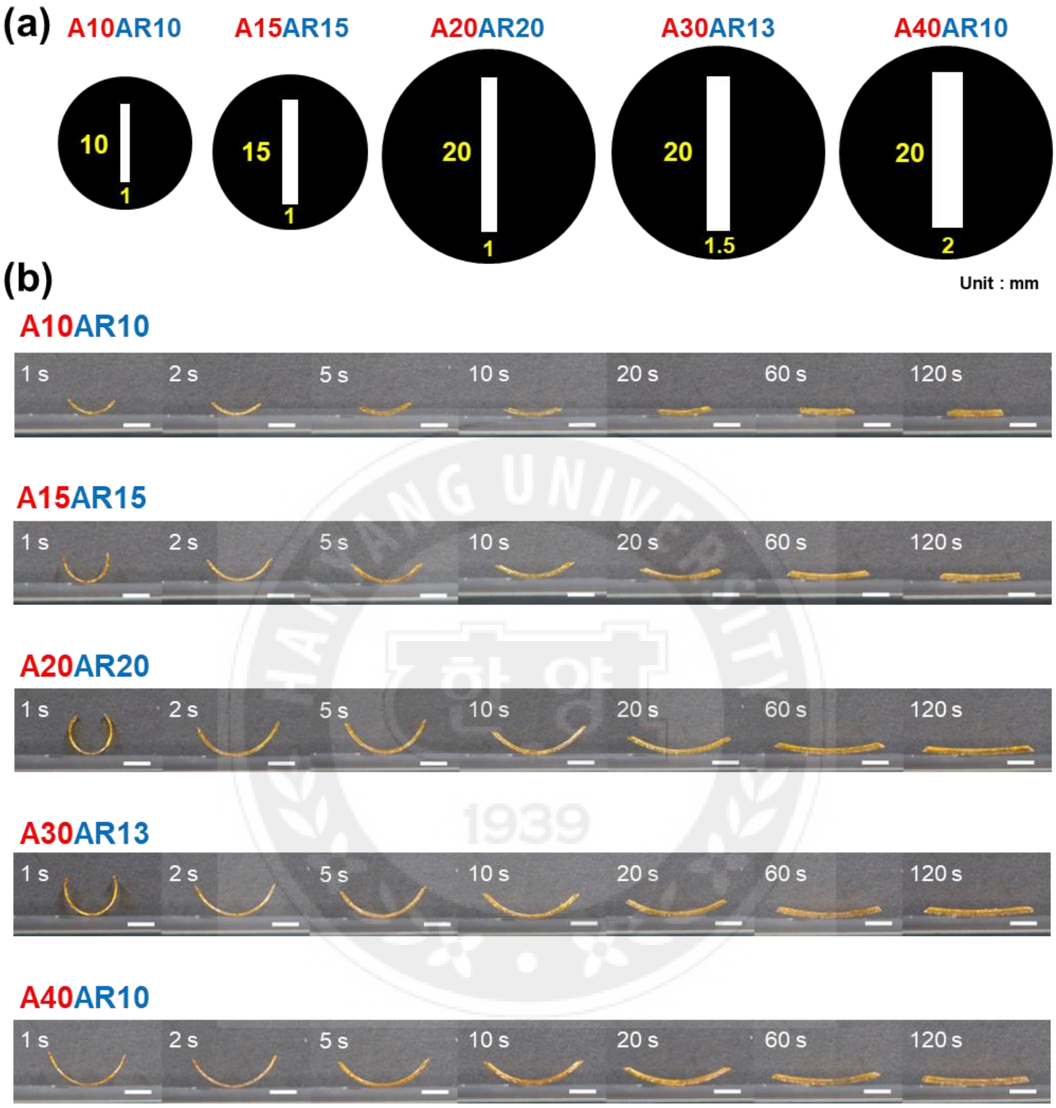

Figure 2.4 Various dimensions of rectangular curvilinear 3D structures depending on controlled spatiotemporal conditions. (a) Schematic diagram of photomask patterns with different areas of 10, 15, 20, 30, and 40 mm^2 and AR of 10, 13, 15, and 20. (b) Curvilinear 3D structures were prepared by using 5 types of 2D pattern design and varied precuring time. Scale bars, 5 mm.

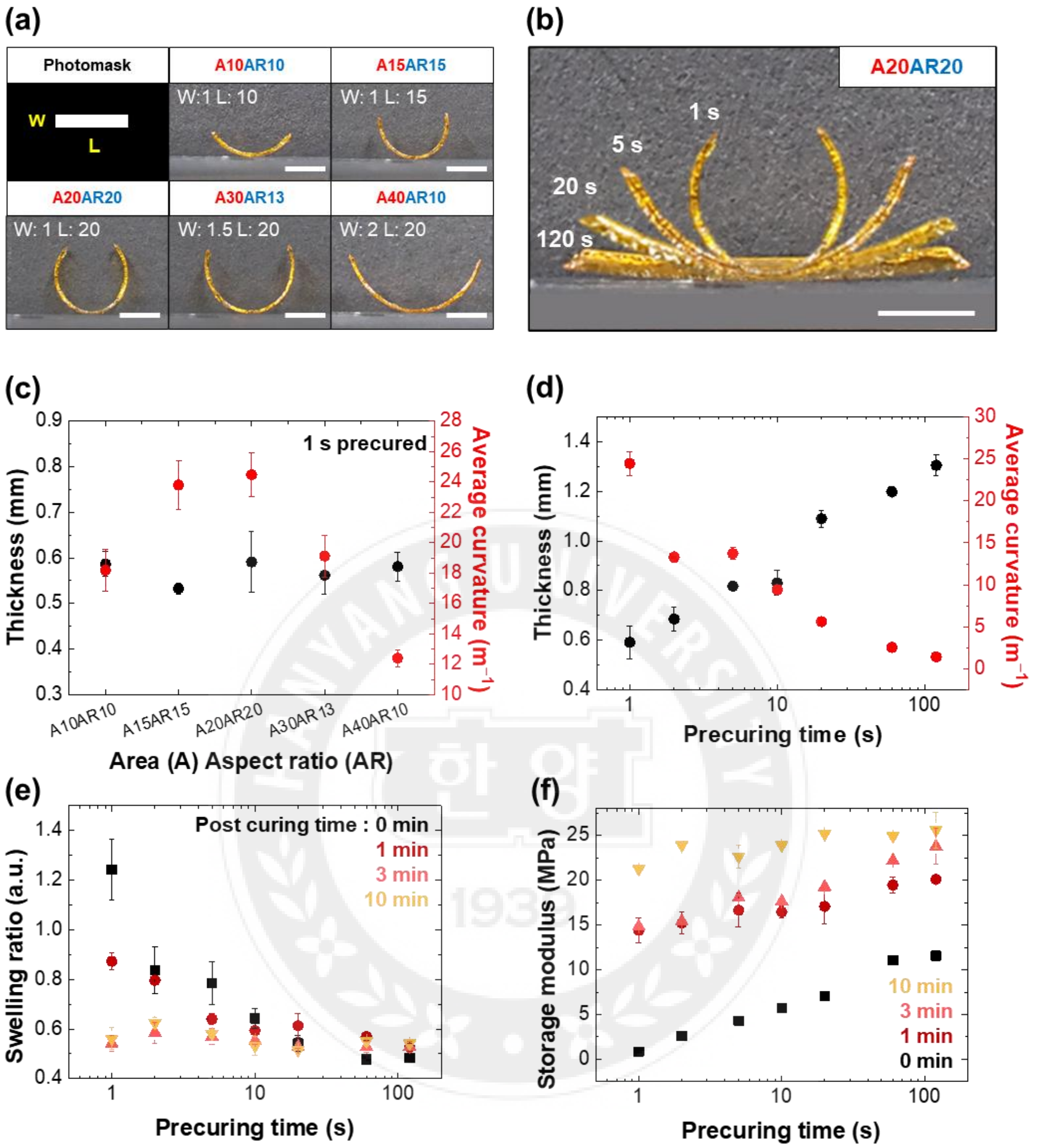

Figure 2.5 Rectangular curvilinear structures. (a) 2D photomask pattern design effect: Cross-section images of 1 s precured five types of rectangular curvilinear structures. W and L inside digital images indicate width and length, respectively. Units: mm. (b) Precuring time effect: A20AR20 rectangular curvilinear structures with different precuring times of 1, 5, 20, and 120 s. (c) Thickness and curvature variation depending on photomask 2D pattern designs. (d) Thickness and curvature of the curvilinear structure using the A20AR20 design were demonstrated as a function of precuring times. (e,f) Postcuring effect: Swelling ratio and mechanical strength were described as functions of precuring times under four types of postcuring times. Scale bars, 5 mm.

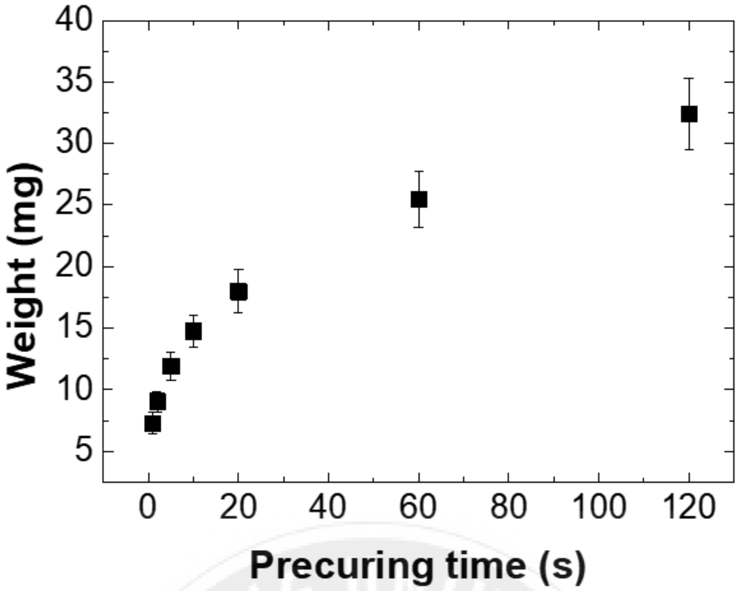

Figure 2.6 The weight change of A20AR20 curvilinear prestructures under varied precuring time from 1 to 120 s.

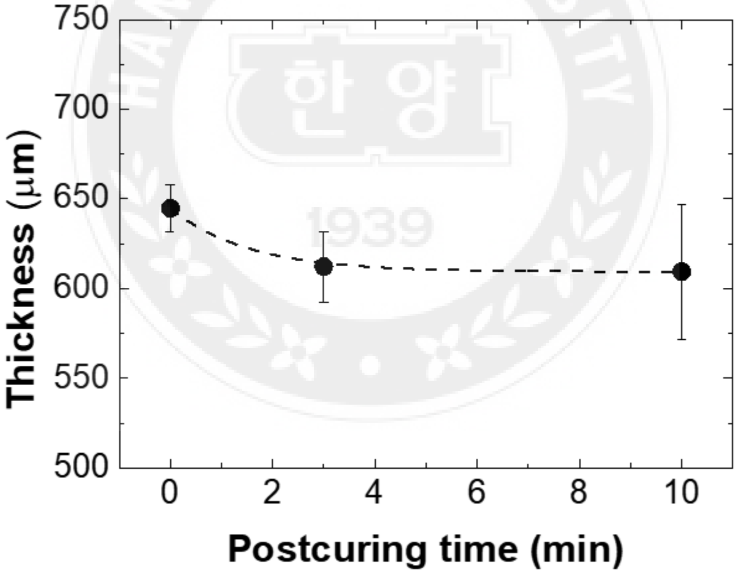

Figure 2.7 Thickness variation of prestructure under 0, 3, and 10 min of postcuring under UV light with an intensity of 16.5 mW cm^{-2}.

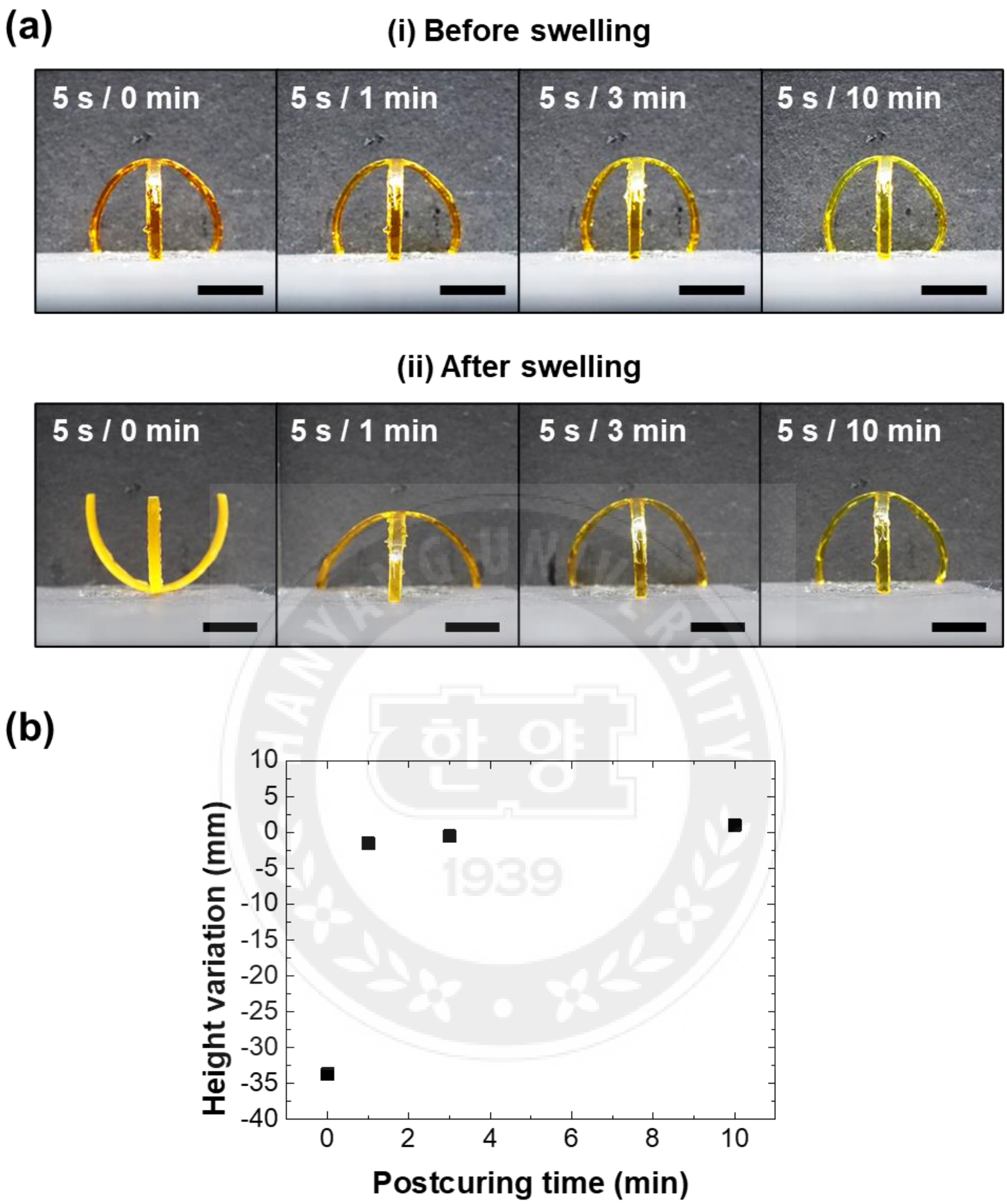

Figure 2.8 Swelling effect. (a) Image of a 3-leg freestanding structure (i) before swelling and (ii) after 1 h of swelling in deionized water. The precuring time was fixed at 5 s, and 4 different postcuring times (0, 1, 3, 10 min) were applied for each prestructure. Scale bar, 1 cm. (b) The height change of swelled structures was demonstrated as a function of postcuring time.

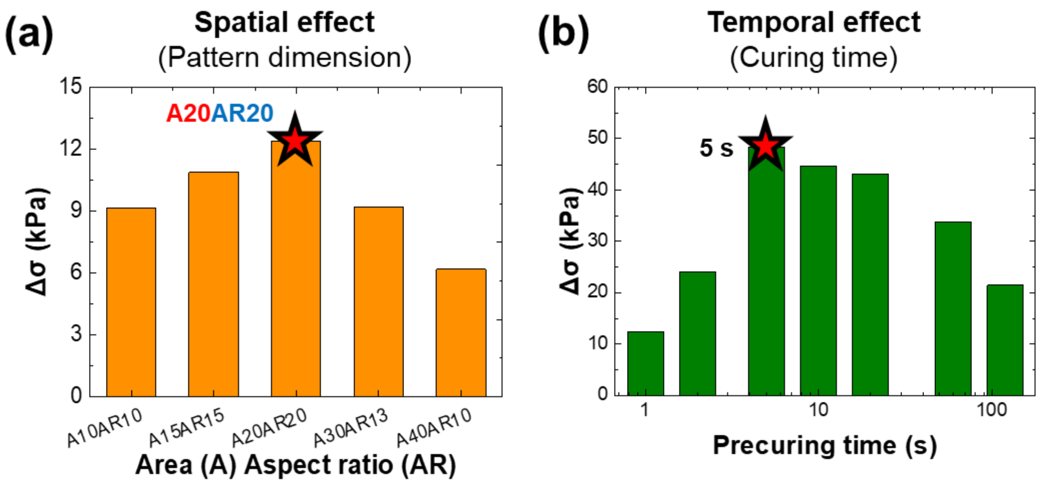

Figure 2.9 Internal stress difference between the bottom and top surface of polymer structure prepared by controlling (a) spatial effect and (b) temporal effect.

To achieve programmable 3D shape-reconfiguration of a polymer monolith, we systematically investigated the effects of spatiotemporal parameters on internal residual stress difference and finally on dimensions of 3D reconfigured structure (**Figure 2.4a and 2.4b**). 2D patterns were designed by adjusting the width (*W*: 1, 1.5, and 2 mm) and length (*L*: 10, 15, and 20 mm) to change areas (*A*) and aspect ratio (AR). The area (*A*) of each 2D pattern was 10, 15, 20, 30, and 40 mm^2 and the AR was 10, 13, 15, and 20. For example, the 2D pattern with an area of 20 mm^2 and an aspect ratio of 20 was named A20AR20.

First, the effect of 2D pattern design on the dimension of reconfigured structure was investigated using 5 types of reconfigured structures which were prepared by applying a uniform precuring condition for 1 s. The degree of bending deformation was governed by the amount of generated internal stress difference between the top and bottom regions (**Figure 2.5a and 2.5c**). A negligible difference in thickness was observed regardless of the 2D photomask pattern design under a constant precuring time of 1 s. However, significant variation in the curvature was observed for five types of rectangular curvilinear 3D structures due to the differently programmed residual stress profile. First, the highest average curvature of 24.5 m^{-1} was achieved from the A20AR20 design showing the largest aspect ratio in our system. The curved structure prepared using a smaller length design than 20 mm exhibited lower average curvature of 23.8 m^{-1} (A15AR15) and 18.2 m^{-1} (A10AR10) than the curved structure prepared using the A20AR20 design. The reduction of AR caused a reduction in the bending curvature of rectangular curvilinear 3D structures due to the short length reducing the magnitude of the longitudinal residual stress. In the A30AR13 and A40AR10 patterns, a further reduction in curvature was observed as the curvatures of each design were measured to 19.1 m^{-1} (A30AR13) and 12.4 m^{-1} (A40AR10). In this case, the compressive force acting in the width direction increased due to the Poisson ratio.[35] Hence, the compressive force acting in the longitudinal direction was interfered with during the 3D shape-reconfiguration with the larger width. In addition, a greater mass of curvilinear 3D structures prepared from larger 2D pattern areas induced a reduction in curvature due to the increased gravitational force (**Figure 2.6**). Therefore, further curvature reductions were observed from the 2D patterns with larger widths.

To investigate the temporal parameter effects, the FPP was conducted with varied precuring times of 1, 2, 5, 10, 20, 60, and 120 s (**Figure 2.5b and 2.5d**). A significant thickness change was observed with a change in precuring time. The thickness of curved structures increased from 0.59 to 1.30 mm as precuring time increased from 1 to 120 s, respectively. On the other hand, the curvature decreased from 24.5 m^{-1} to 1.4 m^{-1} as precuring time increased from 1 to 120 seconds, because greater thickness induced higher bending stiffness, which was proportional to thickness. These results indicated that the thickness of 3D structures was independent of 2D pattern design but predominantly dependent on the curing time, and the curvature was directly affected by the 2D pattern design and thickness.

The crosslinking density of a polymer had a significant effect on its swelling ratio and elastic modulus. First, significant swelling occurred for the curved structures through trapping water molecules inside empty space among network structures owing to the hydrophilic poly(ethylene glycol) moieties. The swelling ratio was described as the ratio of internally trapped solvent weight to the original polymer weight.

$$Swelling\ ratio = \frac{W_s - W_d}{W_d} \quad (2.2)$$

The W_s is the weight after swelling, and W_d is the weight in its completely dried state (**Equation (2.2)**). With increased precuring time, the swelling ratio of the prestructure decreased from 1.24 at 1 s to 0.48 at 60 s due to increased crosslinking density. Additionally, the postcuring process increased the conversion of polymerization and formed more tightly crosslinked networks through further curing of the remaining oligomers inside the prestructures. Above 3 min of postcuring, the swelling ratio of the poststructures became saturated to be 0.55 ± 0.03 regardless of precuring times (**Figure 2.5e**). In contrast to the swelling ratio, the elastic modulus had a positively proportional relationship with curing times. The dimensional stability was consequently increased owing to the tightly crosslinked structure which has higher storage modulus.[36] Typically, prestructures had significantly lower mechanical strength than poststructures. The storage modulus of the prestructures was measured to be 0.9 MPa after 1 second of precuring, which increased with prolonged precuring and eventually reached saturation at 11.3 MPa after 60 seconds of precuring. A drastic enhancement in storage modulus was observed after the postcuring of the prestructure. In particular, the storage modulus was saturated in the range of 20-25 MPa from all precuring conditions when 10 min of postcuring was applied (**Figure 2.5f**). Additionally, when postcuring was performed for more than 3 minutes, it was accompanied by a negligible thickness reduction of approximately 4% compared to the initial thickness, and swelling-induced dimensional change was negligible (**Figure 2.7 and 2.8**). Therefore, all prestructures were postcured for 10 minutes to stably maintain the curved 3D shape of the prestructures against moisture and external impacts.

For quantitative analysis, we calculated the internal stress difference between the top and bottom regions of the curved structure using Hooke's law, since the shape reconfiguration via FPP occurred in the elastic deformation regime without yielding (**Figure 2.9a and 2.9b**).

$$\sigma(x) = E \times \varepsilon(x),\ \ \varepsilon(x) = k \times x \quad (2.3)$$

$$\Delta\sigma(x) = E \times k_{top} \times \frac{t}{2} - E \times k_{bottom} \times (-\frac{t}{2}) \quad (2.4)$$

Here, $\varepsilon(x)$ is the distance-dependent strain, σ is the stress, $\Delta\sigma$ is the accumulated stress difference between the top and bottom regions, E is the elastic modulus, k is the curvature of the top and bottom regions, and t is the thickness. **Equation (2.3)** demonstrated Hooke's law, and the strain was described as the product of curvature and distance from the center. **Equation (2.4)** described the stress difference between the top and bottom regions, which were located at positions +t/2 and –t/2 away from the center, respectively. The center of the polymer was set as the zero point. First, the curved structure from the A20AR20 design with fixed 1 s precuring time demonstrated the highest stress difference between the top and bottom regions, which was well aligned with the results from **Figure 2.5c**. In the fixed 2D pattern design test, curvature and thickness exhibited a trade-off relationship influenced by bending stiffness. Consequently, the optimal precuring time for achieving the highest stress difference was found to be 5 s.

2.2.3 Self-Supporting 3D Structures

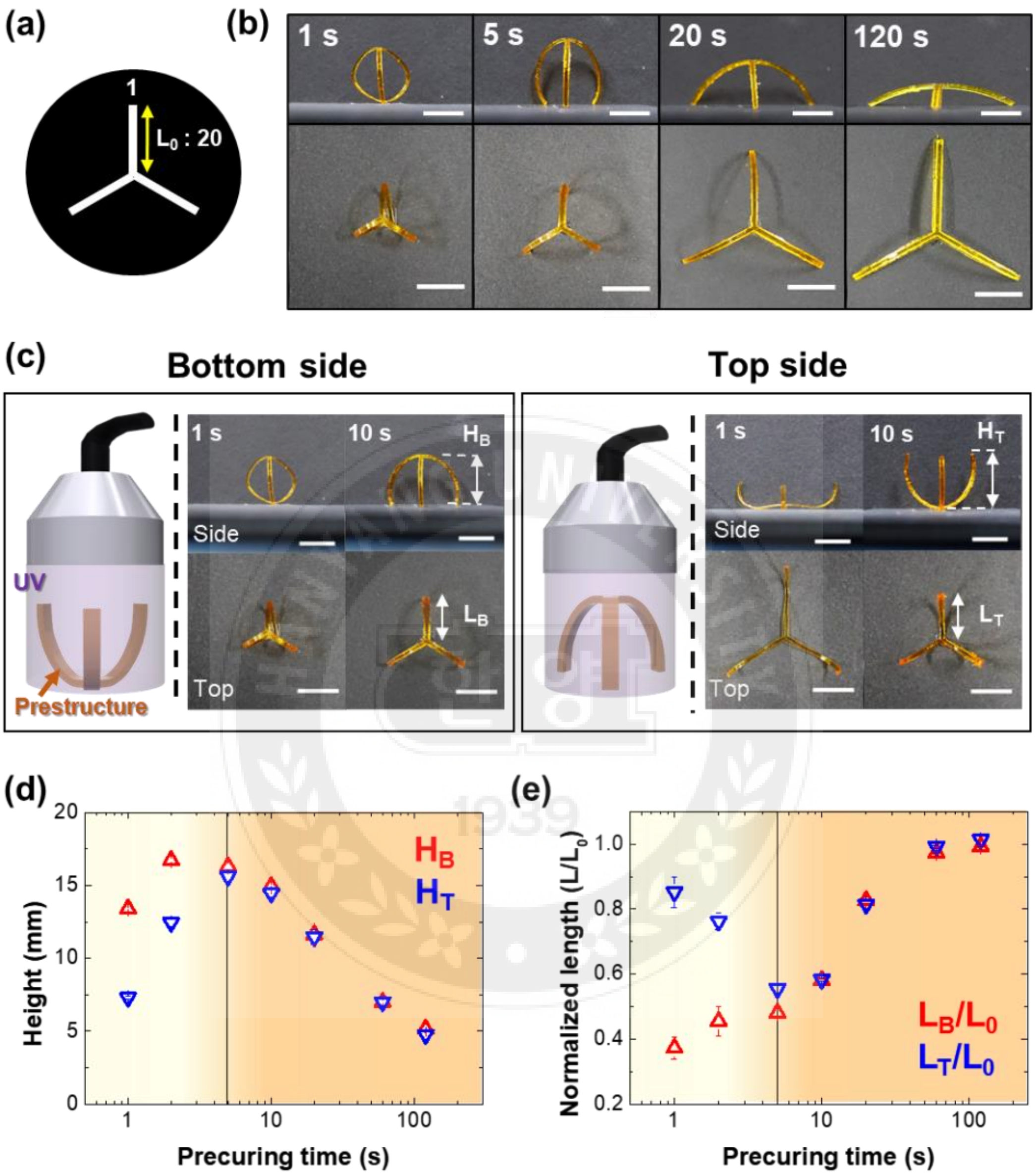

Figure 2.10 Freestanding 3D structures. (a) photomask design consisted of three A20AR20 legs for self-supporting structures. Units, mm. (b) Different top and side views of freestanding 3D structures with varied precuring times. (c) Comparison of the external shape of postcured 3D structures according to the direction of postcuring. (d,e) Postcuring direction-dependent height and normalized length of freestanding 3D structures were shown as a function of precuring time. Scale bars, 1 cm.

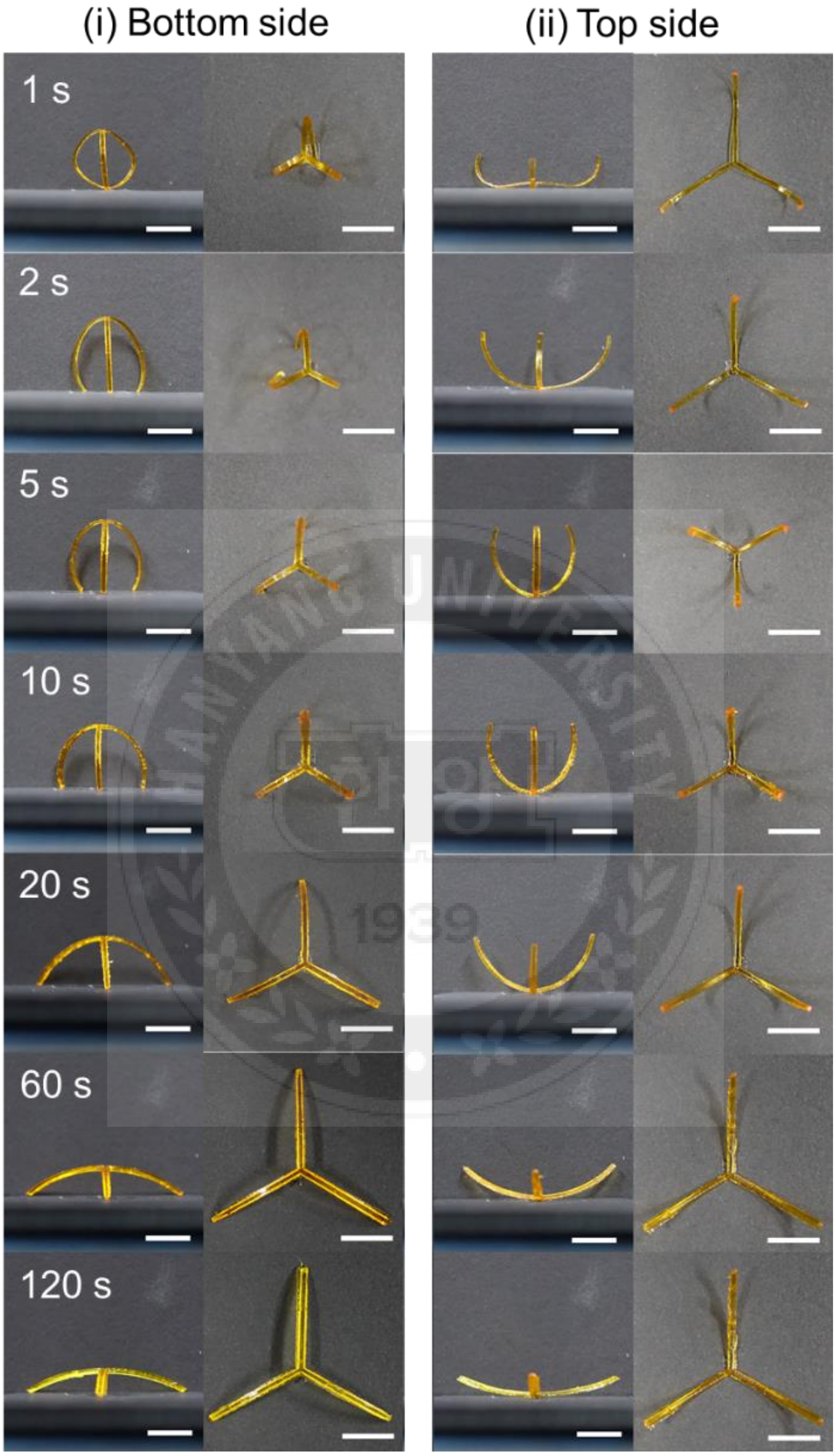

Figure 2.11 Images of 3-leg freestanding structures with 7 types of precuring time (1, 2, 5, 10, 20, 60, and 120 s) and 2 types of postcuring direction: (i) bottom side postcured, (ii) top side postcured. Scale bars, 1 cm

To develop a self-supported structure, we designed a new photomask 2D pattern by radially arranging three A20AR20 patterns at 60° intervals (**Figure 2.10a**). Using this new design, we achieved a three-leg freestanding structure that demonstrated self-supporting capability. First, the dimension of the 3D freestanding structure was precisely controlled by adjusting precuring time (**Figure 2.10b and 2.11**). Furthermore, the postcuring direction which was divided into the bottom side and top side caused a noticeable change in the external shape and dimensions of 3-leg freestanding structures under different precuring time conditions. The height and length of the freestanding structures were determined by the bottom side postcured height (H_B) and length (L_B), and top side postcured height (H_T) and length (L_T). When the bottom of the prestructure was kept directly on the floor, each leg was subjected to gravitational forces, and insufficient precuring time resulted in a low modulus, leading to reduced self-supporting capability. In contrast, when the bottom of the prestructure was inverted to face upwards, the three legs effectively supported the structure, allowing the freestanding structure to be maintained relatively well even with a slightly lower modulus (**Figure 2.10c**). Therefore, the shape of the postructure is determined by postcuring direction. Furthermore, the dimensions (L_T, H_B, H_T, and L_B) of the freestanding structures are also governed by the temporal effect of the precuring process. First, a large height deviation of 6.1 mm was measured between H_B and H_T with a 1 s precuring time (**Figure 2.10d**). The height deviation was reduced to 4.3 and 0.7 mm by increasing precuring time from 2 to 5 s, respectively. Finally, the height difference between H_B and H_T became negligible with a precuring of more than 10 s. In particular, the greatest height measured was 16.7 mm with a leg length of 9.1 mm under the bottom direction postcuring after precuring for 2 s. Since the curvature of the rectangular curvilinear 3D structure that was precured for 1 s represented the highest value, the height of the freestanding structure precured for 1 s was observed to be smaller than that under the condition of a 2 s precuring. Each leg was overlapped by a large, curved shape, and the height and length were relatively small under 1 s precuring conditions.

The length measured in the top view had a trade-off relationship with the height measured from the side view (**Figure 2.10e**). The lengths, L_T and L_B became comparable with a precuring time of more than 5 s. With prolonged precuring, both L_T and L_B drastically increased and converged to the initial L_0 value at the expense of a height reduction. With a precuring time of fewer than 5 s, the fabricated 3D structure had insufficient modulus to establish a self-supporting system due to the short duration available for crosslinking. When the postcuring process was carried out from the bottom, the 3-leg freestanding structures supported the entire weight by evenly distributing the weight over the three legs. Conversely, the individual leg was not subjected to supporting the 3D structure in the case of the top side postcuring. Each leg was being pulled equally by gravitational force. Thus, the H_B value was larger than the H_T and the L_T/L_0 ratio of the top side postcuring case was larger than the L_B/L_0 ratio until the 10 s time mark. With a precuring longer than 10 s, the stiffness of the crosslinked architecture was sufficient to self-support regardless of the postcuring directions.

2.2.4 Multileg Freestanding Structures

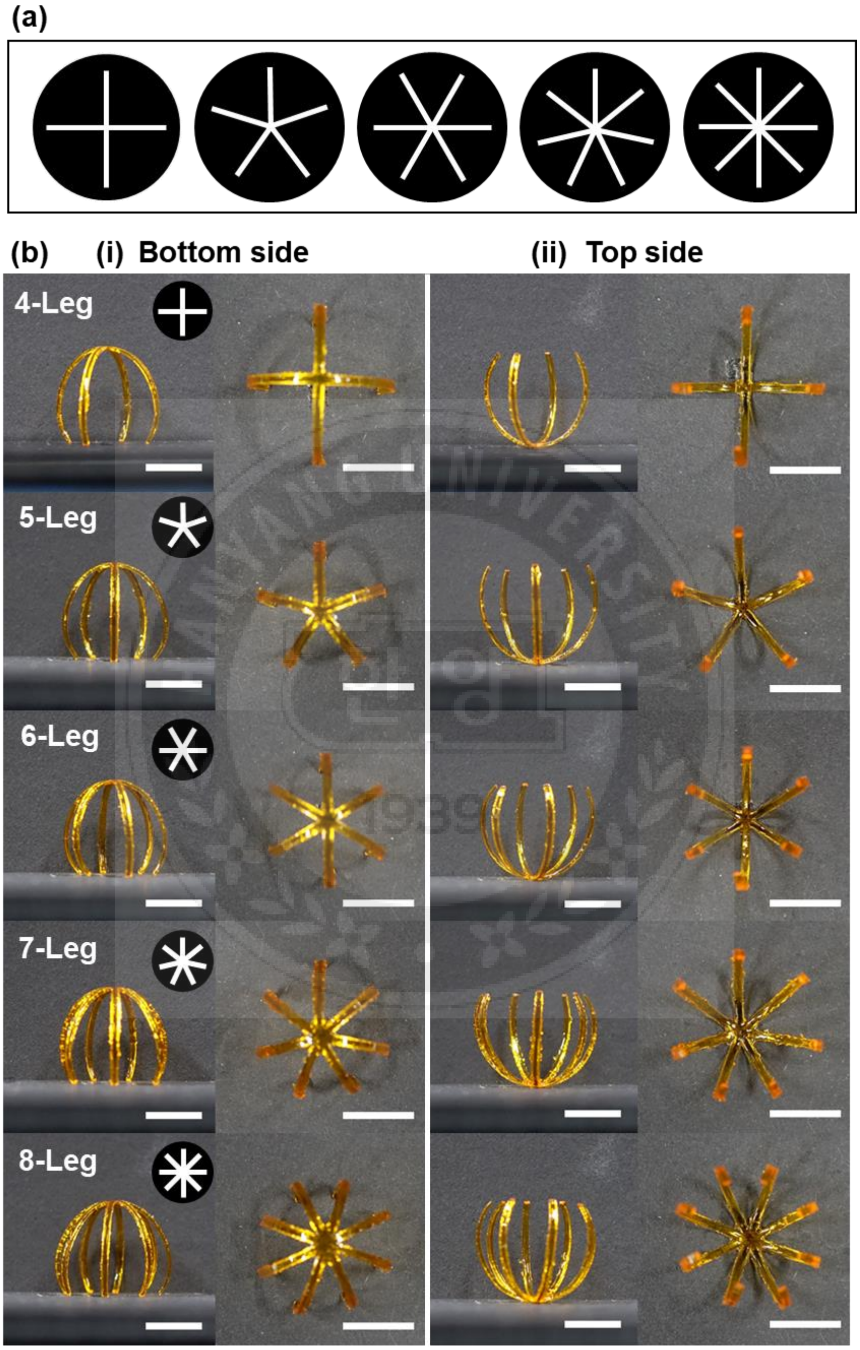

Figure 2.12 (a) Photomask images with multileg design. (b) Multileg 3D structures fabricated by multileg-designed photomasks and different postcuring directions: i) bottom side postcured and ii) top side postcured. Scale bars, 1 cm.

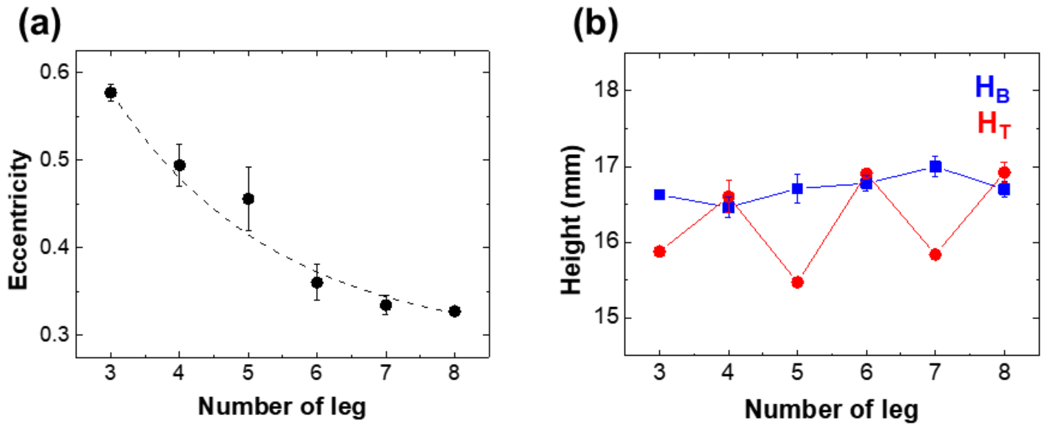

Figure 2.13 (a) Exponential decrease of eccentricity according to increased number of legs in multileg structures. (b) Comparing H_B and H_T differences according to different numbers of legs in multileg structures.

In addition to the effects of the precuring time and postcuring direction, a complex design of 2D patterns was considered to enrich the design strategy for complex 3D freestanding architectures. First, we systematically varied the number of legs from 3 to 8. Again, the legs in all patterns were maintained as A20AR20 and radially arranged with the same interval angle. The regular polygons were drawn by connecting the outer vertices that correspond to the leg numbers (**Figure 2.12a**). To investigate the effect of leg number, 5 s of precuring time was utilized due to the previously discussed self-supporting capability and the comparable H_B and H_T values. The digital images of the multileg freestanding structure were presented from four different directions to quantify their geometry including eccentricity, height, and area of polygons by their vertices (**Figure 2.12b**). The eccentricity of the ellipsoid was expressed as the ratio of the distance from the center to the focus c and the length of the semi-major axis a (**Equation (2.5)**)

$$e = \frac{c}{a} \quad (2.5)$$

For a perfect circle, the eccentricity was zero as the distance from the center to the focus is zero. The multileg structures resembled hollow spheres with unfilled outer areas.

The incomplete sphere-like structures of multileg building blocks caused a difference in eccentricity and height depending on the number of legs. First, when the number of legs increased, the cross-sectional area of the multileg structure approached that of a circle and the eccentricity of the multileg structure decreased simultaneously. The eccentricity declined exponentially, with 0.66 for 3-leg and 0.23 for 8-leg structures, with an increase in the number of legs (**Figure 2.13a**). This was due to increased interactions between adjacent legs caused by a smaller angle between them as the number of legs increased. As the number of legs increased from 3 to 8, the vertex angle between the legs decreased from 120° in an equilateral triangle to 45° in an equilateral octagon. The vertex angle of the multileg structure changed the amount of additional internal stress which subsequently affected the formation of the bending shape.

The number of legs and symmetricity also determined the stress distribution which affected the final 3D multileg freestanding structures. When even-numbered legs were prepared with rotational symmetry, stress components with the same magnitude were paired in the plane. While the odd-numbered geometry provided both rotational and reflection symmetry, this even/odd effect driven by the inversion symmetry generated a phase difference in the stress distribution. As a result, differences between H_T and H_B were observed in the odd-numbered leg structures (3-leg, 5-leg, and 7-leg) (**Figure 2.13b**). The even-numbered multileg structures (4-leg, 6-leg, and 8-leg) presented a negligible difference between H_T and H_B. Two pairs of legs aligned in phase behaved like a leg with double the AR only in even-numbered leg structures. Therefore, residual stresses generated in the longitudinal direction of the legs were theoretically doubled in the even-numbered leg structures and rendered the capability to have an enhanced bent shape and to withstand its weight.

2.2.5 Spirality-Embedded 3D Structures

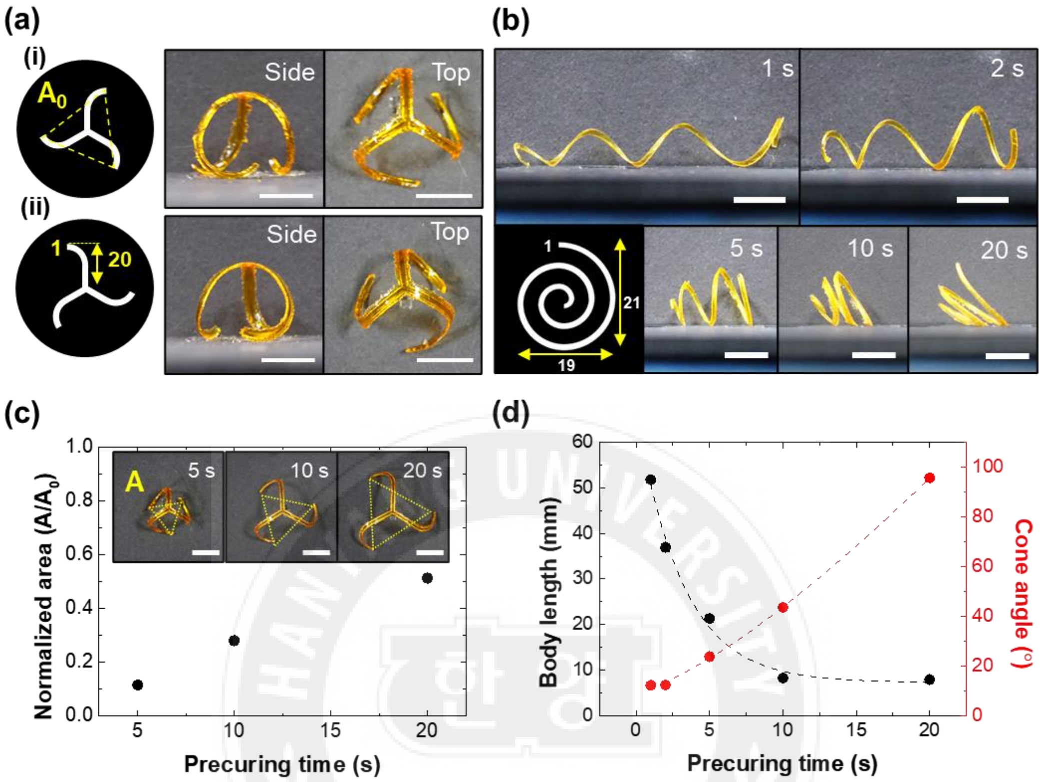

Figure 2.14 Spirality embedded 3D structures. (a) Twisted-leg structures fabricated by chirality applied photomask under 5 s precuring time. Chirality was determined depending on the direction of leg ends: i) right-handed and ii) left-handed. (b) Images of helical 3D structures with prolonged precuring (1, 2, 5, 10, and 20 s) and skim of spiral patterned photomasks. (c) Increased normalized area (A/A_0) of twisted leg structures as prolonged precuring. (d) Body length and cone angle of helical 3D structures were demonstrated as a function of precuring times represented in (b). Scale bars, 1 cm.

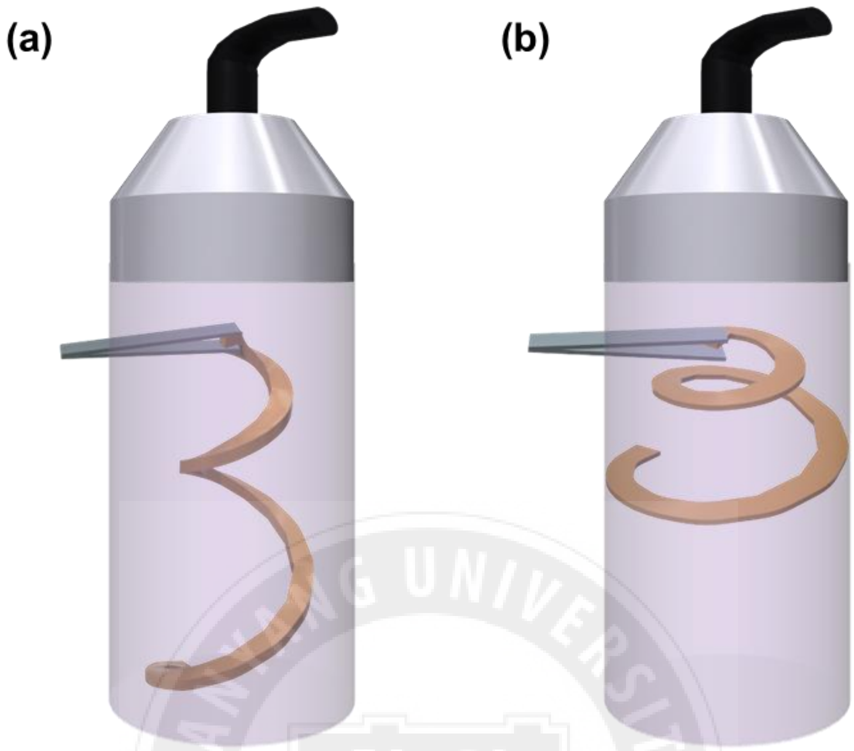

Figure 2.15 Schematic diagram of postcuring process of spiral 3D structures: postcuring processes for prestructure fabricated under the condition of (a) short time precuring and (b) long time precuring

The introduction of spirality in 2D photomask patterns was a distinct design approach from the multileg 2D patterns which have a linear leg design. A 2D pattern with spirality was applied to the fabrication of complex structures containing helically twisted shapes rather than unidirectional bending. First, a twisted-radial leg pattern was designed by applying spirality to each leg of the A20AR20 3-leg 2D pattern (**Figure 2.14a**). The chirality of a twisted-radial leg 2D pattern was controlled by adjusting the end direction of each leg pattern. The A20AR20 3-leg pattern was modified to have an arc shape at the end of each leg pattern. Each arc shape had a radius of 10 mm and an angle of 90°. The rectangular pattern connected to the arc had a width of 1 mm and a length of 10 mm. The pattern where the end of each leg followed a clockwise direction was named "right-handed" and the pattern in the counterclockwise direction was named “left-handed”. In previous 3-leg freestanding structures, the width of the triangle, formed by the imaginary lines between the leg ends observed in the top view, increased from the center point as the precuring time increased. In the twisted-radial leg structure, the area of a triangle, formed by the end of the leg, also increased by the precuring time and simultaneously rotated in the opposite direction of the inherent handedness (**Figure 2.14c**).

Second, we designed a spiral 2D pattern in which the radius decreased by 1 mm with a 90° rotation (**Figure 2.14b**). The structure of a spiral 2D pattern was composed of an outside arc and an inside arc with a 1 mm width difference. The initial radius of each arc was 11 and 10 mm, and each arc radius decreased by 1 mm with a 90° rotation. Postcuring of 3D spiral structures proceeded when the end tip of the prestructure was lifted with tweezers (**Figure 2.15**). The five types of spiral 3D structures were prepared by varying precuring times. The total length and pitch of the spiral 3D structure showed a noticeable difference according to the precuring time changes. In the case of spiral 3D structures, we investigated two comparable structural elements such as body length and cone angle (**Figure 2.14d**). The body length and cone angle had a trade-off relationship, and a spiral 3D structure with a large body length had a relatively small cone angle. The 3D spiral prestructure was stretched down under the influence of gravity during the postcuring process due to insufficient mechanical stiffness under a short precuring time condition. Therefore, the spiral 3D structure produced with a short precuring time had a relatively long body length and a small cone angle.

2.2.6 3D Hierarchical Collective Assembly

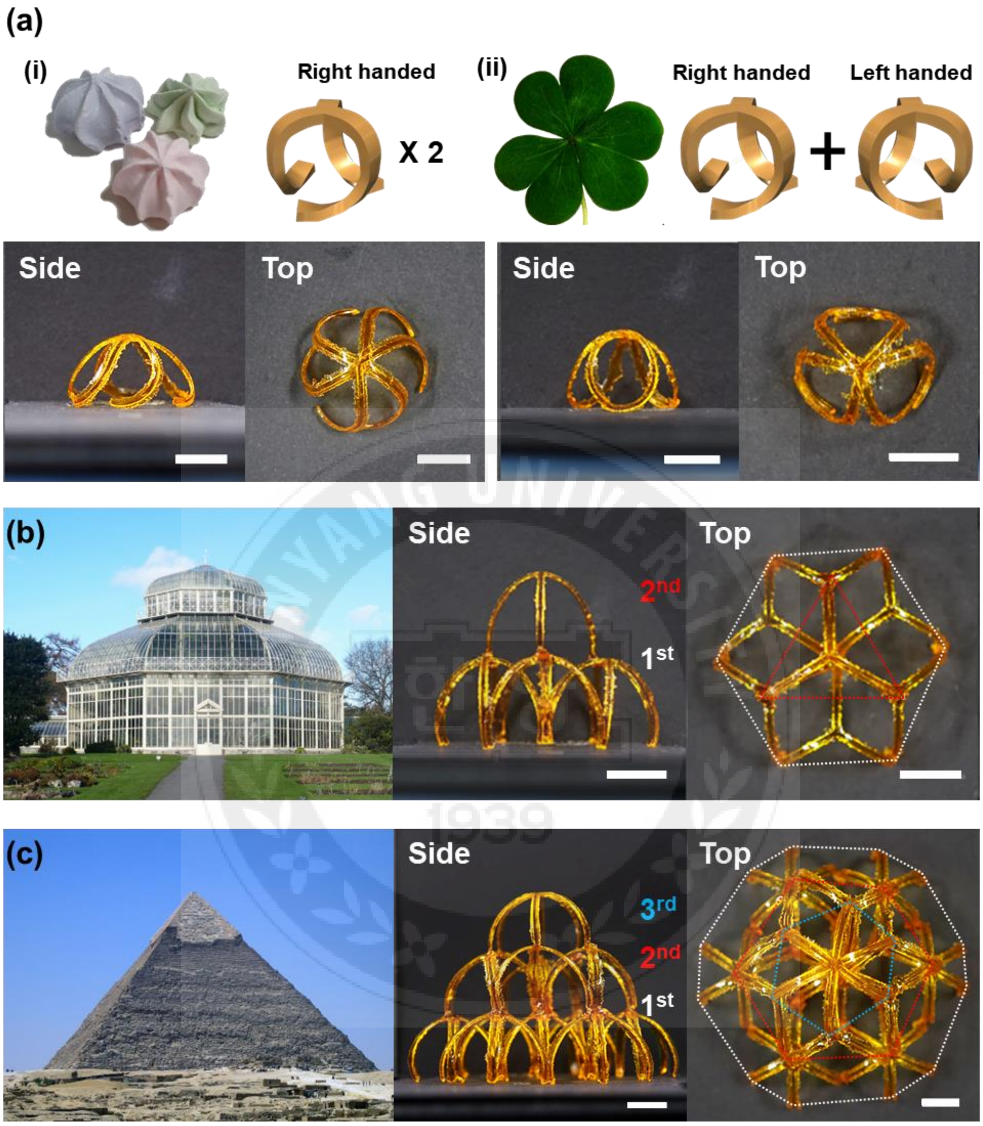

Figure 2.16 Demonstration of collective assembled architectures. (a) Inspired by curve-shaped sundries in our lives, a hollow radial structure mimicking the exterior frame of i) meringue cookies and ii) three-leaf clover were demonstrated by collective assembly of two twisted-radial leg structures with the same or different handedness; (b,c) collective assembled hierarchical 3D structures composed of freestanding building blocks were demonstrated as replicating famous man-made huge architectures. (b) The exterior frame of the National Botanic Garden in Ireland[37] was replicated by the assembly of seven 5 s precured freestanding building blocks. (c) Two types of freestanding building blocks (3-leg and 6-leg) were collectively assembled as three-layered hierarchical architecture inspired by the Great Pyramid in Giza[38]. Scale bars, 1 cm.

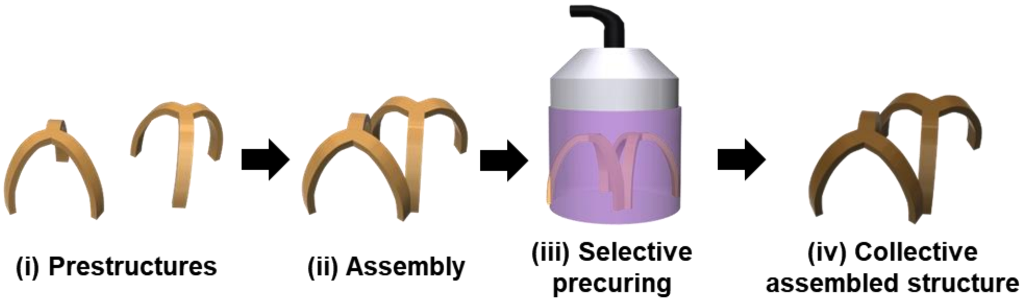

Figure 2.17 Schematic illustration of collective assembly for hierarchical 3D architecture: (i) preparation of building blocks (prestructures); (ii) assembly of building blocks to attach desired points; (iii) selective postcuring at attached point; (iv) scheme of a postcured collective assembled structure

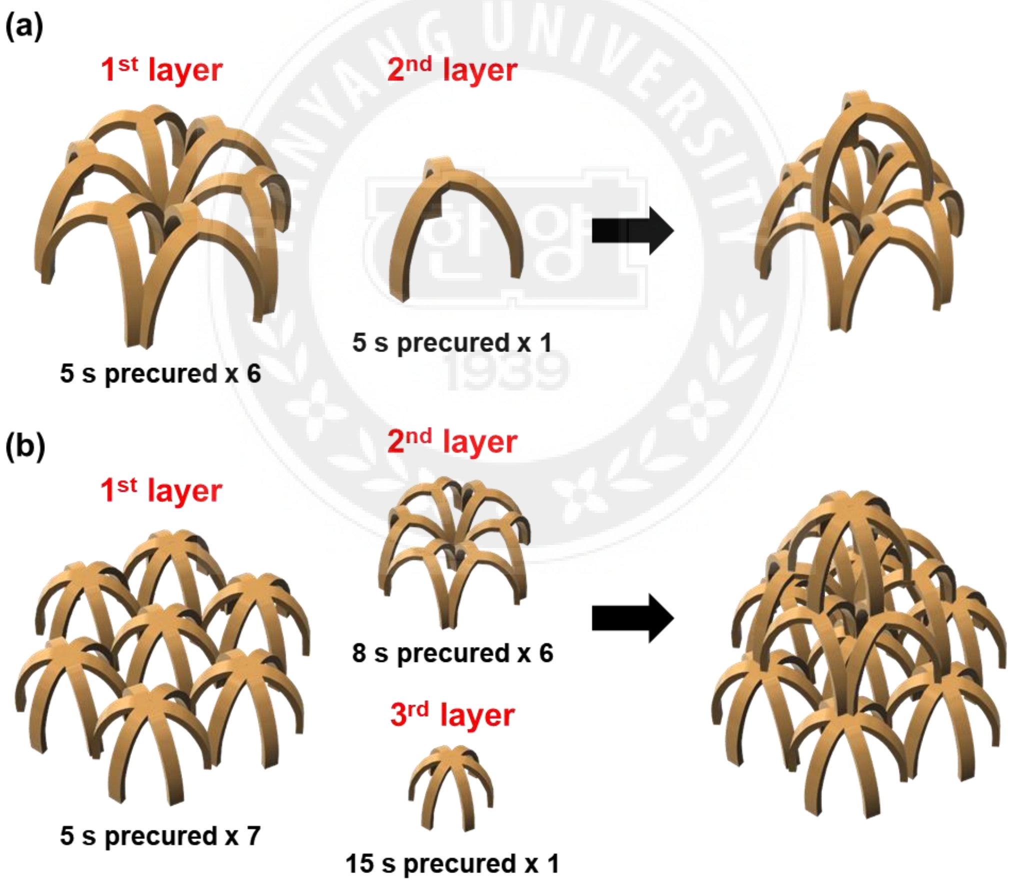

Figure 2.18 Schematic diagram of collective hierarchical 3D structures: (a) illustrations of 3D structure and layer components that replicated the Irish National Botanical Garden; (b) illustrations of 3D structure and layer components that replicated pyramid

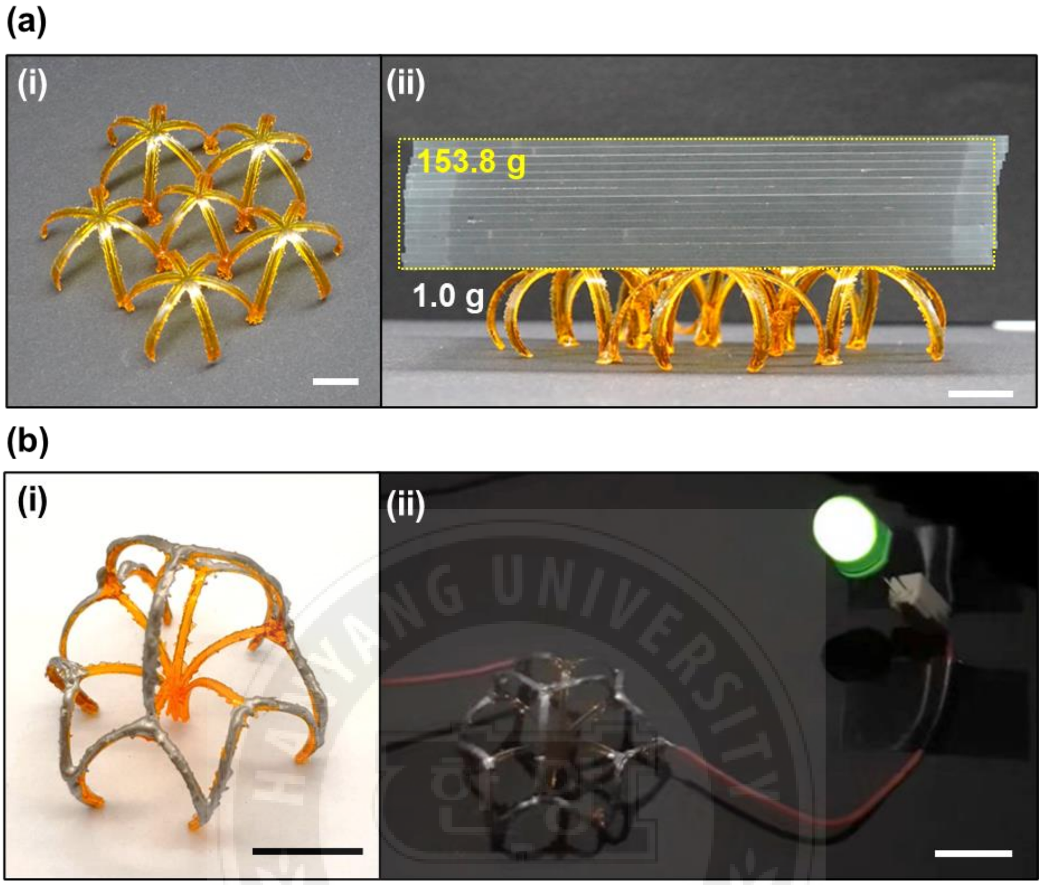

Figure 2.19 (a) i) One-layer assembled structure consisting of one 5-leg, five 6-leg building blocks with a total weight of 1.0 g. ii) The assembled structure withstood 153.8 g, the weight of 16 slide glass. (b) 3D electronic connection via collectively assembled structure. i) Photograph of silver paste-coated collectively assembled structure. ii) Demonstration of turning on LED in a circuit system including 3D assembled structure. Scale bars, 1 cm.

Many design strategies of huge architectures often have introduced the concept of collective assembly of several small building blocks. The assembly of basic building blocks constructed complex shapes of a multifunctional architecture. For example, the Great Pyramid in Giza, an ancient Egyptian architecture, was one of the most famous architectures constructed by applying the collective assembly of heavy bricks. Here, we demonstrated collective assembly by employing 3D shape-morphed structures as building blocks to achieve scaled-up and complex architectures (**Figure 2.16**).

Collective assembly was conducted by attaching building blocks via selective postcuring (**Figure 2.17**). First, two homochiral leg structures or two heterochiral leg structures were assembled by connecting each of the center vertices via photopolymerization. The collective assembly with two homochiral 3-leg structures replicated the exterior feature of meringue cookies described as a hollow radial shape (**Figure 2.16a(i)**). A three-leaf clover-like rotational symmetric structure was also implemented by the collective assembly of two heterochiral 3-leg structures (**Figure 2.16a(ii)**). The structure presented in **Figure 2.16b** was a miniaturized front view frame of the Irish National Botanical Garden manufactured through the assembly of seven 3-leg freestanding structures. The miniaturized Irish National Botanical Garden consisted of two layers and seven freestanding 3D structures prepared by 5 s of precuring. The first layer had six building blocks, and the second layer had one building block: a total of seven building blocks (**Figure 2.18a**). The view of the replicated structure varied according to the direction of observation. In particular, the top view is similar to a feature of snowflakes. The final model structure for the collective assembly was inspired by the Great Pyramid of Giza. As shown in **Figure 2.16c**, the pyramid-mimicking structure consisted of a total of 14 multileg structures assembled as a three-layered geometry. The assembled building blocks applied for each layer were manufactured under different precuring time conditions (5, 8, and 15 s). A total of seven 6-leg freestanding structures fabricated with a 5 s precuring time were assembled as the first layer, and six 3-leg freestanding structures fabricated with a precuring time of 8 s were assembled as the second layer (**Figure 2.18b**). The assembled second layer was located on the first layer as each of the top centers of building blocks fit with those in the first layer. Finally, a 6-leg structure with 15 s precuring condition was located on the top of the pyramid-mimicking structure. The side view of the pyramid-mimicking structure was also like that of a cheerleading human pyramid (**Figure 2.18c**).

The hierarchically assembled 3D structure showed enhanced weight withstanding performance and was applied as a 3D complex framework of electronics. As shown in **Figure 2.19a**, the single 1.0 g layer of assembled 3D structure was able to withstand the weight of a 153.8 g 16-slide glass. The one-layered structure was achieved by placing a 5-leg freestanding structure at the center and assembling 6-leg freestanding structures on each leg of the 5-leg structure. In addition, we demonstrated the potential application in electronic devices by introducing silver paste on the outer part of the collective assembled structure. A silver-coated assembled structure was connected to an electric circuit system composed of two 1.5 V batteries, a switch, and a 10 mm diameter LED and successfully turned on the LED (**Figure 2.19b**).

2.3 Conclusions

In Chapter 2, we reported a facile strategy to prepare complicated 3D architectures through the collective assembly of tailorable polymeric monolithic building blocks. The fast and replicable system was composed of an inexpensive photomask along with unpolarized UV light and enacted with the simple switching on/off UV light. The dimensions of the various designed building blocks were intricately programmed by controlling the spatiotemporal effects of light exposure conditions and 2D elements of patterns including aspect ratio, even/odd condition, and fractal elements. The morphological diversity of the building blocks was demonstrated in the construction of self-supporting 3D structures, the spirality of embedded structures, and multi-curved monolithic structures. The detailed dimension difference of basic building blocks was investigated through reasonable geometric analyses of 3D building blocks. Finally, the collective assembly was introduced to manufacture multi-shaped hierarchical 3D structures. The diversely designed building blocks greatly released the limitations on the size and design of 3D collectively assembled architectures. Inspired by the hollow and interconnected structure of a honeycomb, the scaled-up one-layer assembled 3D structure withstood 150 times its weight. The assembly structure was also applied for a 3D framework of electronics with simple additional treatments. The silver paste-coated 3D assembled structure was demonstrated to perform as an electrical connector within the circuit system. We believed that our creative design strategies had advantages for the rapid prototyping of complex 3D electronics, antennas, and optics.

2.4 Experimental Section

Preparation of patterned photomask: The polydimethylsiloxane precursor (PDMS, Dow Chemical Company) and curing agent were poured into a clean glass dish in a 10:1 weight ratio and then thermally cured at 90 °C in a convection oven for 1 h. The PDMS substrate was placed on the glass dish in a 0.5 mm layer for easy detachment of the prestructures. The paper sticker with the designed 2D pattern was attached to the bottom side of the glass dish.

Photocurable Mixture: The photocurable mixture was composed of 99.28 wt% PEGDA (Mn = 700, Sigma Aldrich) as a photocurable monomer, 0.05 wt% Sudan 1 (Sigma Aldrich) as a photoabsorber, and 0.67 wt% 2-benzyl-2-(dimethylamino)-4′-morpholino-butyrophenone (Irgacure-369, BASF) as a photo initiator. All chemicals were used as received and sufficiently stirred for homogeneous dispersion.

3D structure preparation via frontal photopolymerization (FPP): The photocurable mixtures were poured into the photomask glass dish and bubbles were removed using a vacuum pump. Under the patterned photomask, a collimated UV light (pE-4000, CoolLED, UK) with a 365 nm wavelength was irradiated at 12.3 mW cm^{-2} of center intensity. The light intensity was measured by a power meter (PM100D, Thorlabs, USA). After the light irradiation, the precured structure was generated and peeled from the photomask. Finally, the precured structure was postcured using UV light with 16.5 mW cm^{-2} of center intensity.

Mechanical and swelling properties analysis: The rectangular-shaped pre/poststructure with fixed dimension (1 mm × 20 mm: width × length, AR20) was utilized for DMA measurement and swelling test. The storage modulus was measured by dynamic mechanical analysis (DMA, Q800, TA instrument). The storage modulus of pre/poststructure was measured by utilizing strain sweep mode with a single amplitude of

20 μm at room temperature. For the swelling test, the rectangular-shaped pre/poststructures were immersed in deionized water for 3 h at room temperature. The weight of pre/poststructures was measured before and after swelling at 56% relative humidity.

Preparation of Collective Assembled 3D Structures: The prestructures were assembled by contacting the corresponding joint parts of other prestructures. Then, UV postcuring was conducted on the assembled structure. The silver-coated assembled 3D structure was prepared through a coating of silver paste (ELCOAT, P-100, CANS).

Structural Properties Measurement: The thickness of the rectangular curvilinear 3D structures was measured by optical microscopy. The inverse of the radius was defined as the curvature, measuring the radius of the contact circle on the concave surface of a rectangular curved structure. Other dimensional elements including height, length, area, and angle were measured using the Image J program.

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[38] Egypt Pyramid - Free photo on Needpix

Chapter 3 Photo-Triggered Shape-Reconfiguration in Stretchable Reduced Graphene Oxide-Patterned Azobenzene-Functionalized Liquid Crystalline Polymer Networks

3.1 Introduction

Polymer-based flexible electronics with a high degree of freedom enable the fabrication of 3D curved shapes due to their flexible body, leading to a high level of shape diversity beyond their conventional rigid electronics.[1, 2] However, according to Poisson's principle, the maximum strain capacity of a material under tensile or shear stress is limited even in flexible polymers,[3] which necessitates new engineering approaches for bulk geometries.[4] Kirigami patterns can be applied to increase the deformability of bulk materials without permanent failure.[5, 6] When stress is applied to kirigami-patterned objects, it effectively dissipates the external stress by 2D and 3D shape deformation.[7, 8] Kirigami-structured electronics have been extensively studied for wearable biomedical devices.[9, 10] However, the principle of shape morphing in kirigami-inspired electronics has often been limited to passive mechanical stretching.

The introduction of external stimuli-responsive materials can enable polymer-based actuating systems to directly convert external stimuli into pre-programmed mechanical work without the need for complicated artificial intelligence or power-supply systems.[11-14] Soft actuating systems embedded with stimuli-responsive materials have been widely used to develop intelligent materials, including artificial muscles,[15] antennas,[16] shape-reconfigurable substrates,[17-19] cargo-transport systems,[20, 21] medical devices,[22] and soft robots.[23-25] In particular, molecular machine (e.g., azobenzene) embedded liquid crystalline polymers (LCPs) are great candidates for constructing shape-reconfigurable devices owing to the remotely controllable actuation by programming the alignment of anisotropic liquid crystal (LC) molecules.[26-31] Azo-LCPs also exhibit dual-responsivity to thermal stimuli and photochemical stimuli which induce anisotropic thermal expansion/contraction[32] and trans-cis photoisomerization of azobenzene derivatives, respectively.[33] However, the application of azo-LCP monolith to next-generation electronic devices remains a challenge due to two issues, namely the trade-off relationship between deformability and elastic modulus and their electrically insulating nature.

To achieve high mechanical performance in azo-LCPs, engineering plastics such as polyimide (PI) were used as the backbone.[34-36] Azobenzene-functionalized polyimides (azo-PIs) have high crystallinity[31] and rigid backbones,[35, 36] which imply a high elastic modulus. However, photogenerated strain is suppressed in azo-PIs because their high crystallinity and backbone rigidity increase the required energy for shape-reconfiguration. The high crystallinity of the polymer matrix also interferes with photo-induced actuation by reducing light absorption caused by a larger degree of light scattering.[37] Furthermore, the molecular programming of azo-PIs is limited to uniaxial alignment via hot drawing, which results in a simple bending actuation due to the absence of liquid crystallinity in the fabricated system. Therefore, the trade-off relationship between elastic modulus and deformability is one of the largest limitations of azo-LCPs. Furthermore, typical azo-LCPs are inherent electrical insulators, which makes them unsuitable for electronic applications. An effective strategy to overcome these two issues is the introduction of functional materials in azo-LCPs as fillers.

Herein, we introduce a new process for patterning reduced graphene oxide (rGO) on azobenzene-functionalized liquid crystalline polymer networks (azo-LCNs) to develop shape-reconfigurable electronics. While azo-LCNs can effectively convert light energy to mechanical actuation due to their programmable and tightly crosslinked nature, rGO can reinforce the weak points in azo-LCN owing to its high elastic modulus[38] and electrical conductivity.[39] Patterning rGO on azo-LCN allows the implementation of the intrinsic mechanical/electrical properties of rGO and the actuation characteristics of azo-LCN. The dimensions of patterned rGO, which are directly linked to its mechanical and electrical properties, can be tailored by controlling the number of rGO coating cycles.

The highly conjugated structure in rGO can induce secondary interactions (π–π stacking and hydrogen bonding, respectively) with mesogens of LC monomers.[40] These strong secondary interactions guarantee excellent compatibility between the rGO pattern and azo-LCN, thus enabling rGO-patterned azo-LCN (azo-LCN/rGO) to stably maintain the patterned structure and exhibit good mechanical/electrical properties. When the temperature of the system is increased by light irradiation or heating, a synergistic combination of trans–cis photoisomerization in the azo-LCN and the mismatch between the coefficients of thermal expansion (CTEs) of rGO and azo-LCN result in the facile actuation of azo-LCN/rGO. Hence, the elastic modulus, electrical conductivity, and actuation performance of azo-LCN/rGO can be simultaneously enhanced, despite the conventional trade-off between elastic modulus and deformability. The high absorption coefficient toward a wide range of light (visible to near-IR (NIR))[41, 42] as well as the high thermal conductivity of rGO[43] enable the photothermal actuation of azo-LCN/rGO under multi-wavelength light exposure. Finally, kirigami-patterned azo-LCN/rGO exhibit reversible shape morphing under mechanical strain beyond the limited strain capability of materials without deterioration in electrical conductivity. Thus, these kirigami-patterned structures extend the concept of passive stretchable electronics to realize active and remotely controllable shape-morphing systems.

3.2 Results and Discussion

3.2.1 Preparation of Azo-LCN/rGO

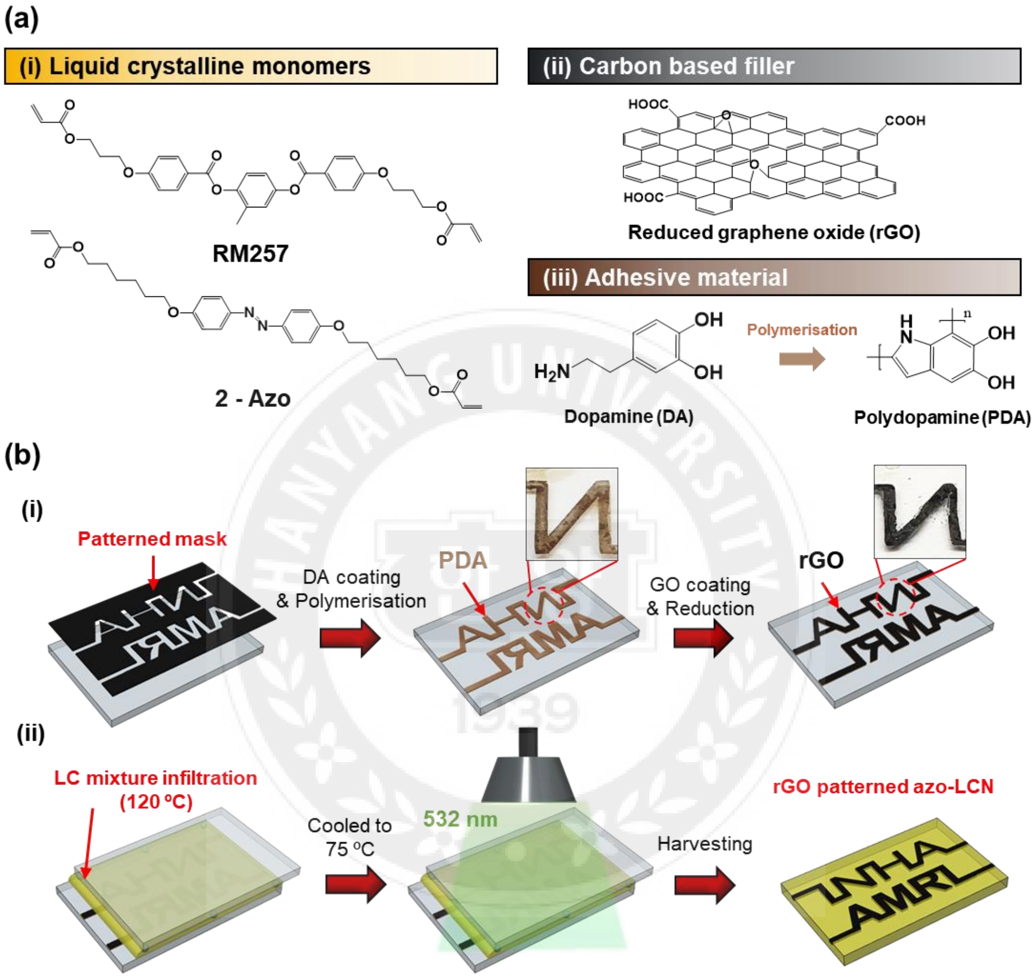

Figure 3.1 Preparation of azo-LCN/rGO. (a) Chemical structures of raw materials used to synthesize azo-LCN/rGO i) LC monomers, RM-257, and 2-Azo; ii) rGO; and iii) DA monomer and PDA. (b) Schematic illustration of the preparation of i) rGO-patterned glass slide and ii) azo-LCN/rGO. The glass cell consisted of polyamide-coated and rGO-patterned glass slides; azo-LCN/rGO was prepared by the photopolymerization of LC monomers within the rGO-patterned glass cell by irradiation with a 532 nm green light (50 mW cm^{-2}) for 1 h at 75 °C.

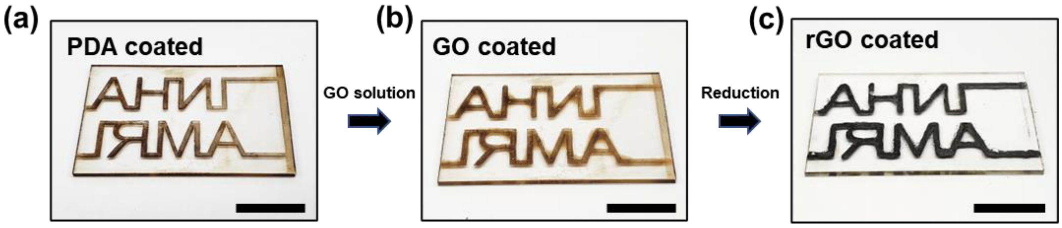

Figure 3.2 Digital images of the pattern INHA AMRL on (a) PDA, (b) GO, and (c) rGO on a glass slide. Scale bar, 1 cm.

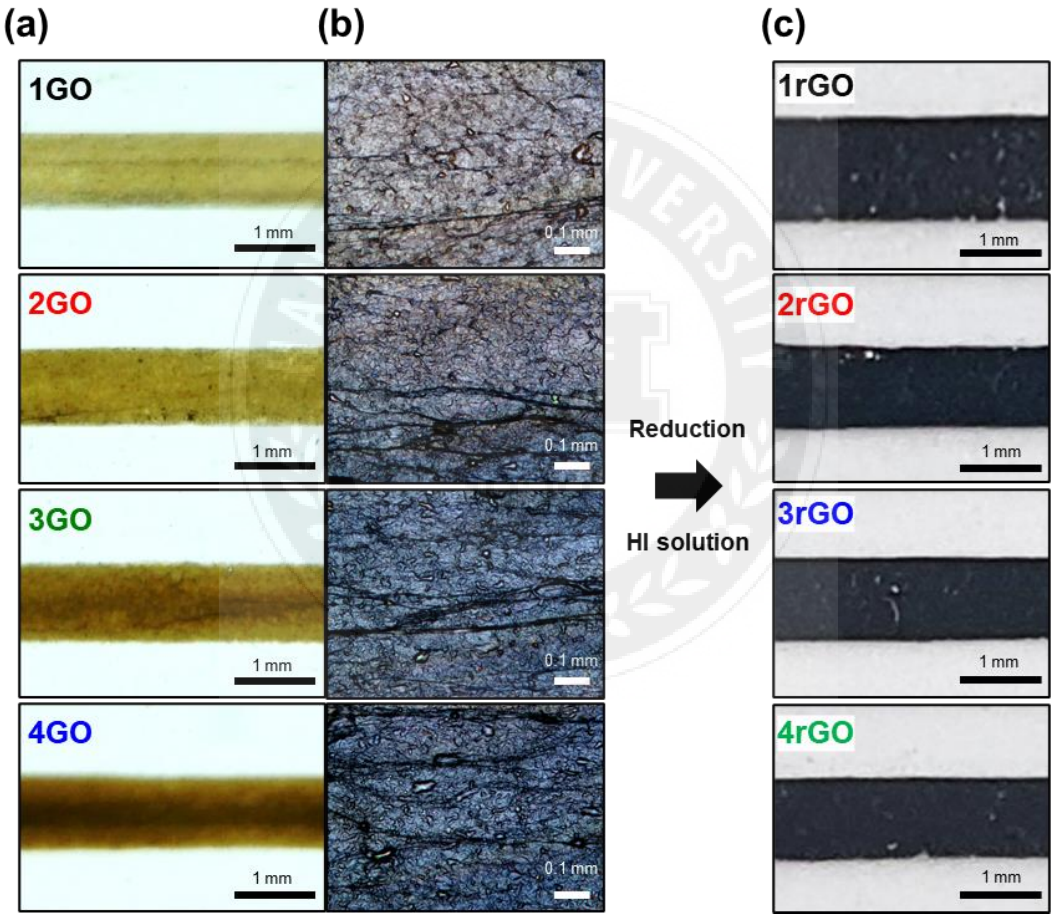

Figure 3.3 (a) Digital images and (b) epi-illumination optical micrographs of GO layers on glass slides. (c) Digital images of rGO layers on glass slides.

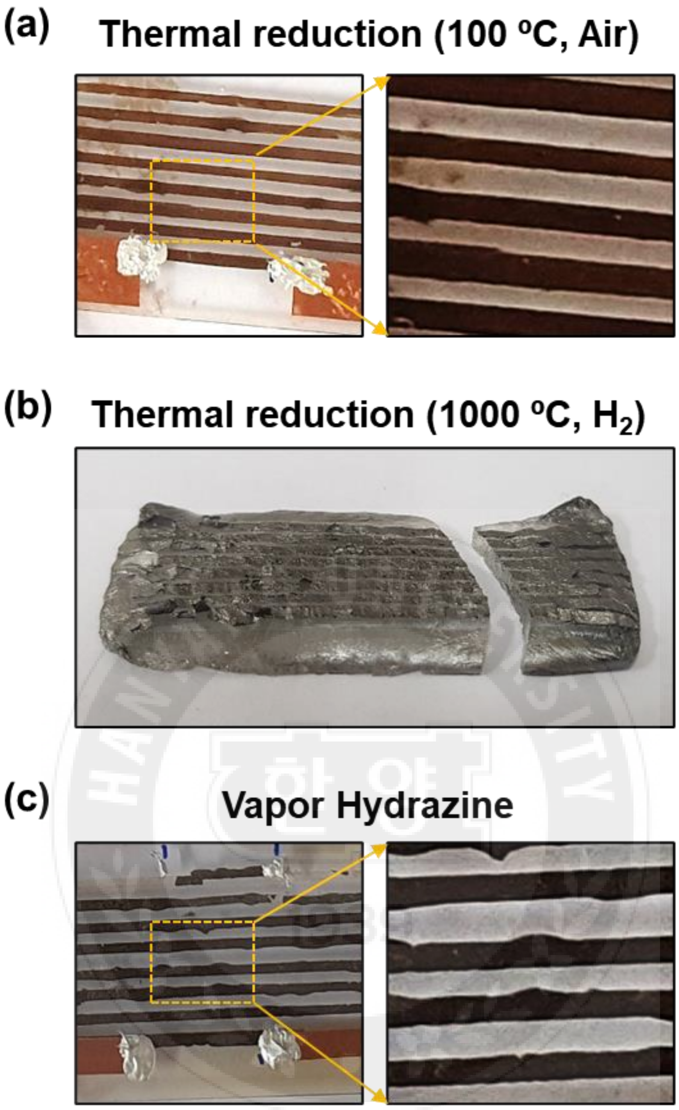

Figure 3.4 Digital images of rGO-patterned glass slides subjected to reduction. (a) Thermal reduction at 100 °C in air for 6 h. (b) Thermal reduction at 1000 °C in H_2 atmosphere for 24 h. (c) Chemical reduction with hydrazine vapor for 6 h.

Multi-stimuli-responsive azo-LCN/rGO were prepared by the photopolymerization of LC monomers in an rGO-patterned glass cell, whereupon patterned rGO was transferred from the glass cell to azo-LCN. Azo-LCN consisted of a reactive LC monomer (RM257) and a photo-responsive azobenzene moiety (2-Azo) as a comonomer (**Figure 3.1a(i)**). Meanwhile, rGO was introduced to enhance the mechanical and electrical properties of azo-LCNs (**Figure 3.1a(ii)**). Polydopamine (PDA) was used as a pre-patterning layer before rGO was patterned on glass slides to make the selective spatial patterning of GO from aqueous solutions feasible. The monomeric dopamine (DA) was polymerized into PDA under basic conditions (**Figure 3.1a(iii)**).

As shown in **Figure 3.1b(i)**, rGO-patterned glass cell was prepared to polymerize azo-LCN/rGO. First, a 2D-patterned adhesive tape was attached to a cleaned glass slide for selective pre-patterning with PDA, which was deposited using a fast PDA coating method.[44] Upon oxidizing DA into quinone using $NaIO_4$ as an initiator, PDA was formed under basic conditions (pH 8.5), which accelerated polymerization kinetics. The pre-patterned PDA layer acted as a selective wetting site for rGO patterning (**Figure 3.2a**). Once the PDA pre-patterned layer was fully dry, it was coated with 5 mg mL^{-1} aqueous GO solution, which formed a stable layer on the PDA layer through π–π interactions and hydrogen bonding.[45] Because of its high aspect ratio, GO underwent a phase transition into the LC phase at high concentrations upon the evaporation of the aqueous medium.[46, 47] The evaporative self-assembly of GO resulted in a solid brown-colored GO pattern with orientational order due to the presence of a lyophilic nematic phase (**Figure 3.2b**). After the evaporative self-assembly process, patterned GO was reduced to rGO by treating it with 30 wt% aqueous hydroiodic acid (HI) for 30 min at 80 °C, which changed the color of the GO pattern from brown to black (**Figure 3.2c**). The rGO patterns produced over 1–4 GO coating cycles were named 1rGO to 4rGO, respectively (**Figure 3.3**). The 1rGO pattern reduced by aqueous HI could stably maintain its original shape with relatively higher electrical conductivity (191 S cm^{-1}) than that of rGO subjected to thermal (1000 °C in H_2 for 24 h, 160 S cm^{-1}, **Figure 3.4a**; 100 °C in air for 6 h, 2.1×10^{-4} S cm^{-1}, **Figure 3.4b**) or chemical (25 °C, hydrazine vapor for 6 h, 4.8×10^{-4} S cm^{-1}, **Figure 3.4c**) reduction.

The rGO-patterned glass was attached to a glass slide coated with a polyamide alignment layer, which was mechanically rubbed along the molecular alignment direction desired in azo-LCN. Finally, the final azo-LCN/rGO was prepared by the photopolymerization of a mixture of LCs containing RM257, 2-Azo, and photoinitiator (**Figure 3.1b(ii)**). The LC mixture was melted in the isotropic phase at 120 °C and then injected into the glass cell by capillary action. The molten LC mixtures were cooled to 75 °C and photopolymerized in the nematic phase using 532 nm green light (50 mW cm^{-2} for 1 h). Solidified azo-LCN/rGO was successfully obtained from the glass cell as a monolithic free-standing film.

3.2.2 Structural Characterization

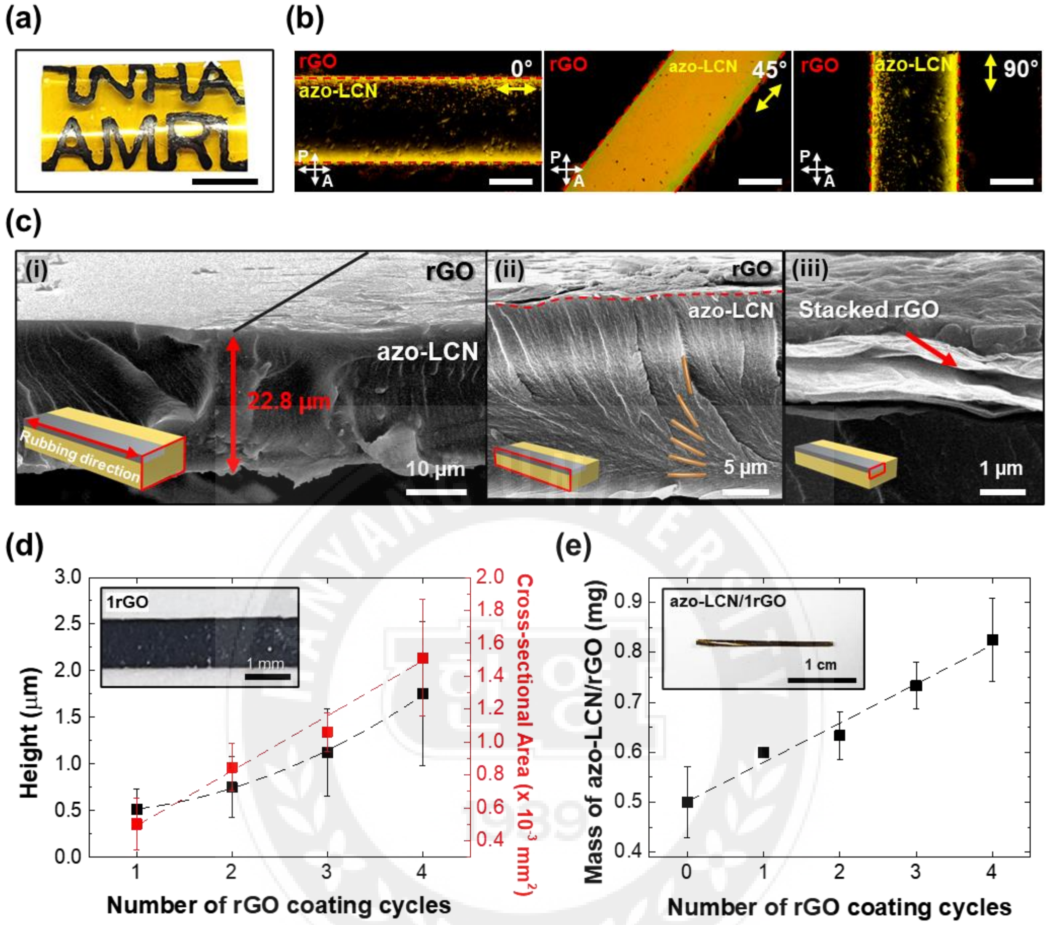

Figure 3.5 Programmability of rGO patterning and molecular alignment of azo-LCN. (a) Digital image of patterned azo-LCN/rGO. (b) POM images of azo-LCN/rGO. The yellow arrows indicate the nematic director in azo-LCN/rGO. Scale bar, 0.5 mm. (c) Cross-sectional SEM images of azo-LCN/4rGO cut i) perpendicular and ii) parallel to the rubbing direction. iii) Stacked rGO layers in the cross-section of patterned rGO. The red boxes in the inset scheme indicate the observation plane on azo-LCN/rGO. (d) Height and cross-sectional area of rGO patterns (1 mm width) as functions of the number of rGO coating cycles. The inset figure showed a digital image of the 1rGO pattern on the glass slide. (e) Mass of azo-LCN/rGO concerning the number of rGO coating cycles.

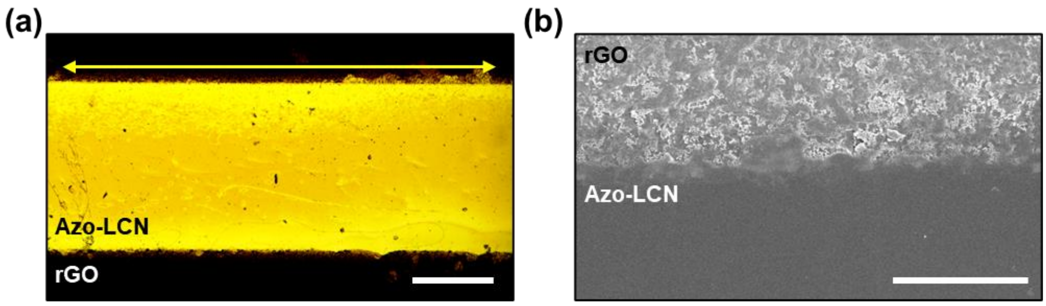

Figure 3.6 (a) Optical micrograph of azo-LCN/rGO. Scale bar, 0.5 mm. (b) Top-down SEM image of azo-LCN/rGO. Scale bar, 0.1 mm.

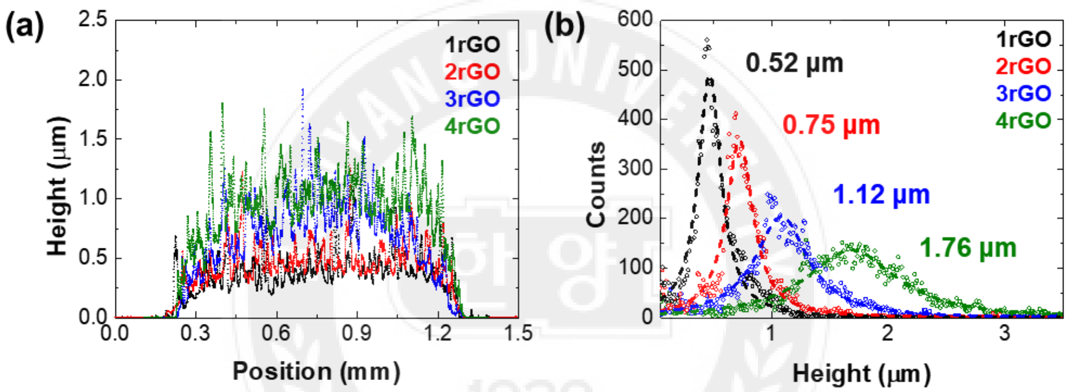

Figure 3.7 (a) Cross-sectional morphology of rGO layers evaluated using a surface profiler. (b) Height histogram of rGO layers. The average heights of rGO layers concerning the number of rGO coating cycles are shown in the graph.

Diverse 2D rGO patterns could be easily applied on azo-LCN films without delamination. To demonstrate this versatile patterning capability, we designed the pattern “INHA AMRL” on azo-LCN/rGO (**Figure 3.5a**). Molecular alignment in azo-LCN/rGO was examined by cross-polarized optical microscopy (POM). When the nematic director of azo-LCN/rGO was perpendicular to the polarizer or analyzer, the color was dark due to light blocking. In contrast, the brightest color was observed when the azo-LCN chains were aligned at 45° concerning both the polarizer and analyzer (**Figure 3.5b and 3.6a**). The morphological and geometric characteristics of azo-LCN/rGO were visualized by scanning electron microscopy (SEM). The boundary between azo-LCN and the patterned rGO region could be clearly distinguished in the cross-sectional and top-down SEM images of vertically cut azo-LCN/4rGO concerning the short axis of the rGO pattern (**Figure 3.5c(i) and 3.6b**). Hybrid alignment could be observed in the azo-LCN layer in the cross-sectional SEM images along the alignment direction (**Figure 3.5c(ii)**). The molecular alignment in azo-LCN was parallel to the long axis of the rGO pattern at the surface opposite to the attached rGO layer. The nematic director was gradually tilted until it was vertically aligned with the rGO layer. Such a hybrid alignment implies T-shaped edge-to-face π–π interactions at the interface between the highly concentrated mesogenic units of the LC and the conjugated structure of rGO.[48] The binding energies of edge-to-face π–π interaction and face-to-face π–π interaction were almost equal and hence, there was a strong adhesion between azo-LCN and rGO at the interface.[49] Nacre-like stacked 4rGO layers were also observed in the magnified cross-sectional SEM images of the rGO pattern at the boundary of the rGO layer (**Figure 3.5c(iii)**).

With an increase in the number of rGO coating cycles, the portion occupied by the rGO patterns in the azo-LCN/rGO increased, which was reflected in the basic geometrical elements including height, cross-sectional area, and weight. The height profile and cross-sectional area were measured using linearly patterned rGO on a glass slide with a fixed width of 1 mm. As the number of rGO coating cycles increased from 1 to 4, the average height of the rGO pattern increased quadratically from 0.52 to 0.75 (2rGO), 1.12 (3rGO), and 1.76 μm (4rGO) (**Figure 3.7a and 3.7b**). Both the cross-sectional area of rGO and the mass of azo-LCN/rGO showed a linear correlation with the number of rGO coating cycles. The cross-sectional area of the rGO pattern was calculated by numerically integrating the height profile measured using a surface profiler and increased uniformly by 0.34 ×

10^{-3} mm^{-2} per coating cycle (**Figure 3.5d**). The mass of azo-LCN/rGO, measured using azo-LCN/rGO specimens with fixed dimensions of 20 mm (length) × 22 μm (thickness) × 1 mm (width), was found to increase linearly by 0.09 mg per coating cycle (**Figure 3.5e**).

3.2.3 Tailorable Electrical and Mechanical Properties

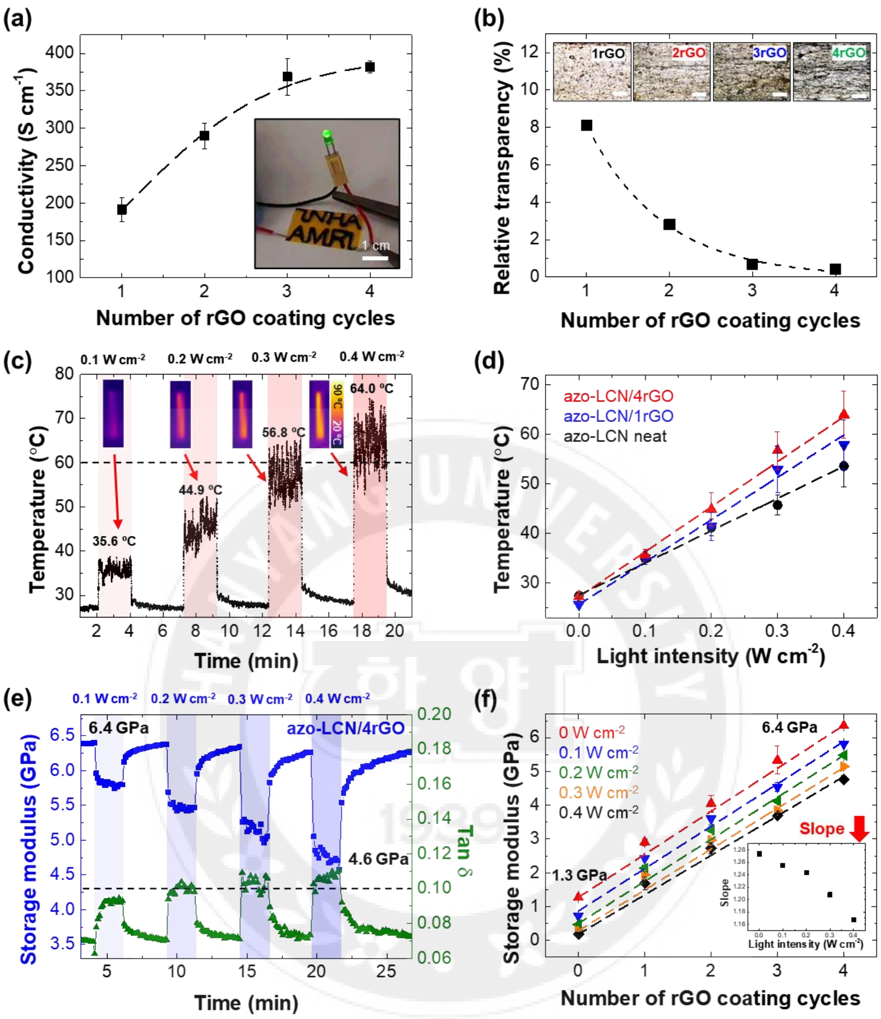

Figure 3.8 Tailorable electrical and mechanical properties of azo-LCN/rGO. (a) Electrical conductivity of azo-LCN/rGO measured as a function of the number of rGO coating cycles. The inset figure shows a green LED (3 V) turned on by the INHA AMRL-patterned azo-LCN/1rGO connected to an electrical circuit. (b) Relative transparency of rGO patterns concerning the number of rGO coating cycles. The inset images show optical micrographs of patterned rGOs in the reflection mode. Scale bars, 0.5 mm. (c) Changes in the temperature of azo-LCN/4rGO as a function of UV-light intensity. The inset figures show thermal images of azo-LCN/4rGO at each intensity. (d) Changes in the temperature of neat azo-LCN, azo-LCN/1rGO, and 4rGO concerning UV-light intensity. (e) Storage modulus and tan δ of azo-LCN/4rGO. The blue regions indicate UV irradiation during DMA testing. (f) Storage modulus of azo-LCN/rGO as a function of the number of rGO coating cycles at different UV-light intensities. The inset graph maps the slope of each plot concerning UV-light intensity.

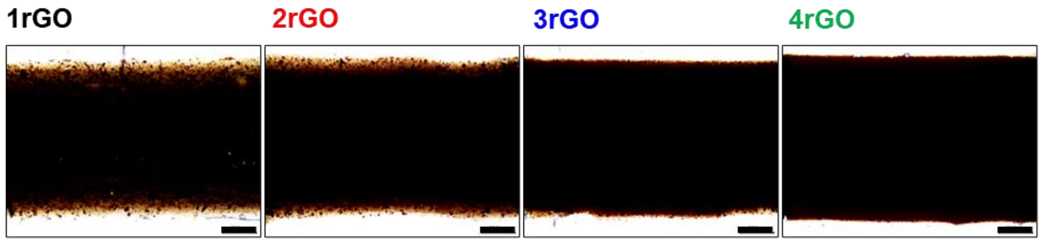

Figure 3.9 Trans-illumination optical micrographs of rGO patterns on glass slides. Scale bar, 0.2 mm.

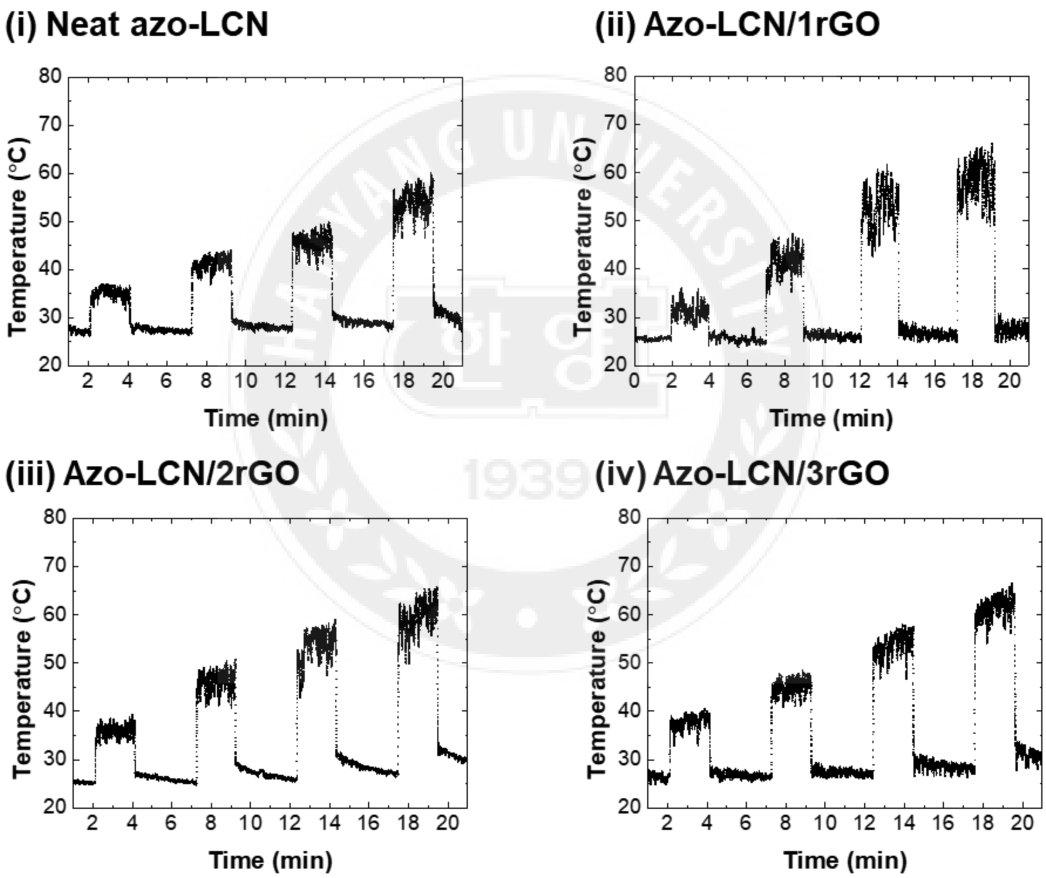

Figure 3.10 Temperature variation in neat azo-LCN, azo-LCN/1rGO, and 3rGO during DMA testing with in-situ exposure to UV radiation of gradually increasing intensity.

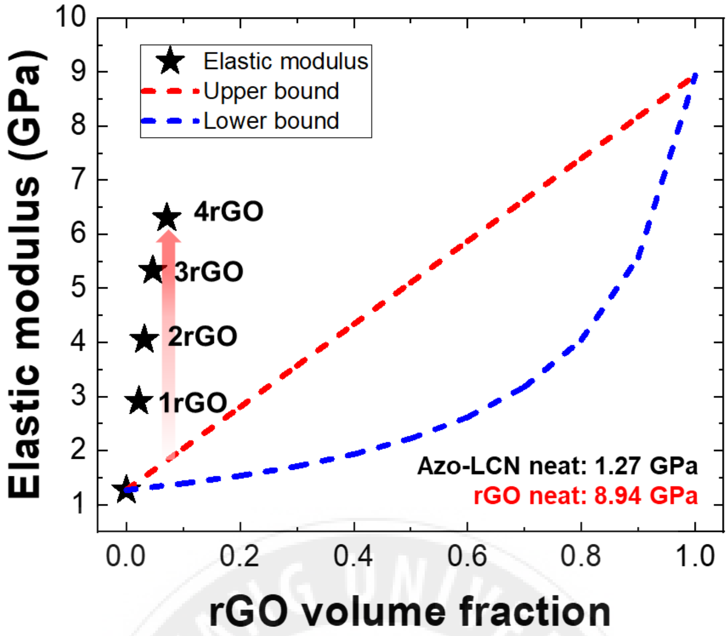

Figure 3.11 Elastic modulus variation according to rGO volume fraction. The red and blue dashed lines indicate the upper bound and lower bound calculated by the rule of mixtures.

Parameters	Measured value
Curvature (κ)	29.6 m^{-1}
Elastic modulus of Azo-LCN ($E_{Azo\text{-}LCN}$)	1.27 × 10^9 Pa
Thickness of Azo-LCN ($t_{Azo\text{-}LCN}$)	22.8 × 10^{-6} m
Poisson ratio of Azo-LCN ($\nu_{Azo\text{-}LCN}$)	0.4
Strain of rGO (ϵ_{rGO})	0.0015
Thickness of rGO (t_{rGO})	1.8 × 10^{-6} m

Table 3.1 Parameters for calculating the elastic modulus of 4rGO using the Stoney equation.

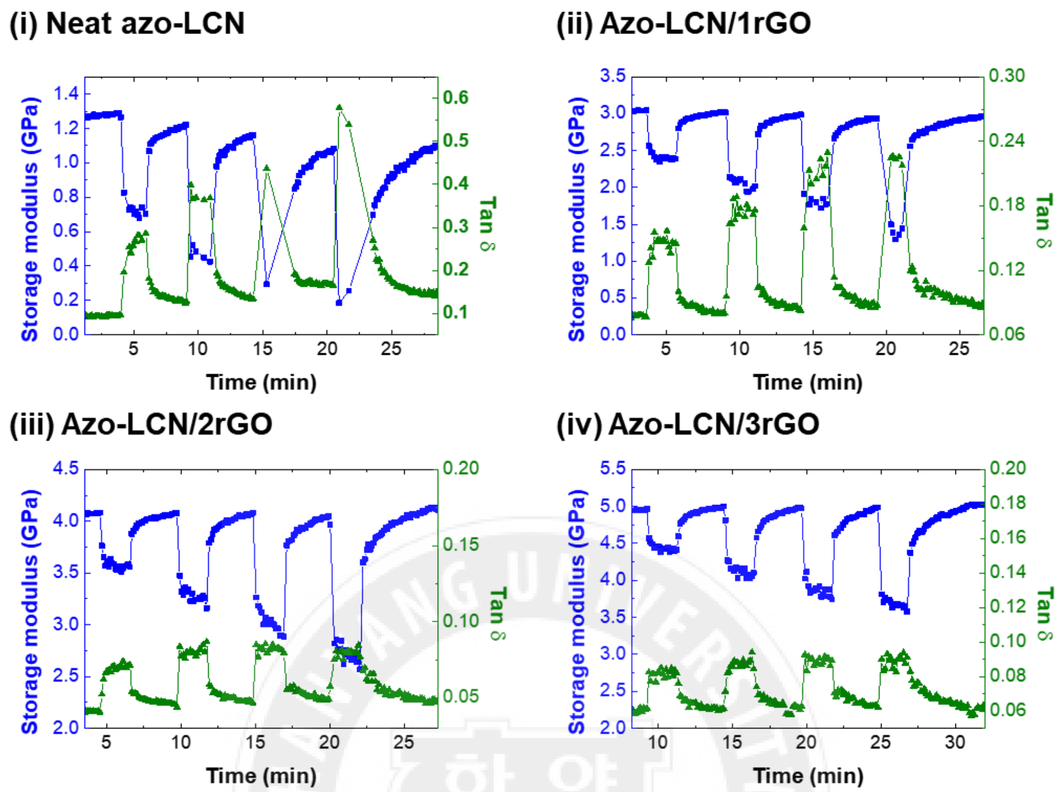

Figure 3.12 Storage modulus and tan δ of neat azo-LCN, azo-LCN/1rGO, and 3rGO during DMA testing with in-situ exposure to UV radiation of gradually increasing intensity.

The quantitative design of rGO patterns has a significant influence on the macroscopic properties, including electrical conductivity, light responsivity, and elastic modulus, of azo-LCN/rGO. Although azo-LCN was initially insulating, it acted as a conductor once it was composited with 1rGO and 4rGO, with an electrical conductivity (σ) of 191.2 and 381.9 S cm^{-1}, respectively (**Figure 3.8a**). This electrical conductivity was calculated using **Equation (3.1)**.

$$\sigma = \frac{L}{A \times R} \quad (3.1)$$

Where A is the cross-sectional area of the rGO pattern (20 × 1 mm^2), L is the length of an rGO layer (20 mm), and R is the measured electrical resistance. A green LED (3 V) connected to the “INHA AMRL”-lettered 1rGO electrical circuit could be successfully turned on (inset image in **Figure 3.8a**), which is evidence of a connected electrical conducting pathway in the rGO pattern. Increasing the number of rGO coating cycles led to a corresponding increase in the connectivity between rGO layers, along with darkening of the color of azo-LCN/rGO as well as reduced transparency (**Figure 3.8b and 3.9**). Over multiple rGO coating cycles, the newly added rGO layers increased the packing density and physical connectivity in rGO, thereby eliminating defects in azo-LCN/rGO and resulting in additional electro-conductive pathways.

In addition to electrical conductivity, the inclusion of rGO enhanced the conversion of photo-induced energy to thermal energy; this phenomenon is called the photothermal effect.[41] Upon UV irradiation, due to the photothermal effect, the temperature of azo-LCN/rGO, which is strongly correlated with UV-light intensity, increased to 35.7±3.3, 44.9±6.9, and 56.9±8.0 °C at 0.1, 0.2, and 0.3 W cm^{-2}, respectively (**Figure 3.8c and 3.10**). At intensities higher than 0.3 W cm^{-2}, the temperature of azo-LCN/4rGO began to exceed the glass-transition temperature (T_g ≈60 °C)[50] of azo-LCN. As shown in **Figure 3.8d**, photothermal actuation was investigated in terms of the number of rGO coating cycles. Under constant UV exposure, changes in the temperature of azo-LCN/rGO systems were analyzed. At a low UV intensity of 0.1 W cm^{-2}, the differences in the average temperatures of neat azo-LCN, azo-LCN/1rGO, and 4rGO were negligible. At 0.4 W cm^{-2}, the temperature of azo-LCN/4rGO was 63.6±4.8 °C while the temperatures of neat azo-LCN and azo-LCN/1rGO were 53.6±4.2 and 57.6±5.1 °C, respectively. The higher temperature observed for azo-LCN/4rGO can be attributed to greater photothermal energy conversion due to its higher rGO content.

The azo-LCN/rGO bilayer initially exhibited a curved structure due to a stiffness mismatch between azo-LCN and rGO. According to the Stoney equation described in

Equation (3.2), the elastic modulus of the rGO layer (E_{rGO}) can be calculated using measured parameters including the initial curvature (κ), the elastic modulus of neat azo-LCN ($E_{azo\text{-}LCN}$), the strain (ϵ_{rGO}) and the thickness (t_{rGO}) of rGO layer (**Table 3.1**).

$$E_{rGO} = \frac{\kappa E_{azo-LCN} t_{azo-LCN}^2}{6(1-\nu_{azo-LCN})\epsilon_{rGO} t_{rGO}} \quad (3.2)$$

$$\epsilon_{rGO} = \frac{t_{azo-LCN} \times \kappa}{2} \quad (3.3)$$

As the azo-LCN/rGO bilayer is a thin film, the strain of the rGO bilayer can be calculated using **Equation (3.3)**. At the azo-LCN/rGO interface, the strain in the rGO layer will be identical to the strain in the azo-LCN layer due to mechanical continuity. Finally, the E_{rGO} was calculated as 8.94 GPa as the Poisson ratio of azo-LCN was set to 0.4 and the initial curvature of azo-LCN/4rGO was utilized. The mechanical strength of azo-LCN was significantly improved by the introduction of rGO; the storage modulus of azo-LCN/4rGO was 6.4 GPa, which was approximately five times higher than that of neat azo-LCN (1.3 GPa). This high value was obtained despite the small thickness of the 4rGO pattern (1.76 μm), which accounted for only 7.8% of the total thickness (22.8 μm). When we compared the elastic modulus of azo-LCN/rGOs with upper and lower bounds from the rule of mixtures, the azo-LCN/rGOs demonstrated remarkably improved elastic modulus (**Figure 3.11**). The upper and lower bounds were calculated using the calculated elastic modulus of the 4rGO layer. As described earlier, secondary π–π interactions at the interface induced a strong adhesion between the azo-LCN layer and the rGO layer, resulting in an effective stress transfer and drastically increasing the modulus.

Furthermore, on-demand mechanical-strength manipulation could be achieved in azo-LCN/rGO by simply varying the UV intensity and number of rGO coating cycles. As shown in **Figure 3.8e and 3.12**, the storage modulus of azo-LCN/4rGO immediately reduced from 6.4 to 4.6 GPa (≈75%) upon exposure to 0.4 W cm^{-2} UV light. Furthermore, the storage modulus of azo-LCN/rGO exhibited a linear correlation with the number of rGO coating cycles (and consequently, with rGO content; **Figure 3.8f**). The decrease in the storage modulus of azo-LCN/rGO upon UV exposure can be attributed to changes in order parameters by photochemical and photothermal effects. First, azobenzene molecules in azo-LCN underwent trans–cis isomerization via UV-light absorption. The photoisomerized cis-azobenzene then broke down the molecular alignment, thus reducing the order parameter.[51] Second, the increase in temperature induced by the photothermal effect promoted active molecular motion in aligned LC molecules and induced chain randomization. Therefore, the mechanical strength of azo-LCN/rGO could be quickly and remotely manipulated by adjusting the intensity of the exposed UV light.

3.2.4 UV-Induced Shape-Reconfiguration

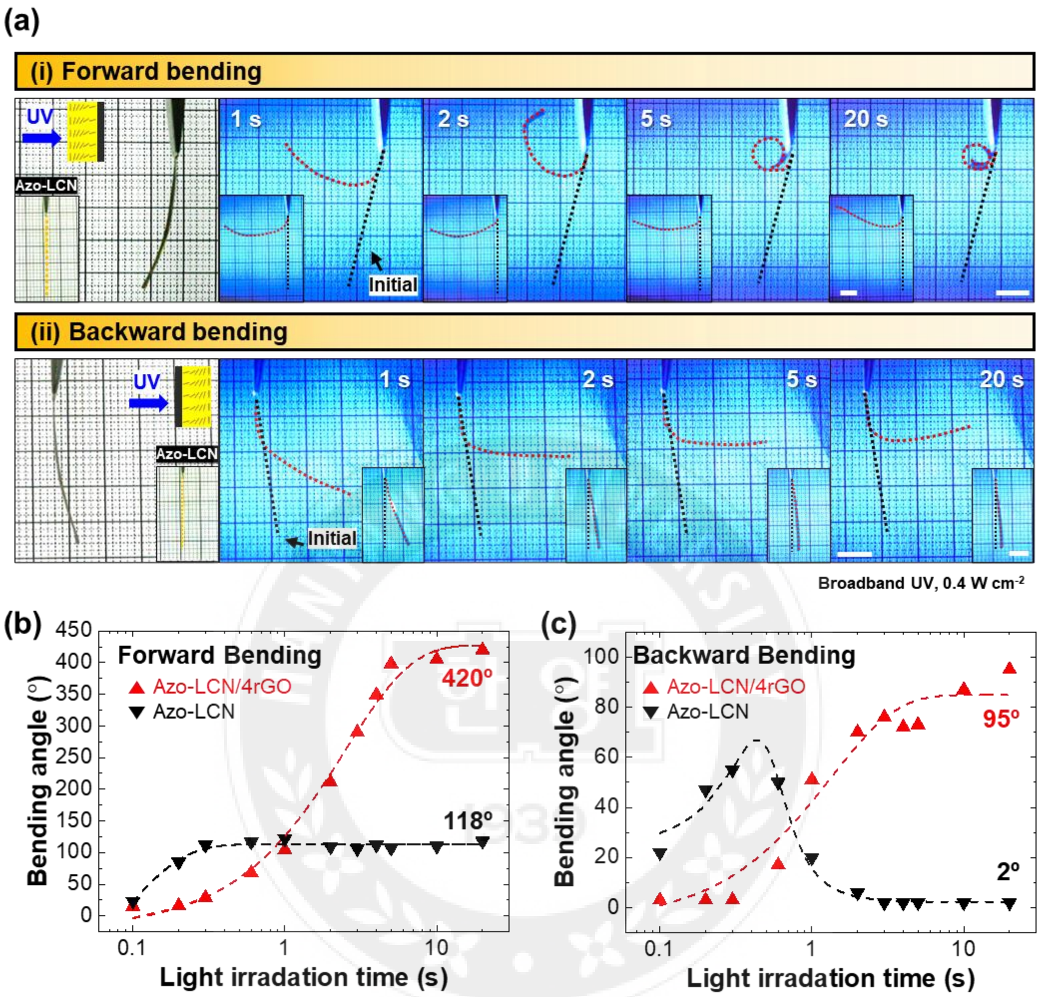

Figure 3.13 UV-induced in-plane bending actuation. (a) (i) Forward and (ii) backward bending of an azo-LCN/4rGO strip at a UV intensity of 0.4 W cm^{-2}. The inset figures in (i) and (ii) illustrate the forward and backward bending, respectively, of hybrid-aligned neat azo-LCN under the same UV exposure conditions. Scale bars, 5 mm. (b) Forward- and (c) backward-bending angles of azo-LCN/4rGO and neat azo-LCN as functions of UV-exposure time

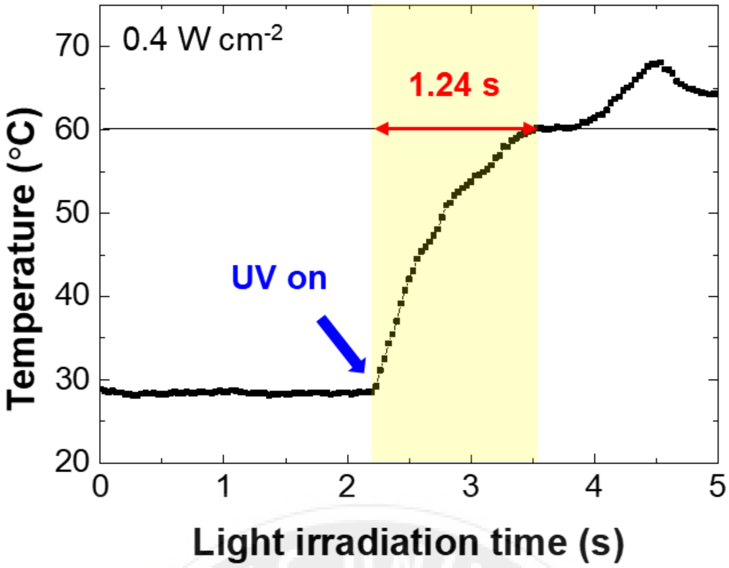

Figure 3.14 Temperature variation in azo-LCN/4rGO under UV light with 0.4 W cm^{-2}

Thickness of soft body, Azo-LCN (L_s, m)	22.8×10^{-6}
Thickness of rigid body, rGO (L_r, m)	1.8×10^{-6}
Thickness ratio (η)	**0.079**
Elastic modulus of rGO, Pa	8.94×10^{9}
Elastic modulus of Azo-LCN, Pa	184×10^{6}
Contrast in stiffness (ϵ)	**48.59**
Temperature change (ΔT, 27 to 63 °C)	**36**
Thermal expansion coefficient of rGO (K^{-1})	1.5×10^{-6}
rGO strain	**5.4×10^{-4}**
Thermal expansion coefficient of Azo-LCN (K^{-1})	-100×10^{-6}
Azo-LCN strain	**-36×10^{-4}**
Strain mismatch (ε_m)	**41×10^{-4}**

Table 3.2 Parameters for calculating the curvature of azo-LCN/4rGO bilayer using the Timoshenko and Stoney equation.

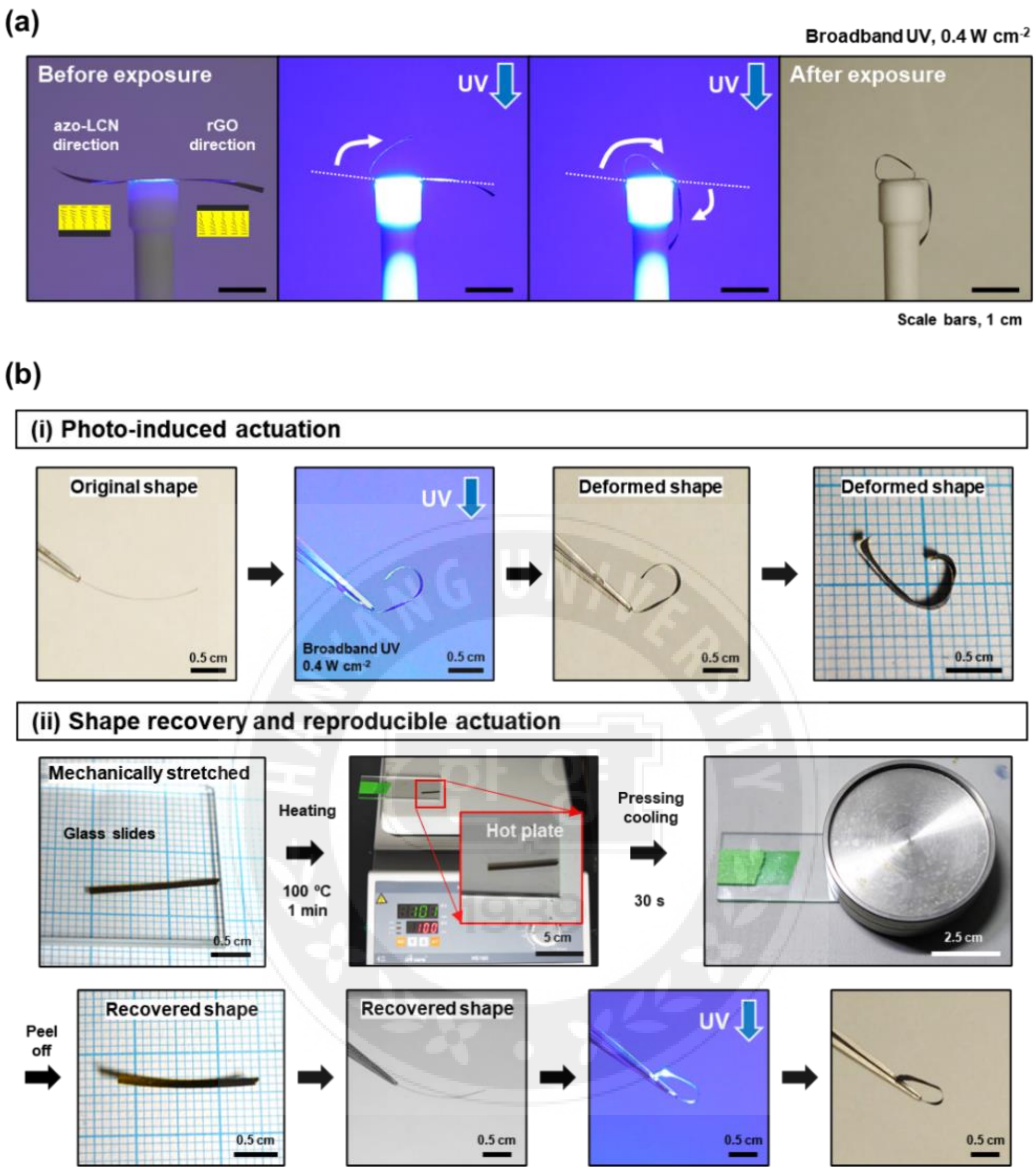

Figure 3.15 (a) Bidirectional bending actuation of the azo-LCN/4rGO strips under UV exposure. (b) The reversible and shape-recovery of the deformed azo-LCN/4rGO strip.

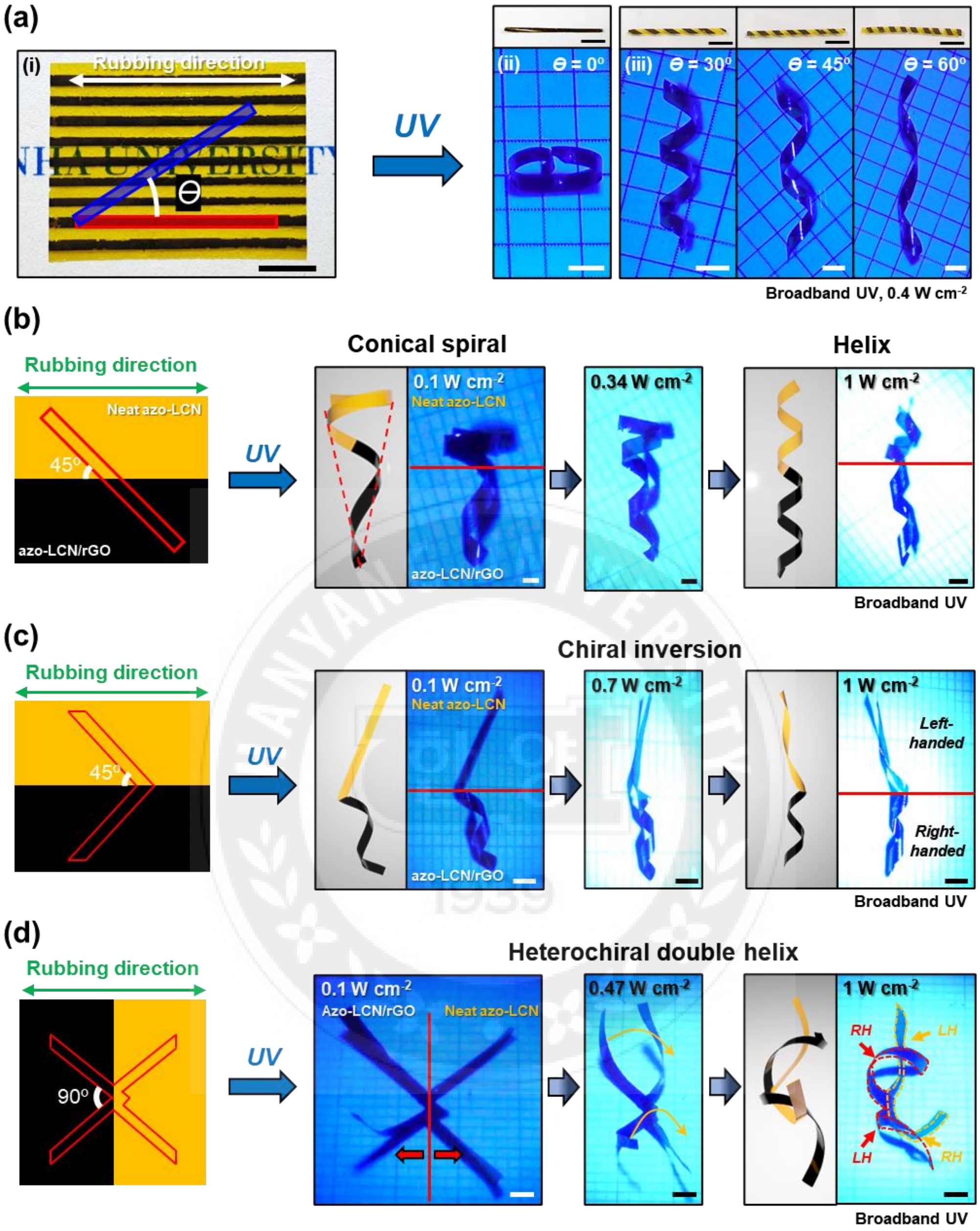

Figure 3.16 Out-of-plane torsional shape morphing of azo-LCN/rGO. (a) (i) Linearly patterned azo-LCN/rGO. (ii), (iii) Various cutting angles (Θ = 0°, 30°, 45°, 60°) applied on linearly patterned azo-LCN/1rGO. Scale bars, 1 cm. The digital images represent the original (scale bars, 5 mm) and deformed spiral shapes (scale bars, 1 mm). (b–d) Shape transition of half rGO-patterned azo-LCN with different cutting designs when exposed to UV light of gradually increasing intensity (0.1 to 1 W cm^{-2}). Scale bars, 2 mm. (b) Transition of neat azo-LCN from a conical spiral structure to a helical structure. (c) Chiral inversion of V-shaped half rGO-patterned azo-LCN. (d) Shape transition of X-shaped half rGO-patterned azo-LCN

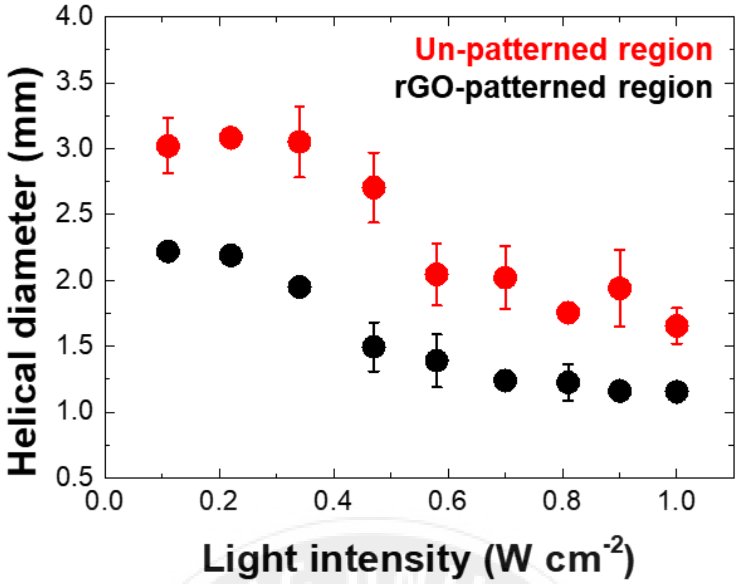

Figure 3.17 Variation in the helical diameter of half rGO-patterned azo-LCN with a 45° cutting angle from the nematic director concerning UV-light intensity.

The photo-triggered actuation capability of azo-LCN was improved by the inclusion of rGO via a synergistic effect of photoisomerization and CTE mismatch between azo-LCN and rGO. As shown in **Figure 3.13a**, bidirectional in-plane bending actuations (forward and backward) were observed in azo-LCN/4rGO, which were controlled by changing the direction of UV irradiation. UV irradiation of the azo-LCN region resulted in forward-bending deformation, while backward bending was achieved by UV irradiation in the rGO pattern direction. The forward-bending capability of azo-LCN/4rGO was 3.6 times higher than that of neat azo-LCN (forward-bending angles of 420° and 118° for azo-LCN/4rGO and neat azo-LCN, respectively; **Figure 3.13b**). A slower bending actuation occurred in azo-LCN/4rGO than in neat azo-LCN due to the much higher modulus of azo-LCN/4rGO. However, the forward-bending angle of azo-LCN/4rGO became larger than that of neat azo-LCN after 1 s.

Photoisomerization was predominant during forward bending after less than 1 s of UV exposure; this is because the photothermal effect requires time to increase the temperature of azo-LCN/4rGO to a value higher than the T_g. As shown in **Figure 3.14**, the temperature of azo-LCN/4rGO was higher than the T_g of neat azo-LCN (≈60 °C)[50] after 1.2 s of UV exposure. After 1 s of UV exposure, the hindered actuation in azo-LCN/4rGO was overcome by the increase in temperature via photothermal heating, which reduced its elastic modulus as well. Furthermore, there occurred a large CTE mismatch between the aligned azo-LCN layer and the rGO layer due to the increase in temperature induced by photothermal heating during forward-bending actuation. The average CTE of multi-layered rGO was 14.9 ppm K^{-1} at 373 K.[52] However, the LCN with a hybrid alignment exhibited an anisotropic CTE depending on the mesogen-alignment direction (perpendicular or parallel). In the previous report, the perpendicular and parallel CTE values of the LCN were measured to be ≈100 and 0 ppm K^{-1}, respectively, at temperatures below its T_g.[53] At temperatures greater than the T_g, the absolute magnitudes of the perpendicular and parallel CTE values were ≈300 and −100 ppm K^{-1} at 400 K, respectively.[53] As the temperature increased, the CTE mismatch between rGO and azo-LCN increased rapidly. This contributed to bending actuation as an additional force along with the direct molecular contraction of azobenzene by photoisomerization. Forward bending in azo-LCN/4rGO was saturated after 20 s resulting in a tightly rolled shape with a curvature of 561.9 m^{-1} (**Figure 3.13a(i)**).

In the case of backward bending (**Figure 3.13a(ii) and 3.13c**), a trend similar to that observed in forward bending concerning UV exposure time could be noted. However, a shading effect occurred due to the existence of the rGO layer, which resulted in a relatively smaller backward-bending angle than the forward-bending angle. The black rGO layer, with its low transmission, screens the UV light propagating toward the azo-LCN layer, thus leading to a photothermally driven bending alone. Hence, the backward bending of azo-LCN/4rGO was only attributed to the effect of photothermal-driven CTE mismatch. In other words, photoisomerization did not have any significant influence on this phenomenon. Notably, the backward-bending actuation of neat azo-LCN was negligible when compared to that of azo-LCN/4rGO. The saturated backward-bending angle of azo-LCN/4rGO was measured to be 95° with a curvature of 259.6 m^{-1}, which was 47 times larger than that of neat azo-LCN (2°). This bending curvature can be quantitatively calculated and compared with measured curvature by using Timoshenko and Stoney equation describing bending actuation induced by CTE mismatch between rigid layer and soft layer (rGO and azo-LCN layers in our case) under temperature variation (**Equation (3.4)**).

$$R_p = t_{azo-LCN} \frac{1+4\eta\epsilon+6\eta^2\epsilon+4\eta^3\epsilon+\eta^4\epsilon^2}{(6\varepsilon_m\eta\epsilon)(1+\eta)} \quad (3.4)$$

Where the R_P is the radius of curvature of the bilayer, the η (= $t_{rGO}/t_{azo\text{-}LCN}$) is the thickness ratio, the ϵ is the elastic modulus ratio (= $E_{rGO}/E_{azo\text{-}LCN}$), and the ε_m (= $\varepsilon_{rGO}-\varepsilon_{azo\text{-}LCN}$) is strain mismatch between rGO layer and azo-LCN layer. The strain of each layer can be calculated by multiplying the CTE value (α) by the temperature variation (ΔT) described in **Equation (3.5)**.

$$\varepsilon = \alpha\Delta T \quad (3.5)$$

Finally, the curvature of the azo-LCN/4rGO bilayer was calculated to be 254.1 m^{-1} when the bending actuation was performed with temperature elevation from 27 to 63 °C (**Table 3.2**). The calculated curvature was comparable to the curvature resulting from the backward bending, as the backward bending was primarily driven by the photothermal effect. However, the calculated curvature was nearly 2 times lower than the forward bending curvature because it did not account for the contribution of photoisomerization. This is strong evidence of the improved bending performance of the azo-LCN/rGO bilayer not only based on photoisomerization but also on CTE mismatch between the azo-LCN layer and the rGO layer.

The bidirectional actuation of azo-LCN/4rGO is demonstrated by attaching the two azo-LCN/4rGO strips in opposite directions (**Figure 3.15a**). The glued film is placed on a cylindrical rod. Upon unidirectional, top-down UV irradiation, the portion exposed to azo-LCN bends toward the light while that exposed to rGO bends away from the light. In addition, azo-LCN/rGO has reversible shape-recovery and is capable of reproducible actuation (**Figure 3.15b**). The shape-recovery of azo-LCN/rGO is demonstrated via the following procedures: mechanical stretching, heating above the T_g, and rapid cooling to room temperature. Following these procedures, the recovered azo-LCN/4rGO can be re-actuated through UV irradiation without deterioration of its actuation capabilities.

Shape morphing in azo-LCN/rGO is not limited to in-plane bending; 3D out-of-plane torsional deformations, including helix angles and/or helical pitch, could also be programmed by manipulating the nematic director. In **Figure 3.16a**, 3D coiled structures with different helical pitches controlled by the nematic director of the azo-LCN/rGO strips are illustrated. Note that the nematic directors could be varied by simply changing the cutting angle of the film concerning the alignment direction. At cutting angles of 30°, 45°, and 60°, the helical pitch values of azo-LCN/1rGO coils were 1.9, 3.2, and 4.7 mm, respectively. The effect of the cutting angle was combined with differences in the UV responsivity of azo-LCN and rGO to demonstrate asymmetric morphing in half-rGO-patterned azo-LCN. In the case of half rGO-patterned azo-LCN with a 45° cutting angle, the 2D strip underwent shape-reconfiguration into a 3D right-handed conical spiral with helical diameters of 2.2 and 3.0 mm for the rGO-patterned region and azo-LCN region, respectively, at a relatively small UV intensity of 0.1 W cm^{-2}. The lower UV responsivity of the neat azo-LCN region resulted in looser coiling than in the rGO-patterned region. When the UV intensity was increased to 1.0 W cm^{-2}, the helical diameters of the rGO-patterned and azo-LCN regions reduced to 1.2 and 1.7 mm, respectively. Consequently, tightly coiled helical structures could be obtained from the conical spiral structure (**Figure 3.16b and 3.17**). In addition to helical diameter, the chirality of a monolith with dual-helical pitches can be manipulated by V-shaped cutting. The rGO-patterned region was selectively coiled with a helical pitch of 5.3 mm at a relatively weak UV intensity of 0.1 W cm^{-2}. When this intensity was increased to 1 W cm^{-2}, chiral inversion occurred along with the development of dual-helical pitches (rGO-patterned region: 4.2 mm, unpatterned azo-LCN region: 9.0 mm) (**Figure 3.16c**). In the case of X-shaped cutting, half rGO-patterned azo-LCN morphed into an X chromosome-like 3D asymmetric geometry at a UV intensity of 0.47 W cm^{-2} and then into a heterochiral double helix at 1.0 W cm^{-2} (**Figure 3.16d**).

3.2.5 Multi-Stimuli-Responsive Shape-Reconfiguration

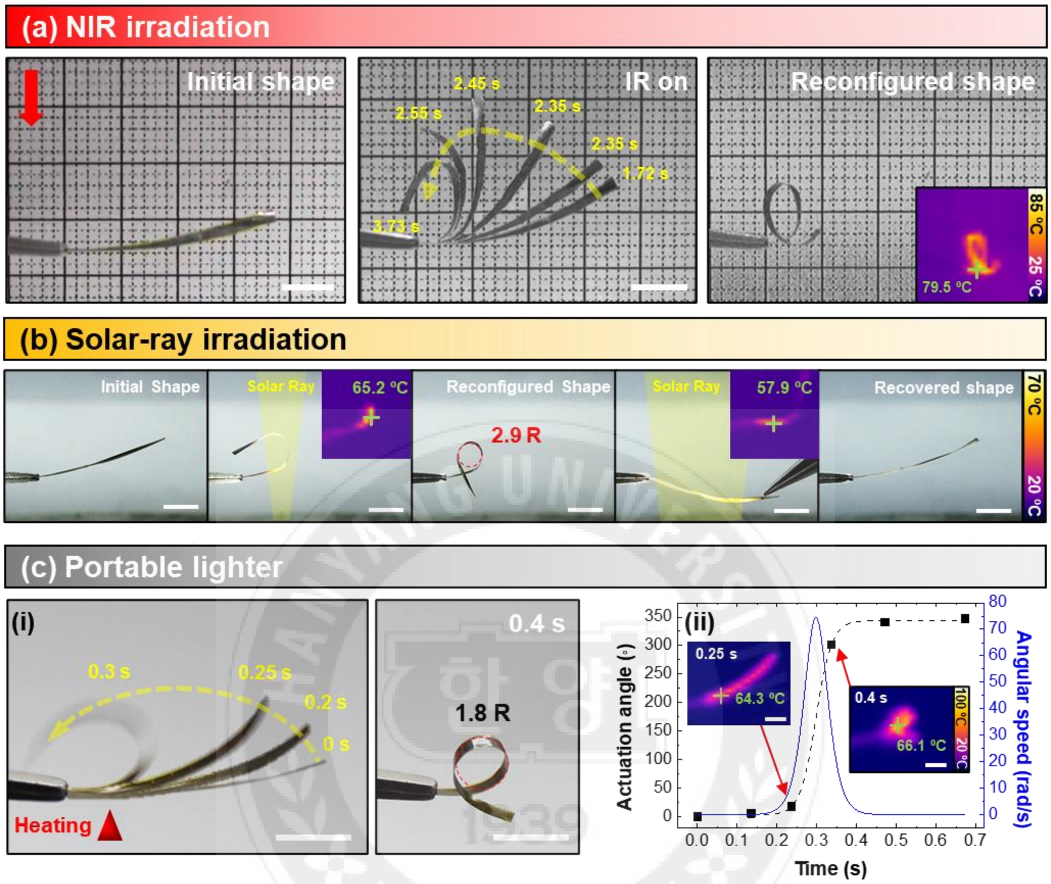

Figure 3.18 Multi-stimuli-responsive azo-LCN/rGO. (a) NIR (0.4 W cm^{-2})-induced bending actuation in azo-LCN/4rGO. The inset is a thermal image of azo-LCN/4rGO. Scale bars, 5 mm. (b) Selective actuation and shape-recovery in azo-LCN/4rGO by local irradiation with a solar ray. The inset figures are thermal images of azo-LCN/4rGO during irradiation. Scale bars, 5 mm. (c) (i) Rapid bending actuation of azo-LCN/4rGO when heated with a portable lighter. (ii) Time-resolved actuation angle and angular speed. Scale bars, 5 mm.

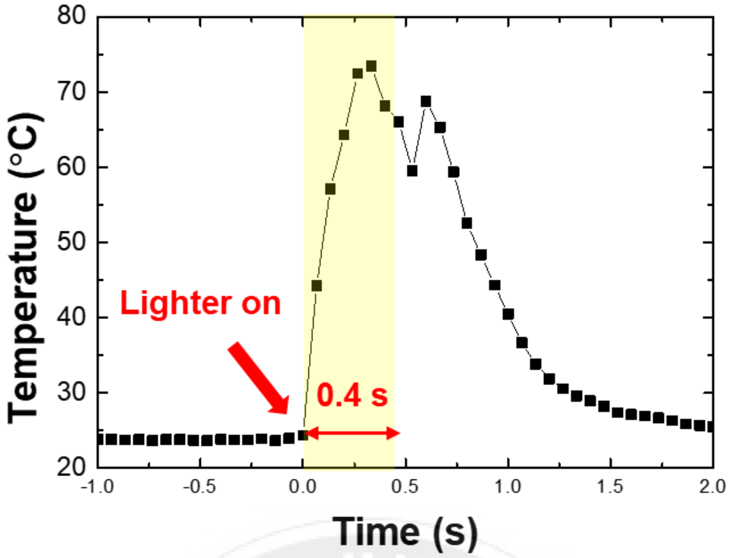

Figure 3.19 Temperature variation in azo-LCN/4rGO due to the direct conduction of thermal energy using a lighter

In addition to UV irradiation, other external stimuli induced rapid shape morphing in azo-LCN/rGO, including NIR light, solar rays focused with a magnifying glass, and a portable lighter. NIR irradiation, which is inexpensive and is often used in medical therapy, is selectively absorbed by the rGO layer, causing a rapid increase in the temperature via the photothermal effect (**Figure 3.18a**).[54] At 0.4 W cm^{-2}, the high NIR absorption in the rGO pattern increased the temperature of azo-LCN/4rGO above its T_g to 78.6 °C in 6 s; this, in turn, resulted in a rapid bending actuation greater than 180°. Notably, solar energy, one of the most abundant natural resources, could also be utilized for spatially regulated direct energy transduction for mechanical actuation in azo-LCN/rGO using a magnifying glass. When irradiated with a focused solar ray, the exposed portion of azo-LCN/rGO experienced a rapid temperature rise to values close to the T_g, resulting in a localized photothermal-driven shape-reconfiguration. After irradiation with solar rays, the deformed temporal shape of azo-LCN/rGO could be maintained stably by quenching to a temperature below its T_g. As shown in **Figure 3.18b**, the middle section of an azo-LCN/4rGO film was irradiated with a focused solar ray and it was selectively actuated to a radius of curvature of 2.9 mm (2.9R). Furthermore, the deformed azo-LCN/4rGO could recover its initial shape once again upon the application of an external force under solar-ray irradiation. Finally, a portable lighter was employed to induce thermal actuation. When the portable lighter was switched on, thermal energy was directly transferred to the azo-LCN/rGO strip due to direct thermal conduction and convection from the flame. Under these conditions, the CTE mismatch between azo-LCN and rGO induced an immediate increase in its temperature to 73.4 °C, leading to bending actuation in only 0.3 s. The maximum angular actuation speed was 74.5 rad s^{-1} during this sub-second scale actuation (**Figure 3.18c and 3.19**).

3.2.6 Kirigami-Inspired Engineering

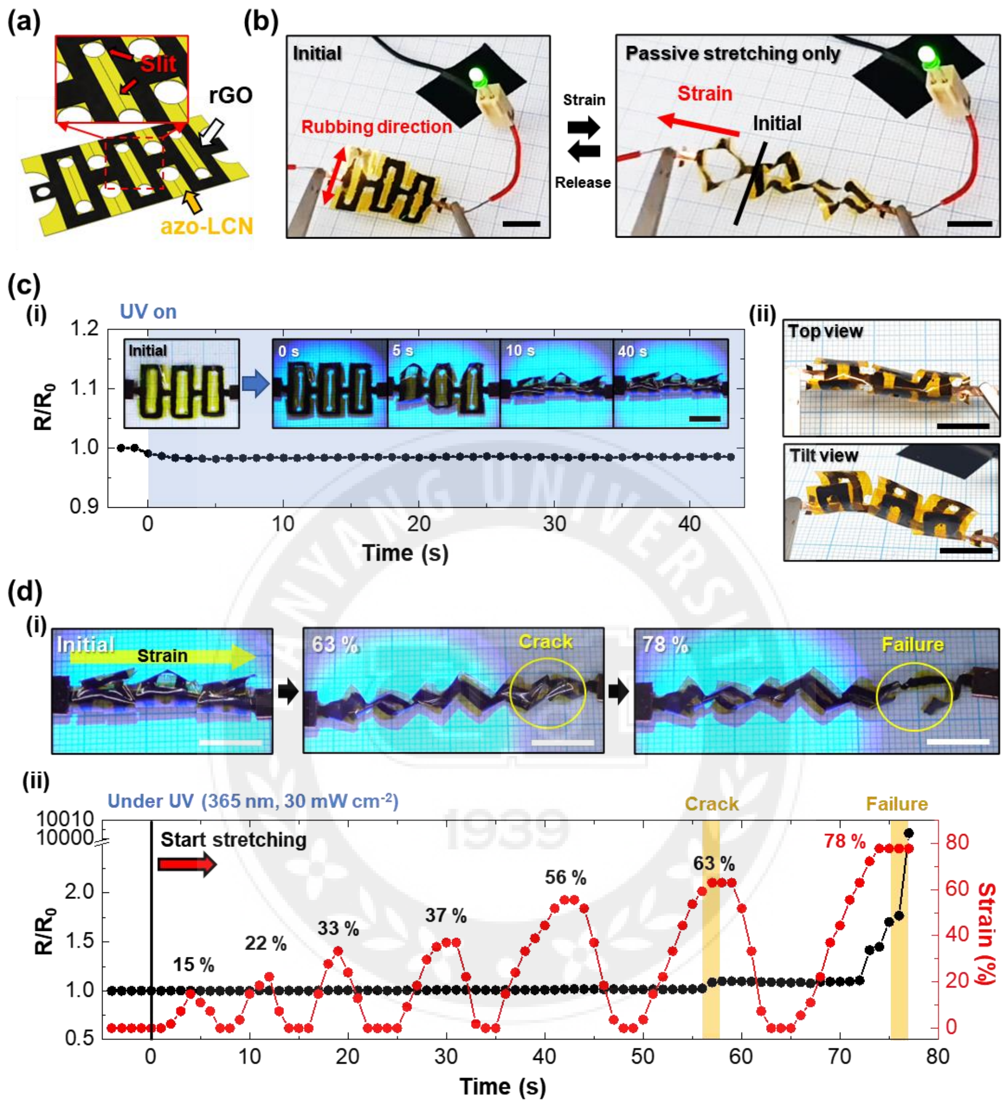

Figure 3.20 Kirigami-inspired engineering of azo-LCN/rGO. (a) Schematic illustration of the kirigami pattern used on azo-LCN/rGO. (b) Photographs of kirigami-inspired azo-LCN/rGO under passive-type stretching. A green LED (3 V) light is turned on during the stretching of the connected kirigami-inspired azo-LCN/rGO. (c) (i) The normalized resistance variation of kirigami-inspired azo-LCN/rGO under UV exposure. (ii) Photographs of UV-induced shape-reconfigured kirigami-inspired azo-LCN/rGO. (d) (i) Photographs of shape-reconfiguration of kirigami-inspired azo-LCN/rGO under UV exposure with in-situ mechanical strain. (ii) The normalized resistance and strain variation of kirigami-inspired azo-LCN/rGO under active/passive dual stimulus. The utilized UV light has 365 nm of wavelength with 0.03 W cm^{-2} of intensity. Scale bars, 1 cm.

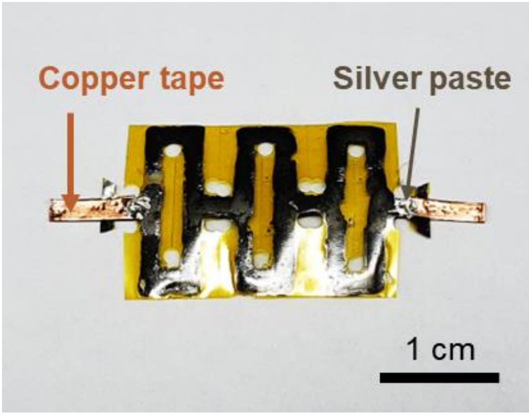

Figure 3.21 A digital photograph of kirigami-inspired azo-LCN/rGO

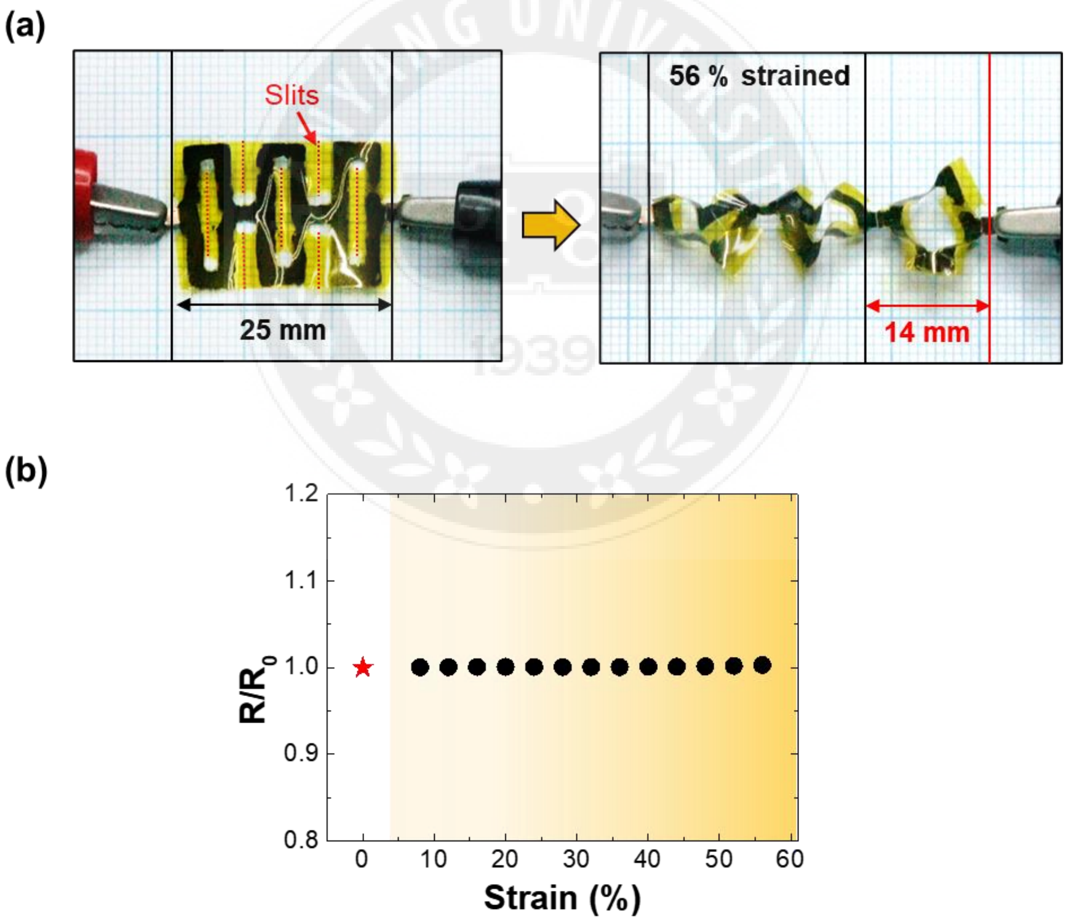

Figure 3.22 (a) Digital images of kirigami-patterned azo-LCN/rGO before and after strain. (b) Resistance variation during passive stretching.

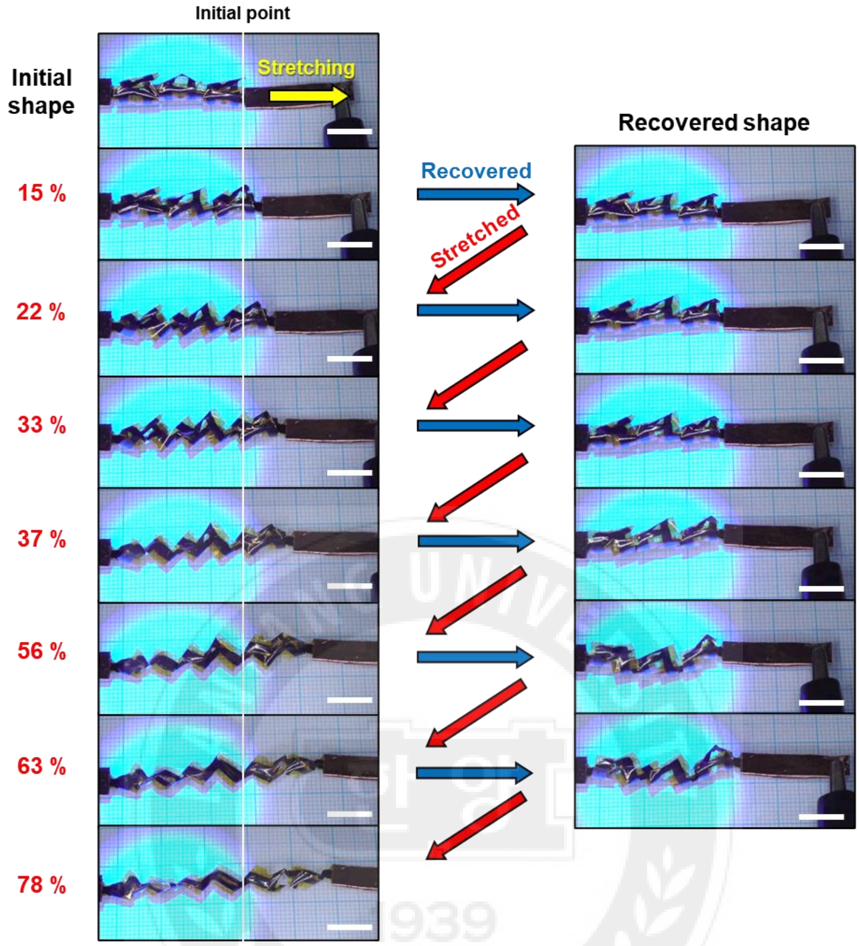

Figure 3.23 Digital images of gradually stretched and recovered kirigami-patterned azo-LCN/rGO under UV light (365 nm, 30 mW cm^{-2}). Scale bars, 1 cm.

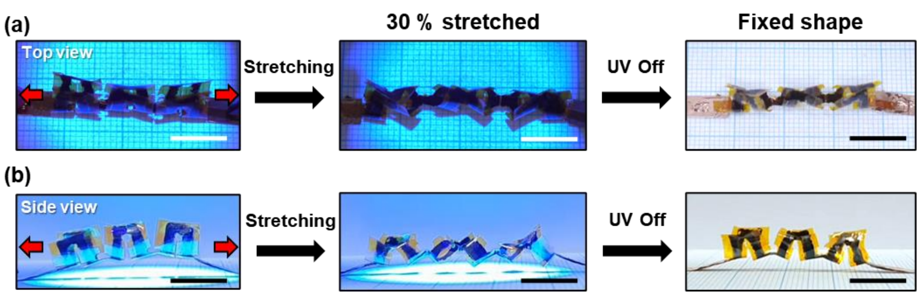

Figure 3.24 Shape fixation of stretched kirigami-patterned azo-LCN/rGO under UV exposure. (a) Top-down and (b) side images. Scale bars, 1 cm.

The concept of "kirigami" has been a focus of interest in numerous fields including soft robotics,[55] energy storage devices[56] in addition to the aforementioned stretchable electronics owing to its ability to increase the degree of freedom in shape-reconfiguration.[57] In this section, we introduce a kirigami-inspired azo-LCN/rGO to present its applicability for stretchable electronics which can be reconfigured simultaneously by passive-type mechanical stretching and active-type UV stimuli. The kirigami-inspired engineering in azo-LCN/rGO ensures its stretchability without sacrificing electrical and mechanical properties.

The kirigami pattern was applied to azo-LCN/rGO without any discontinuity in the rGO pattern (**Figure 3.20a and 3.21**) and we found that this structure could dissipate a large mechanical strain (56%) with negligible changes (≈0.1%) in its electrical resistance under passive mechanical stretching (**Figure 3.22**). This was proven when subjecting an electric circuit with the kirigami-inspired azo-LCN/rGO to an LED light; the LED light remained on under repetitive passive mechanical stretching (**Figure 3.20b**). Furthermore, kirigami-inspired azo-LCN/rGO retained its initial electrical resistance with a negligible loss (≈2%) for 40 s when irradiated with 365 nm UV light at an intensity of 0.03 W cm^{-2} (**Figure 3.20c(i)**). Here, low-intensity UV is employed to exclude the photothermal effect, confirming shape-reconfiguration of the azo-LCN/rGO by photoisomerization. The kirigami-inspired azo-LCN/rGO underwent shape-reconfiguration from a 2D flat sheet shape to a human spine-like curvilinear shape following UV exposure (**Figure 3.20c(ii)**). Hence, the kirigami-inspired azo-LCN/rGO maintained its electrical performance under separated stimuli including active-type UV exposure and passive-type mechanical stretching.

Finally, to achieve passive/active stimulus dual-responsive actuation, the kirigami-inspired azo-LCN/rGO is mechanically stretched under UV irradiation (**Figure 3.20d(i)**). The kirigami-inspired azo-LCN/rGO that had been photomechanically shape-reconfigured is gradually stretched from 15 to 78% of strain and released after each stretching step. During the mechanical stretching and continuous UV irradiation, electrical resistance variation is measured in situ (**Figure 3.20d(ii) and 3.22**). The electrical resistance of stretched kirigami-inspired azo-LCN/rGO maintained its stability before 63% of mechanical strain (≈2%). When it became stretched to 63%, a crack appeared in the body, and the connectivity of the rGO pattern was partially lost. Electrical resistance increased slightly (8–10%) due to the generation of the crack and remained until repetitive stretching proceeded to above 63% of strain. The crack in the kirigami-inspired azo-LCN/rGO denoted failure under repetitive mechanical stretching above 78% of strain. At that point, the rGO pattern became perfectly disconnected and lost electrical conductivity. Nevertheless, the kirigami-inspired azo-LCN/rGO demonstrated high stretchability (≈60%) and durability which was proven through repetitive stretching under UV exposure. In summary, the kirigami-inspired azo-LCN/rGO ensures structural diversity by demonstrating secondary deformation under active/passive type dual-stimulus without deterioration of electrical performance. In addition, the stretched shape can become fixed if the UV light is turned off during the stretching process (**Figure 3.24**).

3.3 Conclusions

In Chapter 3, azo-LCN/rGO demonstrated programmable shape-reconfigurability, multi-stimuli responsivity with a highly enhanced elastic modulus, and electrical conductivity, thus overcoming the traditional trade-off relationship between elastic modulus and strain responsivity. Varying the number of rGO coating cycles helped tune the mechanical and electrical properties of azo-LCN/rGO. Despite the small spatial composition of rGO (≈8% of the total thickness) in azo-LCN/rGO, its elastic modulus was reinforced by up to 500% (6.4 GPa) than that of neat azo-LCN and its electrical conductivity was as high as 380 S cm^{-1}. Such remarkable properties could be achieved by strong secondary interactions (π–π interactions and hydrogen bonding) between azo-LCN and rGO, which enabled a stable patterned structure. Photo-induced actuation was realized by a synergistic combination of photoisomerization in the azobenzene moiety and photothermally driven CTE mismatch between azo-LCN and patterned rGO. Upon UV irradiation, 3D shape-reconfiguration could be achieved, with diverse shapes such as helical coils and double helices, by manipulating the nematic director and rGO patterning. The multi-stimuli-responsive characteristics of azo-LCN/rGO were demonstrated using easily accessible actuation sources such as irradiating NIR and solar rays and using a portable lighter. Finally, the introduction of a kirigami pattern enabled azo-LCN/rGO to undergo active 3D shape transitions without any deterioration in its electrical conductivity. We believe that these kirigami azo-LCN/rGO structures have significant potential for application in contactless shape-reconfigurable stretchable/flexible electronics, antennas, and soft robotics.

3.4 Experimental Section

Materials: The chemicals 1,4-bis-[4-(3-acryloyloxypropyloxy)benzoyloxy]-2-methylbenzene (RM257) and 4,4-bis[6-(acryloyloxy)hexyloxy]azobenzene (2-Azo) were purchased from Synthon Chemicals GmbH & Co. Irgacure 784 was purchased from BASF. Polydimethylsiloxane (PDMS, Sylgard 184) was purchased from the Dow Chemical Company. Dopamine hydrochloride (2-(3,4-dihydroxyphenyl)ethylamine hydrochloride), tris(hydroxymethyl)aminomethane, sodium hydroxide, and sodium periodate were purchased from Sigma Aldrich. Elvamide 8061 was purchased from DuPont. Graphite powder was procured from Asbury Carbon (grade 2012). HI and sulfuric acid (95%) were sourced from Junsei Chemical Co., Ltd., and deionized water (DIW) was obtained using a Millipore (Direct Q3) water-purification system. All chemicals were used as received without additional treatment.

Preparation of the graphene oxide (GO) solution: Graphite and concentrated H_2SO_4 were stirred together and cooled to 5 °C using a circulator. Potassium permanganate was slowly added to the mixture while the temperature was maintained below 12 °C. Subsequently, the mixture was heated to 35 °C and stirred for 12 h after which it was cooled to 5 °C, and DIW was slowly added. H2O2 was poured into the mixture, which resulted in an orange-brown coloration. The synthesized GO was filtered and washed with 1 M HCl. The obtained GO cakes were dried for one week, subsequently redispersed in acetone and then filtered. Later, the filtered GO was dried at room temperature (25 ºC) for 72 h and dispersed in DIW to obtain GO dispersions.

Preparation of LC Mixtures: LC mixtures were prepared by mixing 78.5 wt% RM257, (Synthon Chemicals GmbH & Co. KG), 20 wt% 2-Azo (Synthon Chemicals GmbH & Co. KG), and 1.5 wt% of the photoinitiator (Irgacure 784, BASF).

Preparation of the rGO-Patterned Glass Cell: A patterned adhesive tape was attached to a clean glass slide. Subsequently, PDA was selectively coated on the unpatterned areas of the slide. Tris buffer (29.4 mg mL^{-1}, 12.1 mg of tris(hydroxymethyl)aminomethane, 17.1 mg of sodium periodate, and 0.1 mg of sodium hydroxide) and aqueous dopamine hydrochloride (7.6 mg mL^{-1}) were used for coating the glass slide with PDA. In this process, the Tris buffer and aqueous dopamine hydrochloride were poured on the patterned glass slide and polymerized for 30 min. A GO solution was then spread directly on top of the PDA layer. The number of GO coating cycles was varied based on the required electrical/mechanical performance. After drying, patterned GO was reduced to rGO with 30 wt% HI for 30 min. Patterned rGO was washed with ethanol to remove any residual HI and dried for 24 h in a convection oven at 60 °C. Finally, the rGO-patterned glass cell was prepared by attaching the rGO-patterned glass slide and polyamide-coated glass slide with a spacing of ≈20 μm. A polyamide-coated slide glass was prepared by spin coating with an Elvamide solution (0.125 wt% in methanol) and mechanically rubbing it.

Characterization of rGO Patterns and Azo-LCN/rGO: The surface morphology and thickness profiles of patterned rGO were observed using an optical microscope and a surface profiler (KLA Tencor, D-300). Molecular alignment in these systems was observed by SEM (HITACHI, S-4000) and POM. Trans-illumination optical microscope images were converted to grayscale color codes and normalized to the color code of the brightest glass region to evaluate the relative transparency.

Electrical Conductivity Measurements and Evaluation of Mechanical Properties: The electrical resistance of rGO patterns was measured using a multimeter (Keithley, DMM7510). These measurements were carried out on azo-LCN/rGO strips with a thickness of 22 μm, width of 1 mm, and length of 20 mm. Their mechanical properties were measured using a dynamic mechanical analyzer (DMA; Q800, TA Instruments) under tension. Storage modulus was measured in the strain sweep mode with a single amplitude of 20 μm at room temperature (25 °C). The dimensions of all samples were fixed at 1 mm width, 20 mm length, and 22 μm thickness. During DMA testing, azo-LCN/rGO strips were exposed to collimated broadband UV light (Lumen Dynamics, Omnicure S2000) in situ for 2 min and UV intensity gradually increased from 0.1 to 0.4 W cm^{-2}. The collimator was used to ensure uniform UV intensity over the exposed areas. The thermal images of light-irradiated azo-LCN/rGO were captured using a thermal-imaging camera (FLIR, E40).

Stimuli-Responsive Actuation of Azo-LCN/rGO: Collimated broadband (Lumen Dynamics, Omnicure S2000) and 365 nm monochromatic UV-light sources (CoolLED, pE-4000) were used to induce shape transition in azo-LCN/rGO and neat azo-LCN. To evaluate their bending performance, all the samples (1 mm × 20 mm × 22 μm) were exposed to broadband UV light conditions. An NIR light source (UNIX, UIM-250) was used to demonstrate photothermally induced bending actuation in the samples. Actuation in the kirigami-inspired azo-LCN/rGO samples was induced using a 365 nm monochromatic UV-light source (CoolLED, pE-4000). The light intensity was measured using a power meter (Thorlabs, PM100D).

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Chapter 4 Multi-Functional Locomotion of Collectively Assembled Shape-Reconfigurable Electronics

4.1 Introduction

In miniaturized soft electronics, it is difficult to transform the shape and to transfer the position of the electronics by human hand, without any use of precise equipment. Shape-reconfigurable and locomotive electronics with physical intelligence have great potential to improve the functionality of soft electronics by allowing on-demand programmable actuation which can induce directional motility as well as on-demand 3D shape-morphing.[1-3] As a material encoded with physical intelligence, liquid crystalline (LC) polymers can be introduced as suitable candidates for the main body of shape-reconfigurable electronics that can perform diverse remotely controlled motions without bulky and complicated onboard power sources. The alignment of LC polymers can be programmed at the internal molecular level along the thickness by surface-induced alignment techniques, and the polymers can undergo anisotropic compression/expansion by sensing external thermal energy.[4, 5] In contrast to the typical polymers that have isotropic random coil structures, the LC polymer can perform more dramatic, reversible, and directional shape-reconfiguration due to its highly aligned structures and alignment programmability.[6, 7] Therefore, LC polymers have been widely studied as a key component in actuators[8-10], soft robotics[11-13], shape-reconfigurable micro-structures[14, 15], medical devices[16], antennas[17, 18], and electronics.[1, 2, 19-21] However, there are relatively few attempts to use LC polymers as electronic devices because it is challenging to achieve high electrical conductivity of LC polymers.

To achieve high-performance LC polymer-based shape-reconfigurable electronics, it is critical to obtain high performance in all criteria (electrical conductivity, lightweight, durability, and strain capability) by considering the selection of the materials and methods of integrating both the LC polymer matrix and conductive fillers. As integration methodologies, there are two available strategies for hybridizing LC polymers with conductive fillers; one is embedding the filler inside the LC polymer, and the other is the formation of a bilayer structure. In the case of embedding the filler inside the LC polymer, there are critical problems in that the fillers in the LC polymer interfere with the alignment of the LC molecules and reduce the order parameter.[22, 23] The reduced orientational order suppresses the shape-reconfiguration ability of the LC polymer composites. In addition, the high stiffness of the conductive fillers deteriorates the strain response of the LC polymer composites by increasing the stiffness. However, to achieve high electrical conductivity, high content of conductive fillers is required to satisfy the electrical percolation, which has a trade-off relationship with the strain capability of the LC composites.

On the other hand, introducing conductive fillers into LC polymers as a bilayer structure is a reasonable strategy that is relatively free from the trade-off relationship between the content of the conductive fillers, the strain responsivity, and the electrical conductivity.[1, 24, 25] In the bilayer structure, the conductive fillers should have strong interfacial adhesion with the LC polymer to prevent delamination during repeated actuation and to provide high actuation strain. Hence, the LC polymers are usually hybridized with lightweight, solution-processable, and electrically conductive carbon-based nanomaterials like carbon nanotubes (CNTs)[23, 26-28], reduced graphene oxide (rGO)[1], and graphene[29] to induce strong adhesion with the LC polymer by hydrogen-bonding and π-π interaction. In our previous research, we developed an LC polymer network (LCN)/reduced graphene oxide (rGO) bilayer that demonstrated an electrical conductivity of ~380 S cm^{-1}, and remotely controlled diverse shape-reconfigurations under photonic stimuli[1]. However, the insufficient conductivity and high contact resistance limited the ability to turn electrical diodes and to perform electro-thermally driven actuation. Furthermore, due to the dense chemical crosslinks of LCN, the glass transition occurs when the temperature is near 40 °C, which is above the room temperature[30]. When the photo-thermal effect is removed, the reconfigured shape of the LCN/rGO bilayer is fixed by air quenching at room temperature because the elastic modulus is drastically increased before shape-recovery. Hence, the LCN is limited in achieving a system that requires continuous and reversible motions.

Herein, we introduced a highly conductive $Ti_3C_2T_x$ MXene patterned liquid crystal elastomer (LCE) bilayer via a facile *in-situ* transfer process integrated with photopolymerization for shape-reconfigurable electronics. To achieve high actuation strain and reversible actuation, loosely crosslinked LCE is adapted as a programmable matrix instead of a highly crosslinked LCN. LCE has a longer chain length between the crosslinked points than the LCN, which leads to a shorter relaxation timescale than the timescale of cooling by air quenching. Hence, LCE can be reversibly recovered to its original shape immediately after removing the thermal stimulus. As a conductive filler, we adapted MXene, which is a new family of 2D materials with lightweight, solution processability, significantly high electrical conductivity, and ~99% light-to-heat conversion efficiency in an NIR regime.[31-36] By using a customized MXene patterned glass cell, ~50 μm LCE was integrated with a ~0.37 μm thick MXene layer, which is 133 times thinner than the LCE layer. MXene/LCE bilayer (MLB) demonstrated a remarkable electrical conductivity of ~5,300 S cm^{-1}, 13.4 times higher than the previously reported conductive rGO/LCP composites (380 S cm^{-1})[1] and 14.6 times higher than the previously reported CNTs/LCP composites (350 S cm^{-1})[26]. Electrical performance was stably maintained without delamination of the MXene layer, even after artificially bending to a curvature radius of 1.29R, twisting to 2,160°, and repeating photo-thermally driven actuation for 1,000 cycles by NIR irradiation. The structural durability of the MLB originates from the strong interfacial adhesion, induced by the hydrogen bonding between the terminal groups in MXene such as fluorine (–F), oxide (–O), hydroxyl (–OH), and carbonyl groups of mesogen in the LCE. MLBs demonstrated photo- and electro-thermally driven rapid and large in-plane bending or torsional twisting along the pre-programmed alignment direction. As a proof-of-concept, an MLB was applied in an electrical circuit as a shape-reconfigurable wire, which can remotely switch on/off the light-emitting diodes (LEDs) by photo-thermally driven actuation. Finally, we introduced collectively assembled MLBs (Co-MLBs) by connecting several MLB strips, to expand the diversity of reversible shape-reconfiguration towards multi-curvilinear structures such as S-, W-, Flower-shape, and chiral structures. Geometrical diversity of the Co-MLBs is achieved by varying the direction of molecular alignment and body length of each MLB unit. As a result, the Co-MLBs perform on-demand soft-robotic actions such as crawling and directional rotation under repetitive NIR irradiation. Furthermore, inspired by snap-through behavior, Co-MLBs attached to a rigid frame demonstrated photo-thermally driven jumping and slingshot motion through storing elastic energy and explosively releasing stored energy.

4.2 Results and Discussion

4.2.1 Preparation and Characterization of $Ti_3C_2T_x$ MXene Patterned LCE Bilayer (MLB)

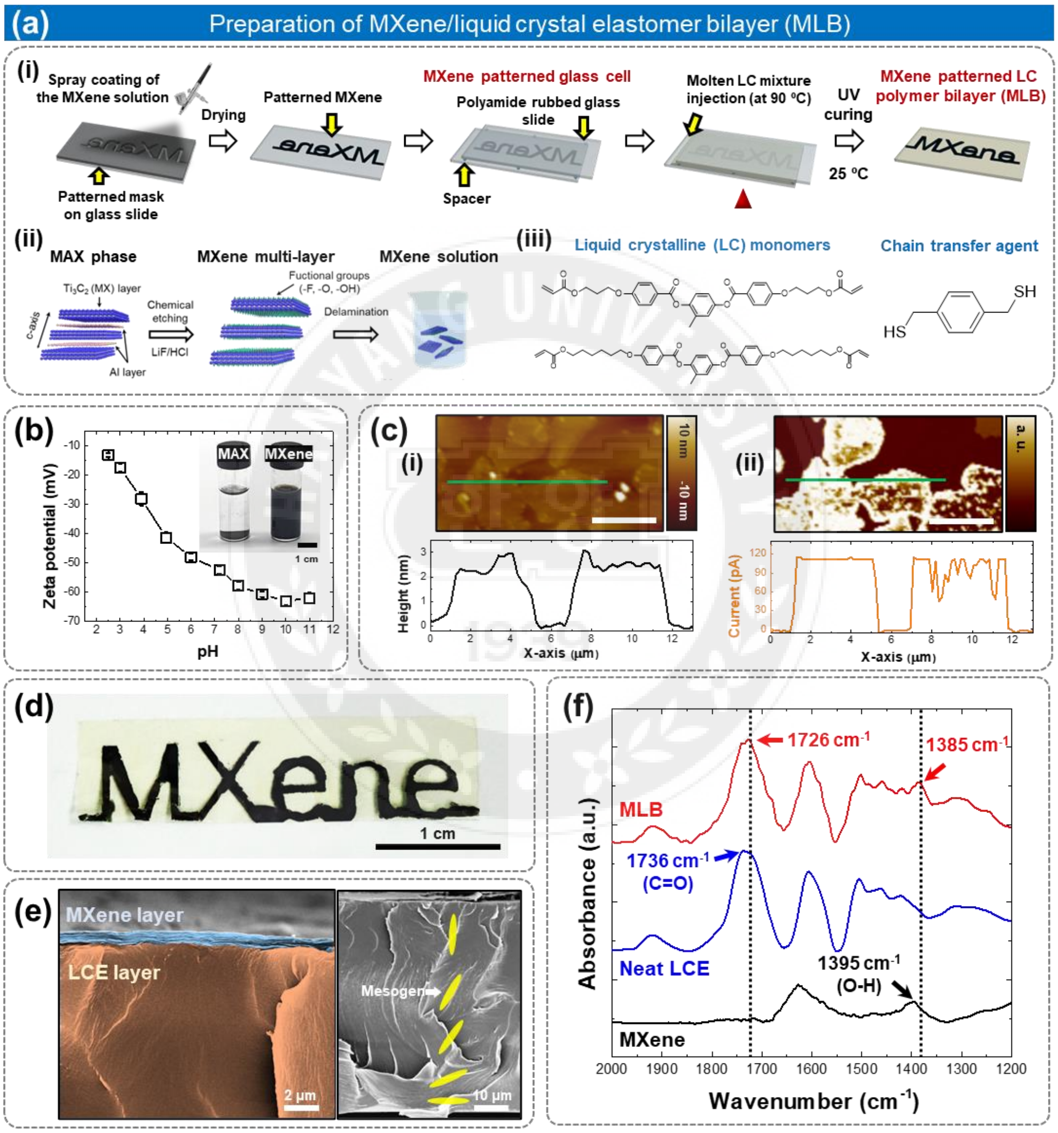

Figure 4.1 Preparation and characterization of the MXene-patterned LCE bilayer (MLB). (a) (i) Preparation of MLB, (ii) preparation of MXene aqueous solution, and (iii) chemical structures of applied liquid crystalline (LC) monomers and dithiol chain transfer agent. (b) (i) Zeta potential of MXene aqueous solution according to pH. The inset digital image indicates 10 mg mL^{-1} MAX aqueous suspension (left) and 10 mg mL^{-1} MXene aqueous solution after 4 h of mixing (right). (c) (i) Atomic force microscopy (AFM) topological

images and corresponding conductive-AFM current map of MXene single sheets coated on SiO_2 wafer. (ii) Height and current profile along the green line in Fig. 1c (i). (d) A 'MXene' letter-patterned MLB was prepared by utilizing 10 mg mL^{-1} MXene aqueous solution. (e) Cross-sectional SEM images of MLB. The inset yellow rod images indicate the mesogens. (f) Fourier transform infrared (FT-IR) absorption spectra of MLB, neat LCE, and MXene powder.

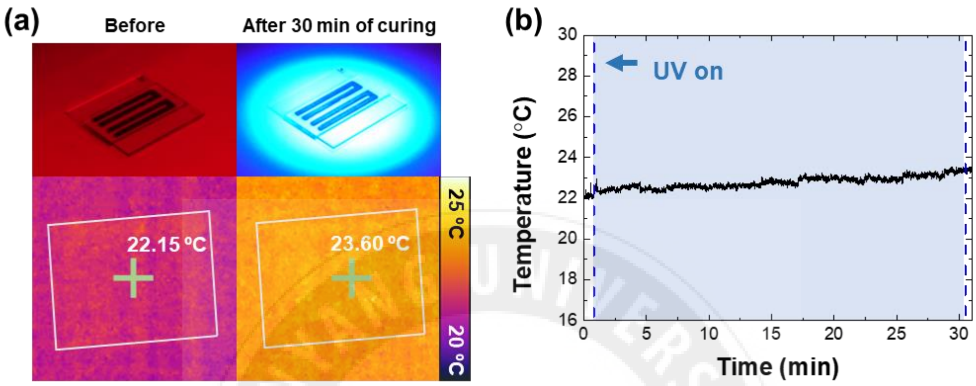

Figure 4.2 (a) Digital images and thermal images of MXene patterned glass cell filled with LC mixture before photo-polymerization, and after 30 min of the *in-situ* photo-polymerization. (b) Temperature variation is plotted as a function of polymerization time during photo-polymerization under UV irradiation with an intensity of 30 mW cm^{-2}.

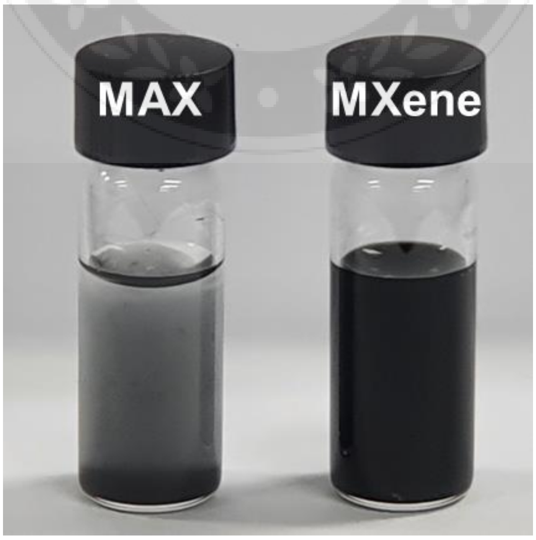

Figure 4.3 Digital image of MAX phase suspension and MXene dispersion immediately after dissolution in deionized water.

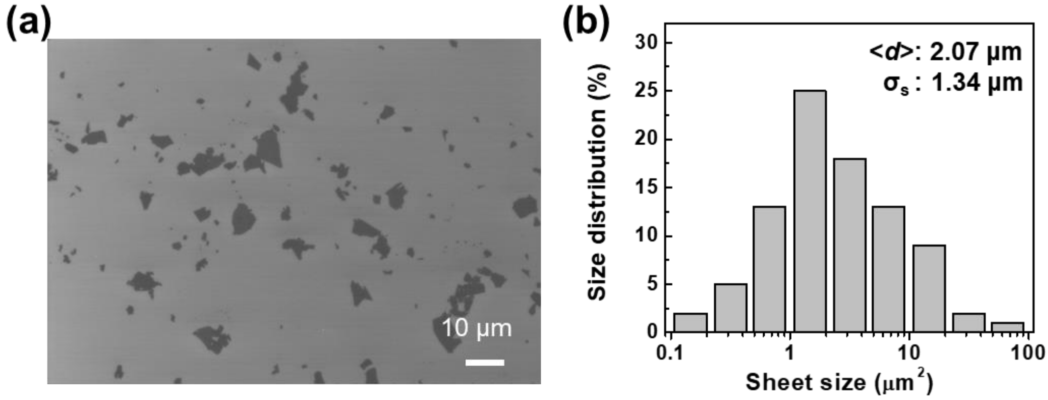

Figure 4.4 (a) SEM image of MXene sheets and (b) sheet size distribution data for MXene.

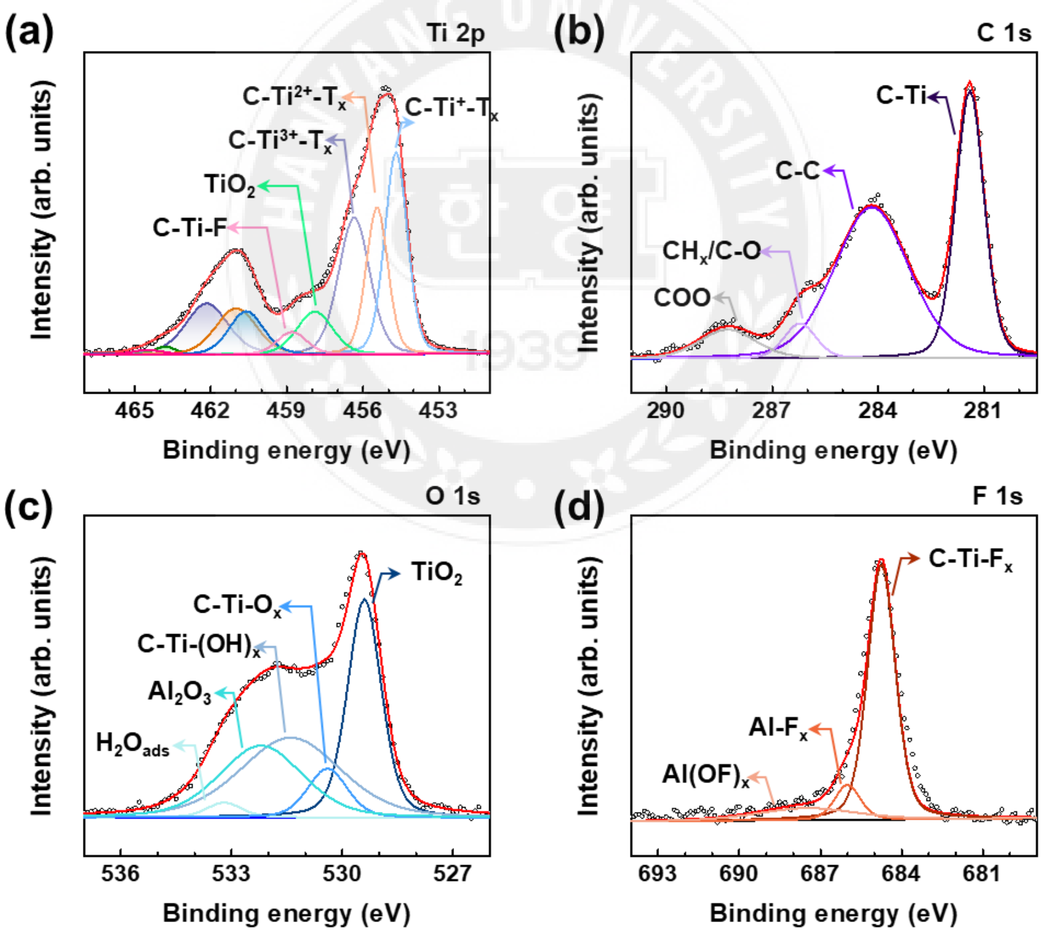

Figure 4.5 XPS data for patterned MXene, (a) Ti 2p, (b) C 1s, (c) O 1s, (d) F 1s. Ti $2p_{3/2}$ spectra are labeled with arrows while Ti $2p_{1/2}$ spectra are filled with color.

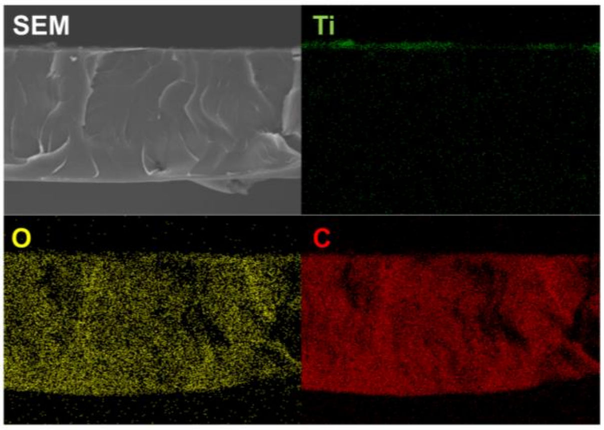

Figure 4.6 A cross-sectional SEM image of MLB and its corresponding Ti, O, and C elemental mapping images

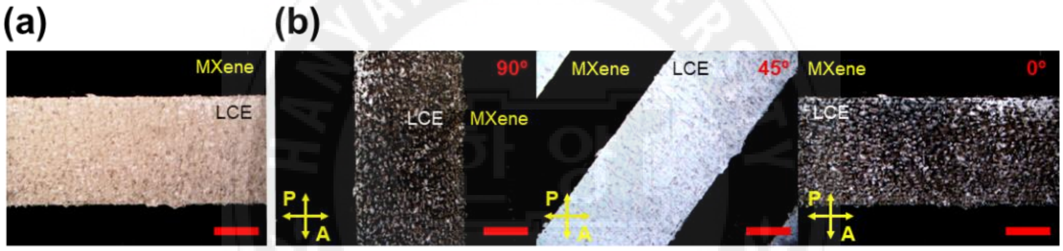

Figure 4.7 (a) Optical microscopy image of LCE region in MLB. (b) Polarized optical microscopy images of the LCE region in MLB according to the tilt angle. Scale bars: 0.5 mm

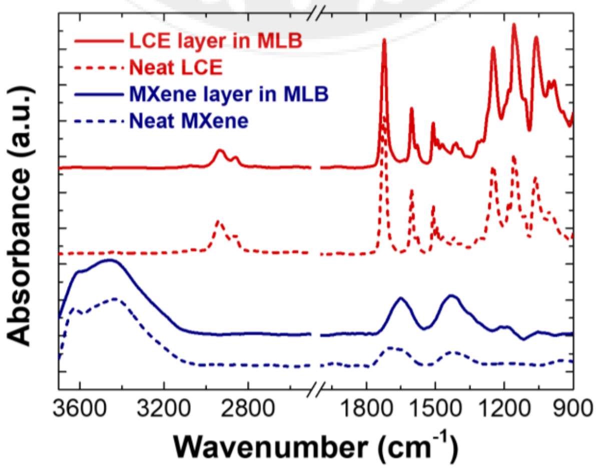

Figure 4.8 Attenuated total reflectance (ATR) mode FT-IR spectra of both sides of MLB (LCE layer and MXene layer), neat LCE, and neat MXene.

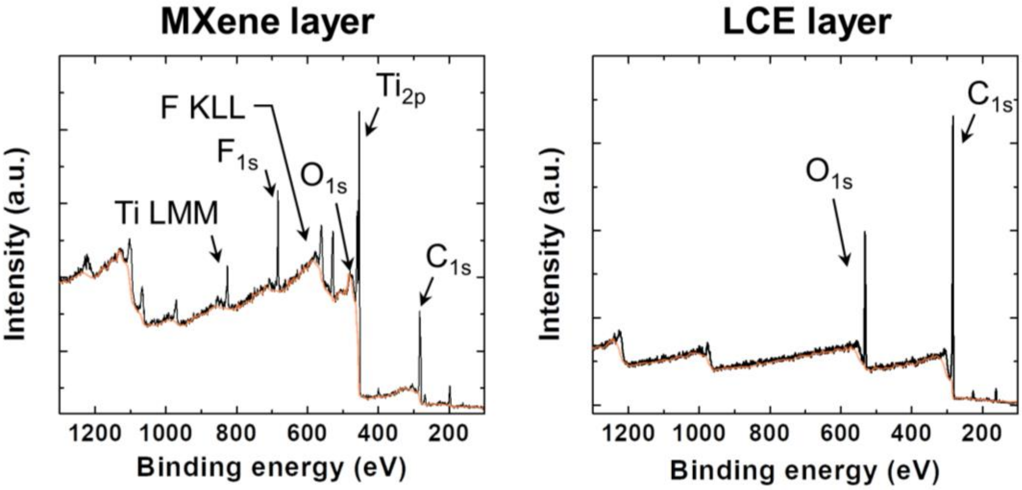

Figure 4.9 XPS survey spectra of both sides of MLB. (a) MXene layer and (b) LCE layer. The black solid line represents survey scan lines while the orange solid lines are background peaks

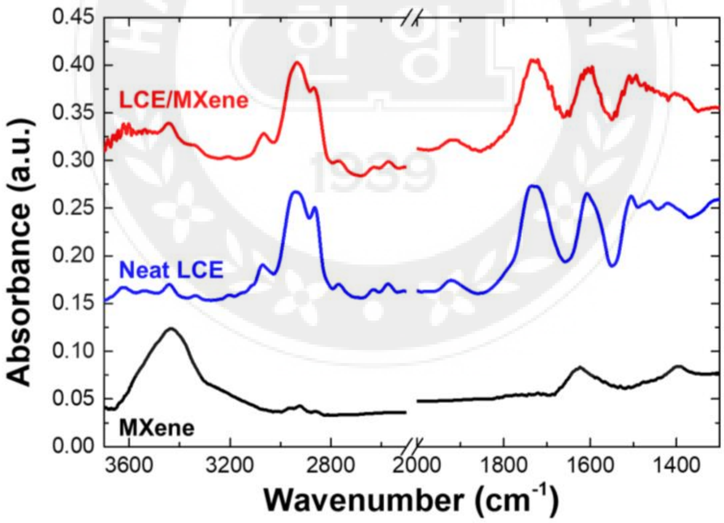

Figure 4.10 Transmission mode FT-IR spectra of MLB, neat LCE, and neat MXene.

Element	Area CPS.eV	Atomic percent (%)
C_{1s}	67638.38	39.57
Ti_{2p}	268979.53	26.04
F_{1s}	70293.55	11.96
O_{1s}	99804.22	22.43

Table 4.1 The atomic percentage of components in the MXene ($Ti_3C_2T_x$)

The highly conductive MLB with programmed alignment was prepared by facile transfer of the MXene layer to the LCE via a strong interfacial interaction between these heteromaterials (**Figure 4.1a(i)**). The $Ti_3C_2T_x$ MXene patterned glass cell was constructed by attaching two glass slides, each coated with a patterned MXene layer and mechanically rubbed polyamide-based alignment layer. To prepare the glass slide coated with the MXene layer, the multi-layer MXene was exfoliated as a single layer and stably dispersed in deionized (DI) water by utilizing Ti_3AlC_2 ($M_{n+1}AX_n$ phase) powder with a LiF/HCl etchant that can selectively remove the Al in Ti_3AlC_2 (**Figure 4.1a(ii)**).[36] The MXene-patterned glass slide was obtained by spray-coating the MXene aqueous solution on the masked glass slide and peeling off the mask. Another glass slide with the alignment layer was attached to the MXene patterned glass slide, with a physical spacing of 50 μm, and the MXene patterned cell was obtained. Microgrooves are generated on the surface of the alignment layer along the rubbing direction that can guide the molecular alignment of the LC monomers.[37] The MXene-patterned cell acts as a thickness-controlled planar reactor for synthesizing the MXene/LCE bilayer, named MLB. The LC mixture is composed of 68.0 wt % RM82 as a monomer, 19.5 wt% RM257 as a comonomer, and 1.0 wt% I-369 as a photoinitiator, respectively (**Figure 4.1a(iii)**). At this composition, the LC monomers can prevent the nematic-crystallization phase transition upon quenching to room temperature (25 ◦C) which is below the crystallization temperature, and a meta-stable nematic phase can be maintained for ~1 h.[5] A dithiol chain transfer agent (1,4-benzenedimethanethiol (BDMT); 11 wt%) was employed with the LC monomers to reduce the crosslinking density via chain transfer and chain extension mechanisms, and antioxidant (butylated hydroxytoluene (BHT); 0.5 wt %) was also employed to prevent side-reaction by radical scavengers.[38] In the experiment, the molten LC mixture was injected into the MXene patterned glass cell by capillary action at 90 °C, which is above the nematic to isotropic transition temperature (T_{NI}) of 85 °C. Then, the glass cell filled with the LC mixture was cooled down to room temperature and the nematic phase was obtained. In this step, the LC monomers in the glass cell generate hybrid alignment because the LC monomers near the MXene layer form a homeotropic alignment while the LC monomers near the polyamide-based alignment layer induce a parallel alignment. Then, we conducted *in-situ* photo-polymerization by irradiating ultraviolet (UV) light to a glass cell at room temperature, and the hybrid alignment is maintained as the acrylate functional groups in LC monomers become connected. A negligible temperature change of less than 1.5 °C (22.2–23.6 °C) was observed during 30 min of in-situ photopolymerization, indicating no significant decrease in the order parameter (**Figure 4.2a and 4.2b**). Finally, we opened the glass cell to obtain the MLB. Then, the MXene layer on the inner surface of the LC cell migrates to the LCE because the adhesion between the MXene layer and LCE is stronger than that between the MXene layer and the glass slide due to the large number of hydrogen bonds at the interface between MXene layer and LCE layer. The detailed method of preparing MLB is described in the Experimental Section.

The dispersity of the MXene nanosheets in DI water was measured by zeta-potential analysis (**Figure 4.1b**). At pH 4.95, the zeta potential of the MXene nanosheets was −41.4 mV, exceeding the standard zeta potential (−40 mV) for highly stable dispersion.[39] Hence, the zeta potential data indicate that the neutralized MXene nanosheets were stably dispersed in the aqueous medium due to electrostatic repulsion among the MXene nanosheets.[32] The charge stabilization was further enhanced under the basic conditions, and the zeta potential converged to a value of −62.6 mV when the pH exceeded 10. As expected, the MXene nanosheets were stably dispersed in DI water at pH ~7 (**Figure 4.3**). The topology and electrical conductivity of the MXene nanosheets were visualized by utilizing AFM and conductive AFM (C-AFM), respectively (**Figure 4.1c(i) and 4.1c(ii)**). The electrical conductivity of the MXene nanosheets was distinctively visualized as a bright region in the current map, and electrical conduction was confirmed by the current profile. The measured height and lateral size of the MXene nanosheets were 2.3±0.5 nm and 2.1±1.3 μm, respectively, as extracted from the AFM topological image and scanning electron microscopy (SEM) images (**Figure 4.4**). In particular, the measured height of the MXene nanosheet corresponds to the height of the single-layer MXene, indicating that the synthesized MXene was successfully exfoliated.[36,40] The functional groups of the MXene sheets were classified by using X-ray photoelectron spectroscopy (XPS) (**Figure 4.5 and Table 4.1**). From the deconvoluted peaks of MXene, abundant terminal groups containing Ti–O and Ti–F were detected; these groups enhance the colloidal stability of the MXene solution via electrostatic repulsion.

The alignment-programmed MLB with the “MXene” letter pattern was prepared without any delamination of the MXene layers by employing an MXene solution with a concentration of 10 mg mL^{-1} (**Figure 4.1d**). The LCE region in the MLB was optically clear, and the measured thickness of the MLB (~50 μm) was the same as the physical gap applied in the MXene-patterned LC cell. The hybrid alignment of the MLB was visualized by cross-sectional SEM images (**Figure 4.1e**). The MXene layer showed a thickness of 370 nm with a nacre-like layered structure, which is ~133 times thinner than the LCE layer which showed a thickness of ~50 μm (**Figure 4.1e(i)**). At the interface of MLB, the LC molecules are vertically aligned due to the fluorine groups located at the surface of the MXene. Small anchoring stress is induced near the MXene layer, originating from the low surface energy caused by the fluorine terminal groups at the surface of the MXene layer.[41,42] However, the LCE located far away from the MXene layer appeared to have horizontal alignment, induced by the physical confinement of the LC molecules on the grooves of the mechanically rubbed polyamide-based alignment layer in the cell (**Figure 4.1e(ii)**). The cross-sectional elemental mapping images of MLB are provided in **Figure 4.6.** The alignment of the MLB was also demonstrated by crossed-polarizer optical microscopy (POM) images (**Figure 4.7**). When the alignment direction was parallel to the polarizer or the analyzer of the POM, the LCE region was observed as a dark image because the aligned LC molecules absorbed the polarized light transmitted by the POM. The hybrid alignment of MLB is essential for inducing directional actuation under an external stimulus.

Both surfaces of MLB were completely composed of pure MXene and LCE confirmed by attenuated total reflection (ATR) mode FT-IR and XPS measurement indicating that the LCE layer and MXene layer were physically attached without changes in chemical composition (**Figure 4.8 and 4.9**). At the interface of MLB, the presence of hydroxide (O–H), oxide (-O_x), and fluorine groups can induce hydrogen bonding with the carbonyl groups (C=O) of the mesogens in the LCE layer. Hydrogen bonding between the MXene and LCE was confirmed by measuring FT-IR spectra with transmission mode (**Figure 4.1f and 4.10**).

First, the 1395 cm^{-1} peak of neat MXene, which corresponds to the stretching vibration of the O–H group, was red-shifted to 1385 cm^{-1} in the MLB spectrum.[43,44] Secondly, the 1736 cm^{-1} peak of neat LCE, which corresponds to the stretching vibration of the C=O group of the mesogens in RM257 and RM82, was shifted to 1726 cm^{-1} in the spectrum of MLB.[45] These two types of redshifts suggest intermolecular hydrogen bonding between the LCE and MXene layers, which contributes to strong interfacial adhesion between the LCE and MXene.

4.2.2 Electrical Performance

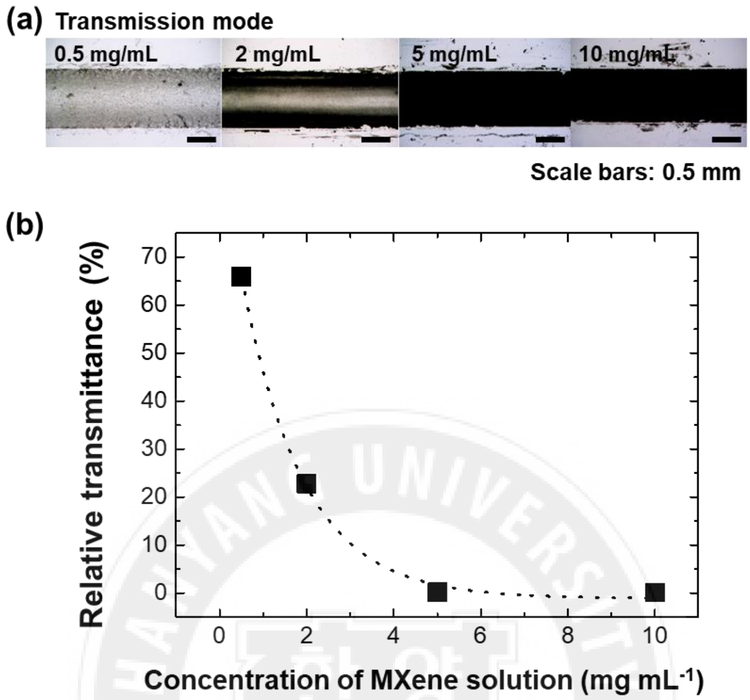

Figure 4.11 (a) Optical microscopy images of solidified MXene layers on glass slide according to the applied concentration of MXene aqueous solution. (b) Relative transmittance of MXene layers as a function of the concentration of MXene solution.

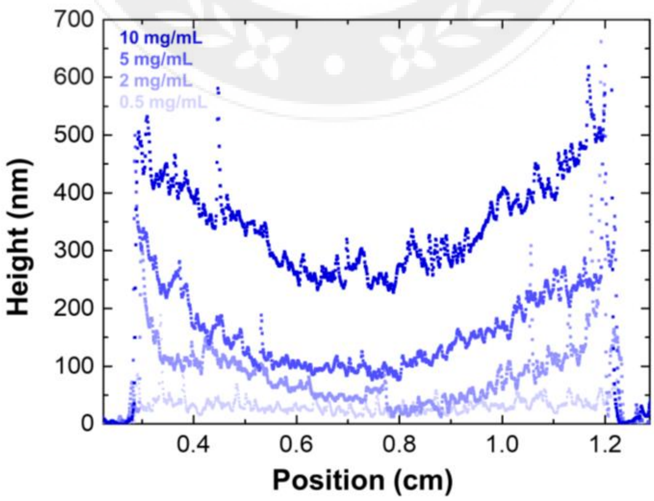

Figure 4.12 Height profiles of solidified MXene layers on glass slide according to the concentration of applied MXene solution.

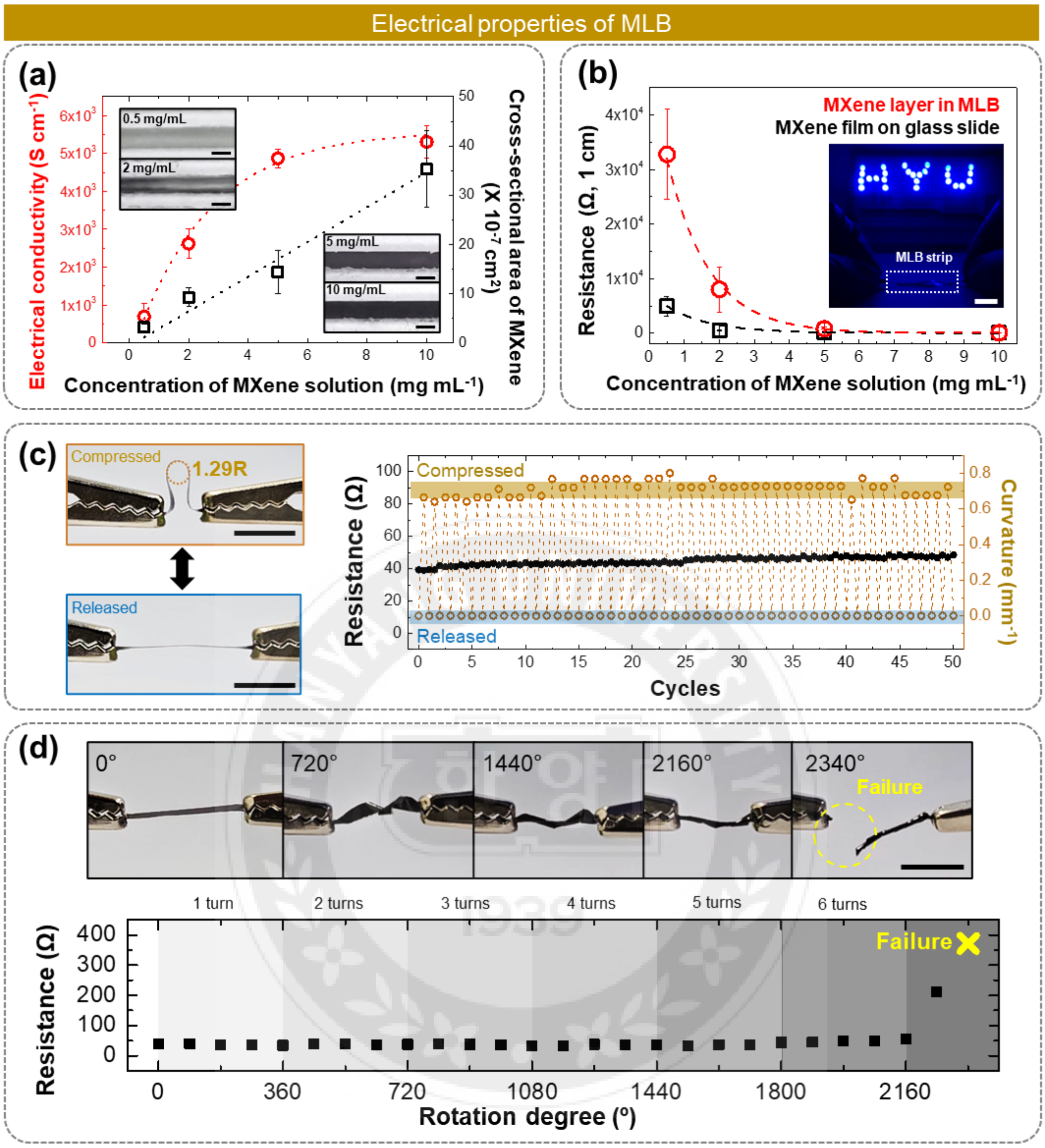

Figure 4.13 Electrical performance of MLB. (a) Electrical conductivity and cross-sectional area of MXene films on glass slide according to the applied concentration of the MXene aqueous solution. (b) The electrical resistance of the MXene layer before and after transfer onto the LCE. The inset figure demonstrates the turning on of blue LEDs connected in an electrical circuit with an MLB strip. Scale bar: 1 cm. (c-d) Negligible variation of electrical resistance of MLB (L: 20 mm and W: 2 mm) under mechanical stimulus (c) Digital images of the MLB compression and relaxation. The graph showed electrical resistance and curvature variation of MLB during repeated mechanical compression. (d) Digital images of the mechanically twisted MLB according to the twist angle. The graph showed the electrical resistance variation of the MLB under twisting. Scale bars: 1 cm.

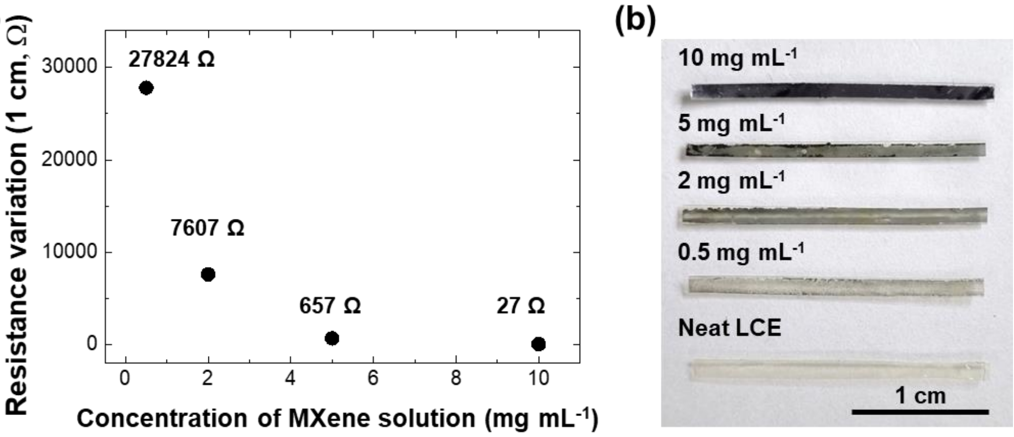

Figure 4.14 (a) Resistance variation between before and after transfer of MXene layers. (b) Digital images of MLB according to the applied concentration of MXene solution.

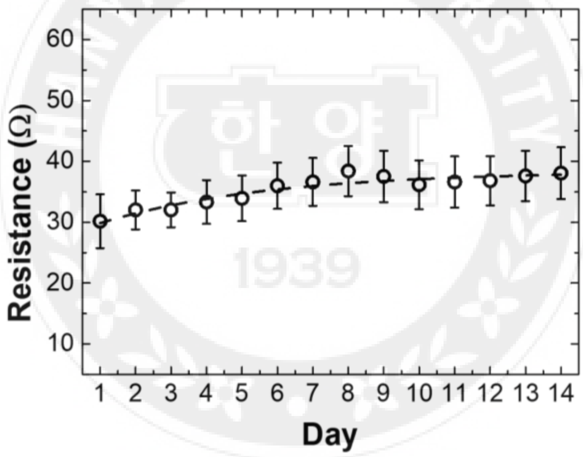

Figure 4.15 Resistance variation of MXene layer in MLB film under ambient air condition (43.1±9.1 %RH and 19.4±2.0 °C) for 2 weeks.

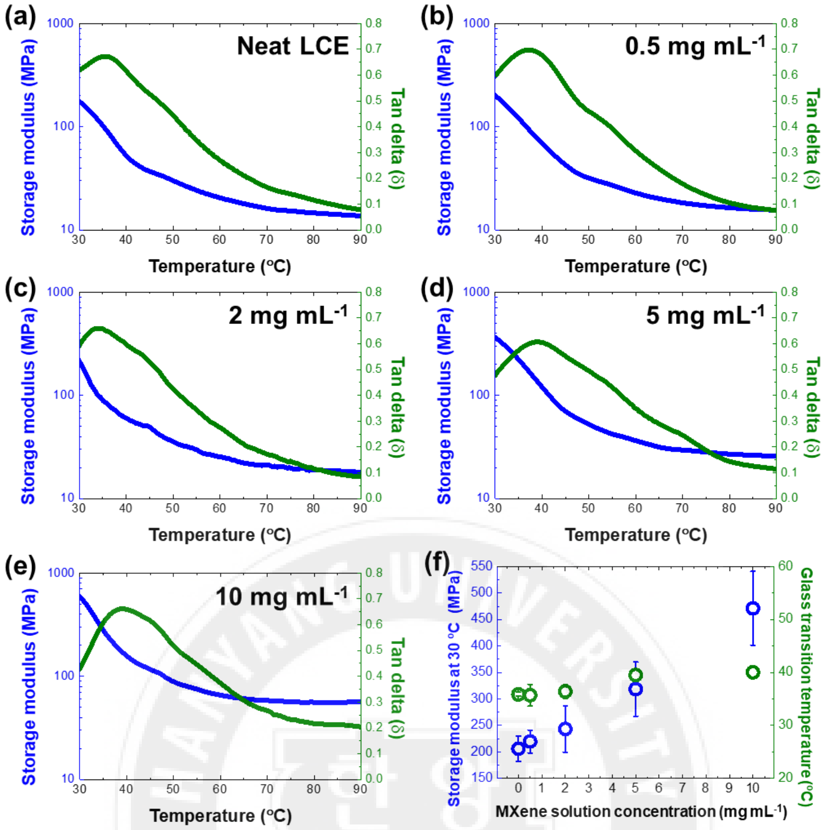

Figure 4.16 (a–e) Temperature sweep data for MLBs with different applied concentrations of MXene aqueous solution (0.5–10 mg mL^{-1}). (f) Storage modulus and glass transition temperature of MLBs (from temperature sweep data) as a function of MXene solution concentration.

Parameters	Measured value
Curvature (κ)	59.6 m^{-1}
Elastic modulus of Azo-LCN (E_{LCE})	205 × 10^{6} Pa
Thickness of Azo-LCN (t_{LCE})	50 × 10^{-6} m
Poisson ratio of Azo-LCN (ν_{LCE})	0.4
Strain of rGO (ϵ_{MXene})	0.0015
Thickness of rGO (t_{MXene})	0.37 × 10^{-6} m

Table 4.2 Parameters for calculating the elastic modulus of the MXene layer using the Stoney equation.

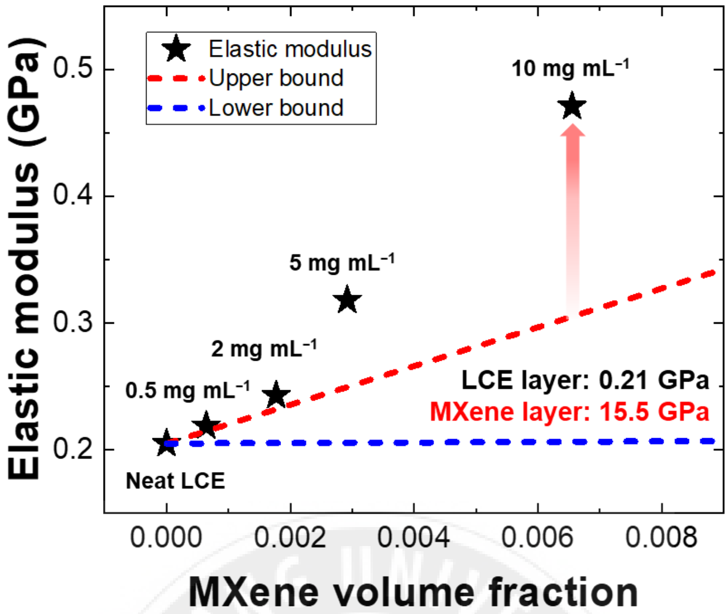

Figure 4.17 Modulus variation of MLB according to the MXene volume fraction.

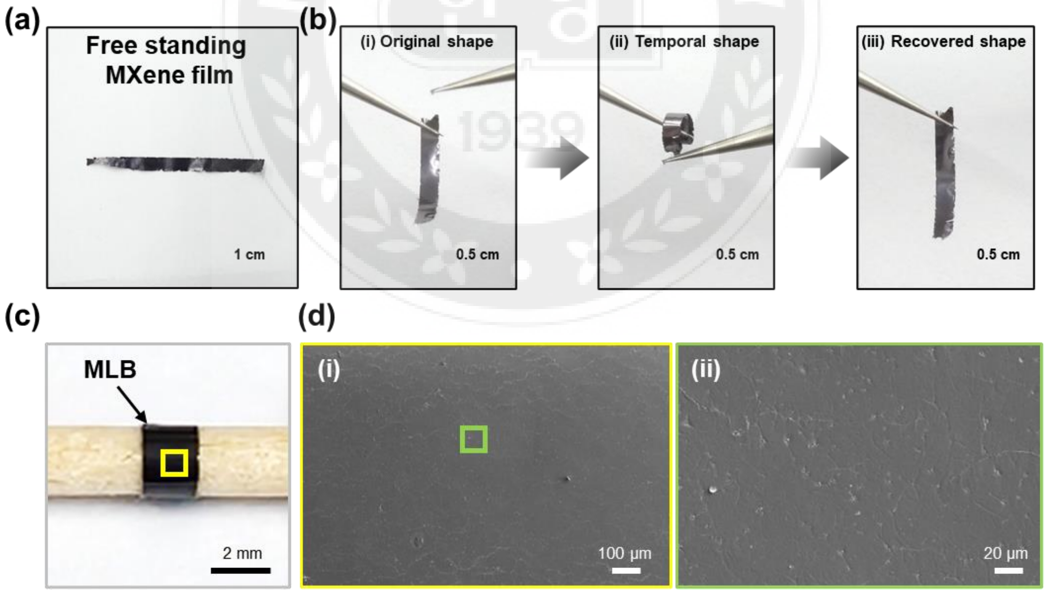

Figure 4.18 (a) An optical image of pristine MXene freestanding film. (b) (i) Original shape, (ii) temporally bent shape under mechanical stress and (iii) recovered shape after removing the mechanical stress. (c) An optical image of MLB rolled up on a bar with a radius of curvature of 1R. (d) (i) Low-magnification and (ii) high-magnification SEM images of MXene layer in bent MLB.

In the formation of the solidified MXene film via spray coating on the glass slide, the electrical conductivity is affected by the number of defects in the solidified MXene film, which interfere with percolation among the MXene nanosheets. To ensure high electrical conductivity, the threshold concentration of the MXene solution was determined by examining the optical microscopy images and calculating the electrical conductivity of the solidified MXene films on the glass slide, prepared by employing various concentrations of MXene solution (0.5, 2, 5, and 10 mg mL^{-1}) with a fixed volume. In the optical microscopy images, the solidified MXene film had a darker color when the applied solution contained a higher concentration of MXene (**Figure 4.11a**). The relative transmittance exponentially decreased as the concentration of the MXene solution increased, saturating at ~0.1% at a concentration above 5 mg mL^{-1} (**Figure 4.11b**). The electrical conductivity of the MXene films on the glass slide was calculated using Equation (4.1).

$$\sigma = \frac{1}{\rho} = \frac{L}{A \times R} \quad (4.1)$$

Here, σ is the electrical conductivity, ρ is the resistivity, L is the length, A is the cross-sectional area, and R is the measured resistance. The L was fixed at 1 cm and the A was determined by integrating the cross-sectional 1D height profile determined by using the surface profiler (**Figure 4.12**). With increasing concentration of the MXene solution, the average cross-sectional area of the MXene film increased linearly from 3.2×10^{-7} cm^2 at 0.5 mg mL^{-1} to 35.3×10^{-7} cm^2 at 10 mg mL^{-1}, and the electrical conductivity of the MXene film increased exponentially from 698.9 S cm^{-1} at 0.5 mg mL^{-1} to 5,307.4 S cm^{-1} at 10 mg mL^{-1}, then became saturated (**Figure 4.13a**).

The cross-sectional area of the MXene film is dependent on the amount of MXene nanosheets that participated in self-assembly. Hence, the cross-sectional area of the MXene films has a linear relationship with the applied concentration of the MXene solution. The electrical conductivity is highly related to the electrical percolation, induced only when there is physical contact among the solidified MXene nanosheets [46]. Initially, the electrical conductivity increased exponentially and then reached saturation when the concentration of the MXene solution exceeded the threshold for inducing complete volumetric percolation among the solidified MXene nanosheets. Notably, the electrical performance of the MLB is strongly related to the quality of transferring the MXene layer to the LCE layer. The change in the electrical resistance is demonstrated as a function of the MXene solution concentration, before and after the transfer of the MXene layer to the LCE layer (**Figure 4.13b**). The electrical resistance deviation before and after the transfer of the MXene layer declined exponentially from 27,824 Ω to 27 Ω as the applied MXene concentration increased from 0.5 mg mL^{-1} to 10 mg mL^{-1} (**Figure 4.14**). The highly percolated MXene film has a more regular monolithic film structure than the less percolated MXene film and enables the intact transfer of the MXene film to the LCE layer with fewer defects. The electrical resistance of the MXene layer in MLB slightly increased from 30.16±4.42 Ω to 38.07±4.24 Ω, and saturated following the exponential decay function during 2 weeks of measurement under ambient air conditions (**Figure 4.15**). The temperature sweep tests in the dynamic mechanical analysis (DMA) showed an increase in the storage modulus of the MLB from 205.9 MPa at 0.5 mg mL^{-1} concentration to 471.0 MPa at 10 mg mL^{-1} concentration due to the enhanced percolation of the MXene film (**Figure 4.16**). The elastic modulus of the MXene layer was calculated using the Stoney equation and measured parameters described in Table 4.2. A detailed explanation of the Stoney equation is provided in **Equation (3.2) and (3.3)** in Chapter 3. As a result, the elastic modulus of the MXene layer was determined to be 15.5 GPa and this value was subsequently applied to calculate upper and lower bounds using the rule of mixtures. In particular, the elastic modulus of the MLB, prepared by using 10 mg mL^{-1} MXene solution, is higher than the upper limit of the rule of the mixtures even though the MLB has a bilayer structure (**Figure 4.17**). This result implies that there was a strong interfacial interaction due to the molecular-level hydrogen bonding.[47] Finally, we selected the 10 mg mL^{-1} MXene solution to achieve a saturated conductivity of ~5,300 S cm^{-1} with a stably transferred state to the LCE layer.

The stable electrical performance of the MLB was demonstrated by turning on LEDs and maintaining the electrical resistance of the MLB under mechanical stimuli, respectively. As shown in the inset digital image of **Figure 4.13b**, blue LEDs were turned on by composing an electrical circuit with an MLB strip. The MLB exhibited structural and electrical durability under repetitive mechanical stimuli such as compression/relaxation cycles and torsional twisting. The electrical resistance of the MLB changed negligibly, and there was no physical delamination over fifty mechanical compression and relaxation cycles or with harsh torsional deformation of 2160° (**Figure 4.13c and 4.13d**). As we pointed out in SEM images in **Figure 4.1e**, the thickness of the MXene layer was much thinner than the LCE layer, assuring the flexibility of the MXene layer during bending and twisting tests. The bending stiffness (*EI*) is proportional to the third power of the thickness of the strip, indicating that the thin MXene layer shows high flexibility (Equation (4.2)).[48]

$$EI \sim Eh^3 \quad (4.2)$$

Where *EI* is the bending stiffness of the film, *E* is the elastic modulus, and *h* is the thickness of the film. Hence, the pristine MXene film demonstrated reversible bending when the mechanical stress was applied and removed (**Figure 4.18a and 4.18b**). However, the pristine MXene film is fragile due to its intrinsically low elongation at a break of 1 ~ 3 %.[49,50] Since LCE is one of the excellent damping materials with high flexibility as previously reported, the fragile MXene layer is supported by the LCE layer and can stably retain its original shape.[51,52] Hence, the LCE can effectively release the tensile stress transferred through the tightly attached interfaces between the LCE layer and the MXene layer. As shown in **Figures 4.18c and 4.18d**, the MXene layer in MLB stably retains its shape without damage even when the MLB is artificially rolled up on a bar with a curvature radius of 1R. As a result, the bilayer structure and electrical performance of the MLB were durable and stable under various mechanical stimuli.

4.2.3 Multi-Stimuli-Responsive Shape-Reconfigurations

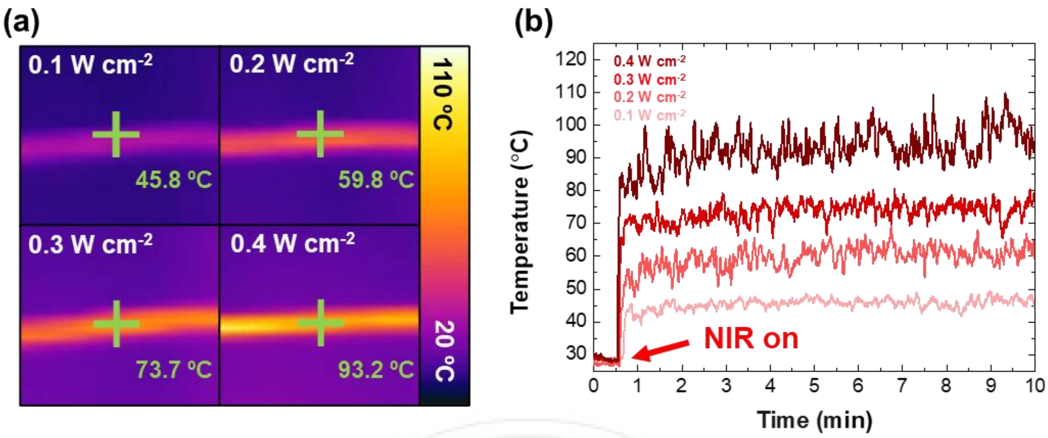

Figure 4.19 (a) Thermal images of MLB under NIR light with different intensities. (b) Temperature variation of MLB with 10 mg mL^{-1} MXene solution under irradiation with NIR light of different intensities.

Programming the molecular alignment of the LC enabled the MLB to perform on-demand controlled actuation under various external stimuli, including NIR light and direct current (DC) power, which can increase the body temperature of MLB. First, when exposed to NIR light, the MLB underwent a rapid temperature increase owing to the presence of the MXene layer, which induces a photo-thermal effect with high efficiency.[31] The temperature of the MLB was dependent on the intensity of the incident NIR light. When the MLB was irradiated with NIR of various intensities, the temperature of MLB was saturated to 45.6±1.4 °C at 0.1 W cm^{-2} and 93.1±5.1 °C at 0.4 W cm^{-2} (**Figure 4.19a and 4.19b**).

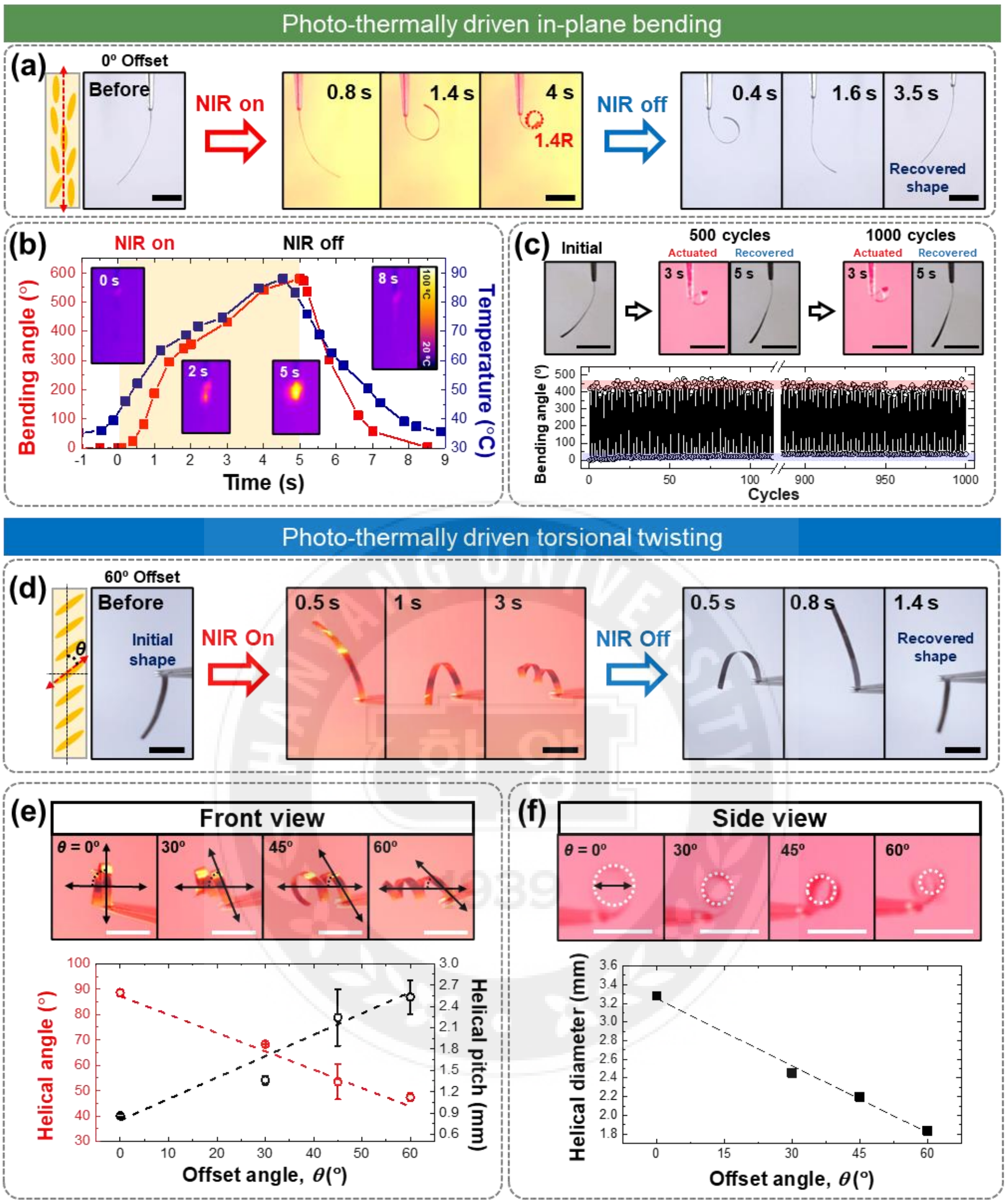

Figure 4.20 Electro-thermally driven actuation of the MLBs. (a) Bending actuation of the MLB with different magnitudes of applied voltage. Scale bars: 1 cm. (b) Variation in temperature and bending angle of the MLBs when different voltages were applied. Inset figures indicate thermal images of the MLBs with different magnitudes of applied voltage. (c) Front-view digital images of the MLB before and after electrothermal actuation at different offset angles of ±60º and 0º. Scale bars: 1 cm.

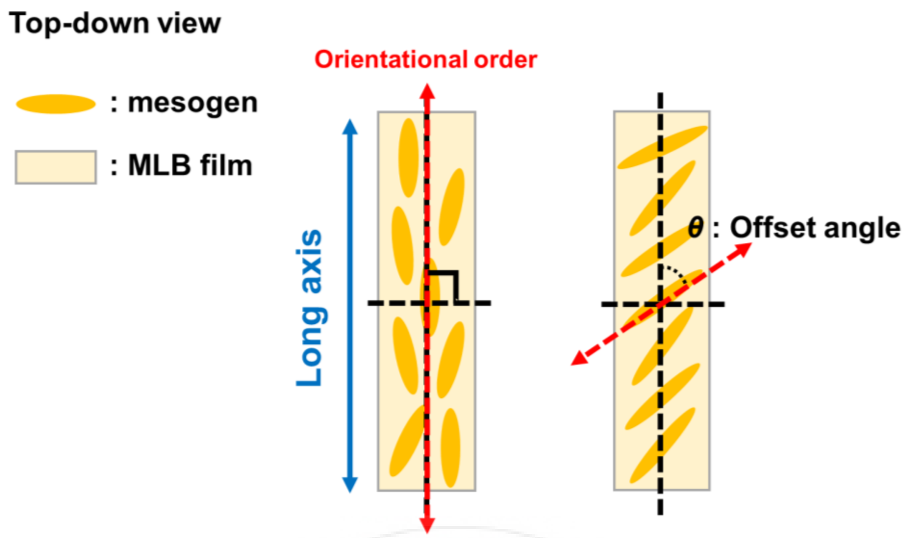

Figure 4.21 Schematic of a top-down view of MLB film. The offset angle is determined as the angle between the long-axis direction of the MLB and the programmed orientational order of the mesogens.

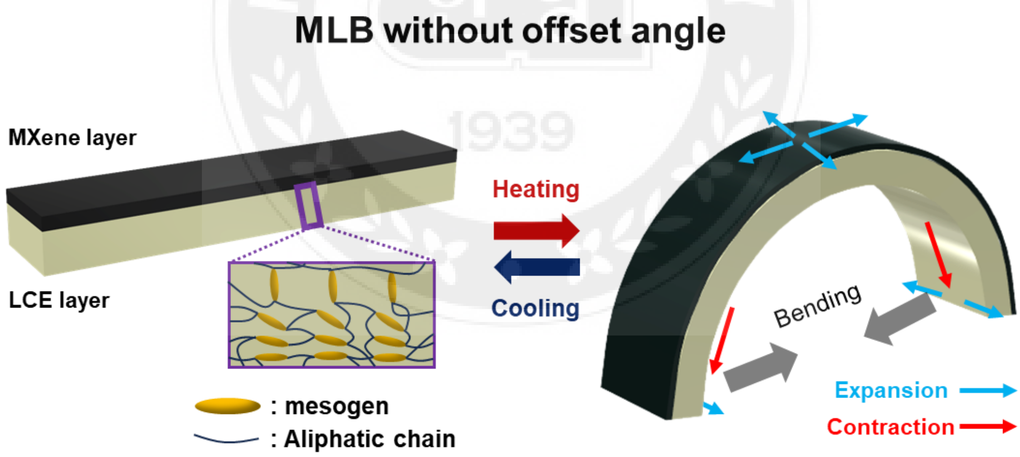

Figure 4.22 3D illustrations of the actuation mechanism of the MLB film with hybrid alignment.

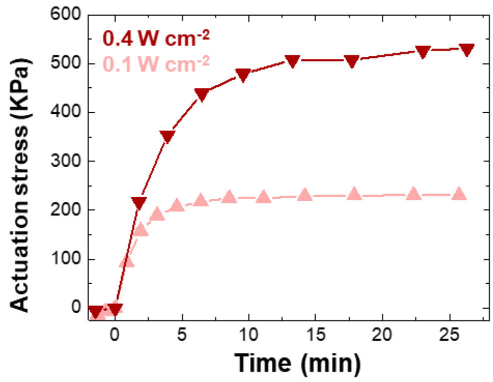

Figure 4.23 Photo-induced actuation stress of MLB under irradiation with NIR light of different intensities.

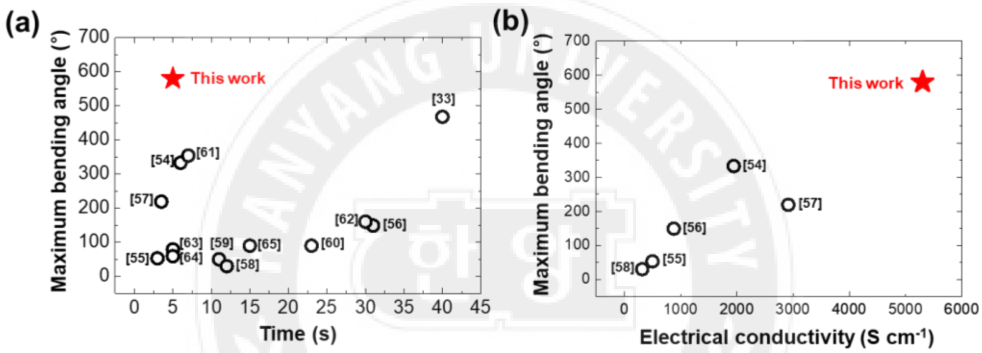

Figure 4.24 Maximum bending angles of previously reported NIR light-driven MXene-based soft actuators were demonstrated as a function of **(a)** actuation time[33, 54-65] and **(b)** electrical conductivity of the actuators.[54-58] Detailed parameters are described in Table 4.2.

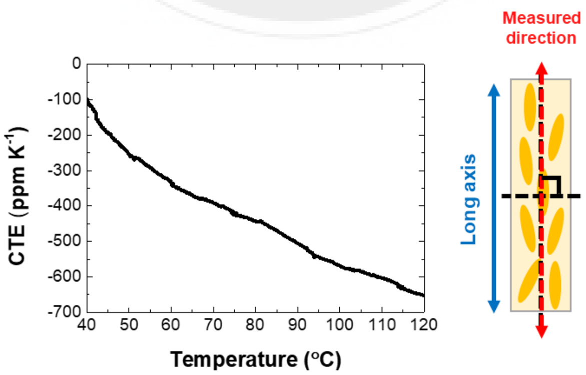

Figure 4.25 Temperature-dependent coefficient of thermal expansion (CTE) of neat LCE film with 0° offset angle, parallelly measured along the alignment direction. The schematic on the right side indicates the measured direction and LCE with a 0° offset angle.

Thickness of soft body, LCE (L_s, m)	50×10^{-6}
Thickness of rigid body, MXene (L_r, m)	0.37×10^{-6}
Thickness ratio (η)	**0.0074**
Elastic modulus of MXene, Pa	15.5×10^{9}
Elastic modulus of LCE, Pa	56.5×10^{6}
Contrast in stiffness (ϵ)	**274.34**
Temperature change (ΔT, 35 to 88 °C)	**53**
Thermal expansion coefficient of MXene (K^{-1})	-5×10^{-6}
MXene strain	$\mathbf{-2.7 \times 10^{-4}}$
Thermal expansion coefficient of LCE (K^{-1})	-504×10^{-6}
LCE strain	$\mathbf{-268 \times 10^{-4}}$
Strain mismatch (ε_m)	$\mathbf{265 \times 10^{-4}}$

Table 4.3 Parameters for calculating the curvature of LCE/MXene bilayer using the Timoshenko and Stoney equation.

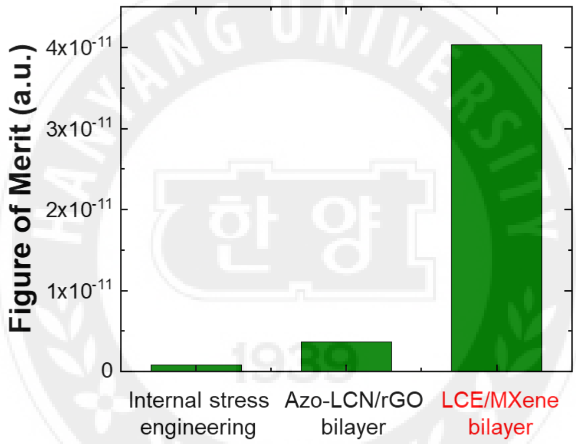

Figure 4.26 Figure of Merit in shape-reconfiguration of systems including internal stress engineering (Chapter 2), azo-LCN/rGO bilayer (Chapter 3), and LCE/MXene bilayer.

	1. Internal stress engineering	2. Azo-LCN/rGO bilayer	3. LCE/MXene bilayer
Light intensity (W cm^{-2})	0.012	0.4	0.4
Light irradiating time (s)	1	0.6	1
Area (cm^2)	0.2	0.2	0.2
Input energy (J)	0.0025	0.048	0.08
Thickness (m)	591×10^{-6}	25×10^{-6}	52×10^{-6}
width (m)	1×10^{-3}	1×10^{-3}	1×10^{-3}
weight (kg)	7.28×10^{-6}	0.83×10^{-6}	1.9×10^{-6}
Moment of inertia (kg m^2)	8.18×10^{-13}	6.88×10^{-14}	1.59×10^{-13}
Max actuation speed(rad/s)	0.036	1.925	6.213
Rotational kinetic energy (J)	5.23×10^{-16}	1.27×10^{-13}	3.23×10^{-12}
Figure of merit	4.03×10^{-5}	1.20×10^{-4}	4.87×10^{-4}

Table 4.4 Parameters for calculating figure of merit in shape-reconfiguration

#	Maximum bending angle (°)	Time (s)	Maximum temperature (°C)	Light Intensity (W cm^{-2})	Electrical conductivity (S cm^{-1})	Materials
1 [54]	333	6	50.0	0.08	1,945	Graphene oxide (GO)/ MXene bilayer
2 [55]	53	3	72.0	0.50	500	Biaxially oriented polypropylene (BOPP)/MXene/ Bacterial cellulose (BC) trilayer
3 [56]	149	31	42.9	n.a	880	MXene/Silver nanoparticles (AgNPs) (MS) encapsulated Ecoflex
4 [57]	219	3.5	102.0	0.50	2,915	MXene-CNT composites/ Polydimethylsiloxane (PDMS) bilayer
5 [58]	30	12	79.2	0.50	318	MXene/Polyethylene terephthalate (PET) bilayer
6 [59]	50	11	40.3	0.90	0.12	γ-Methacryloxypropyl trimethoxy silane surface-functionalized MXene /PEDOT:PSS composites
7 [60]	90	23	120.0	1.00	n.a.	4-(6-acryloxy-hex-1-yloxy) benzoic acid (6OBA) functionalized MXene/LCE copolymer
8 [33]	468	40	56.8	0.30	n.a.	MXene/PNIPAM composites
9 [61]	354	7	n.a.	0.08	n.a.	MXene/low-density polyethylene (LDPE) bilayer
10 [62]	160	30	47.0	0.12	n.a.	MXene embedded delignified wood (TDW) composite/LDPE bilayer
11 [63]	79	5	47.0	0.35	n.a.	MXene-coated paper (MCP)/ polyethylene (PE) film bilayer
12 [64]	59	5	59.0	0.60	n.a.	MXene/BOPP bilayer
13 [65]	90	15	40.6	0.10	n.a.	MXene coated Polyamide (PA) filament/BOPP bilayer
This work	**580**	**5**	**88.1**	**0.40**	**5,300**	**Collectively assembled MXene/LCE bilayers (Co-MLBs)**

Table 4.5 Summary of NIR light-driven bending performance of MXene-based soft actuators.

For high-performance actuation, we demonstrated photo-thermally driven reversible in-plane bending of MLB without an offset angle by irradiating the NIR light with an intensity of 0.4 W cm^{-2} (**Figure 4.20a**). The offset angle is determined as the angle between the nematic director of the mesogens and the long axis of the MLB film (**Figure 4.21**). Under NIR irradiation, the MLB with 0° offset angle (0° MLB) was heated to 88 °C in 5 s and was bent towards the LCE layer direction due to the hybrid alignment as confirmed by the SEM images, with a maximum bending angle of 580° and curvature of 714.9 m^{-1} (**Figure 4.20b and 4.22**).[6] Furthermore, the MLB showed actuation stress of 531.1 KPa under NIR light with a power of 0.4 W cm^{-2}, thereby inducing rapid photo-thermally driven actuation (**Figure 4.23**). Under repetitive NIR irradiation, the MLB exhibited stable in-plane bending without deterioration of the actuation performance over 1,000 cycles (**Figure 4.20c**) and demonstrated a negligible change in the electrical resistance. The electrical resistance of MLB was measured as 32.5 Ω before the cycle test and 34.2 Ω after 1,000 cycles of bending. Compared to other previously reported photo-thermally driven MXene-based actuators, the MLB showed a significantly higher maximum bending angle and a faster response time (**Figure 4.24a**).[33, 54-65] This distinctive bending performance of MLB originates from the synergistic effect between the anisotropic contraction of the LCE layer and the CTE mismatch between the LCE layer and the MXene layer. Both effects occur simultaneously during stimuli-responsive actuation. As shown in **Figure 4.25**, the neat LCE showed a higher CTE value of −504 ppm K^{-1} at 90 °C while the MXene layer had a negligible thermal expansion of −5 ppm K^{-1}.[53] The bending performance of the LCE/MXene bilayer was calculated using the measured parameters provided in **Table 4.3** and the Timoshenko and Stoney equation, which describe the relationship between bending curvature and CTE mismatch between LCE and MXene layers. A detailed explanation of the Timoshenko and Stoney equation was provided in **Equation (3.4) and (3.5)** in Chapter 3. Consequently, the calculated bending curvature was 708.3 m^{-1} with a remarkably low error of 0.93%, as the bending actuation of the LCE/MXene bilayer was based on the photothermal effect.

Furthermore, figure or merit in shape-reconfiguration was calculated and compared of the systems, including internal stress engineering via frontal photopolymerization (FPP), azo-LCN/rGO bilayer, and LCE/MXene bilayer, dealt with in Chapters 2, 3, and 4, respectively. The figure of merit (FOM) in shape-reconfiguration was defined as a ratio between input energy and generated maximum kinetic energy via bending actuation (**Equation (4.3)**).

$$Figure\ of\ Merit\ (FOM) = \frac{Kinetic\ energy\ (T)}{Input\ energy} \quad (4.3)$$

$$Kinetic\ energy = \frac{1}{2} I_{total}\omega^2 \quad (4.4)$$

$$Input\ energy = Pt \quad (4.5)$$

The kinetic energy and input energy of each system were calculated by **Equation (4.4) and (4.5)**. Where I_{total} is the total mass moment of inertia, ω is the maximum angular velocity, P is light power, and t is irradiation time. Consequently, the LCE/MXene demonstrated the highest figure of merit in shape-reconfiguration compared to other systems (**Figure 4.26 and Table 4.4**). This means that using LC polymers provides significant advantages in shape-reconfigurability compared to using isotropic polymers due to their anisotropic thermal expansion and contraction-based large deformation. Furthermore, the new material combination for the LCE/MXene bilayer provides a synergistic effect in shape-reconfiguration as MXene was applied with a significantly thin thickness of 370 nm, effectively reducing bending stiffness. In particular, the MLB showed the highest electrical conductivity among other MXene-based actuators (**Figure 4.24b**). The detailed parameters of previously reported MXene-based actuators are described in **Table 4.5.**

3D torsional shape-reconfiguration was also performed by adjusting the offset angle in the MLB. The MLB with an offset angle of 60° (named 60° MLB) generated a coil shape under NIR irradiation at an intensity of 0.4 W cm^{-2} (**Figure 4.20d**). During torsional actuation, we can control the dimensional parameters of the coil shape including the helical angle, pitch, and diameter by adjusting the offset angle in the MLB (**Figure 4.20e and 4.20f**). As the offset angle increased, the helical pitch of the MLB coil increased from 1.36 mm at 30° offset to 2.53 mm at 60° offset, and the helical angle and helical diameter were correspondingly reduced from 68.2° and 2.46 mm at 30° offset to 47.4° and 1.84 mm at 60° offset, respectively.

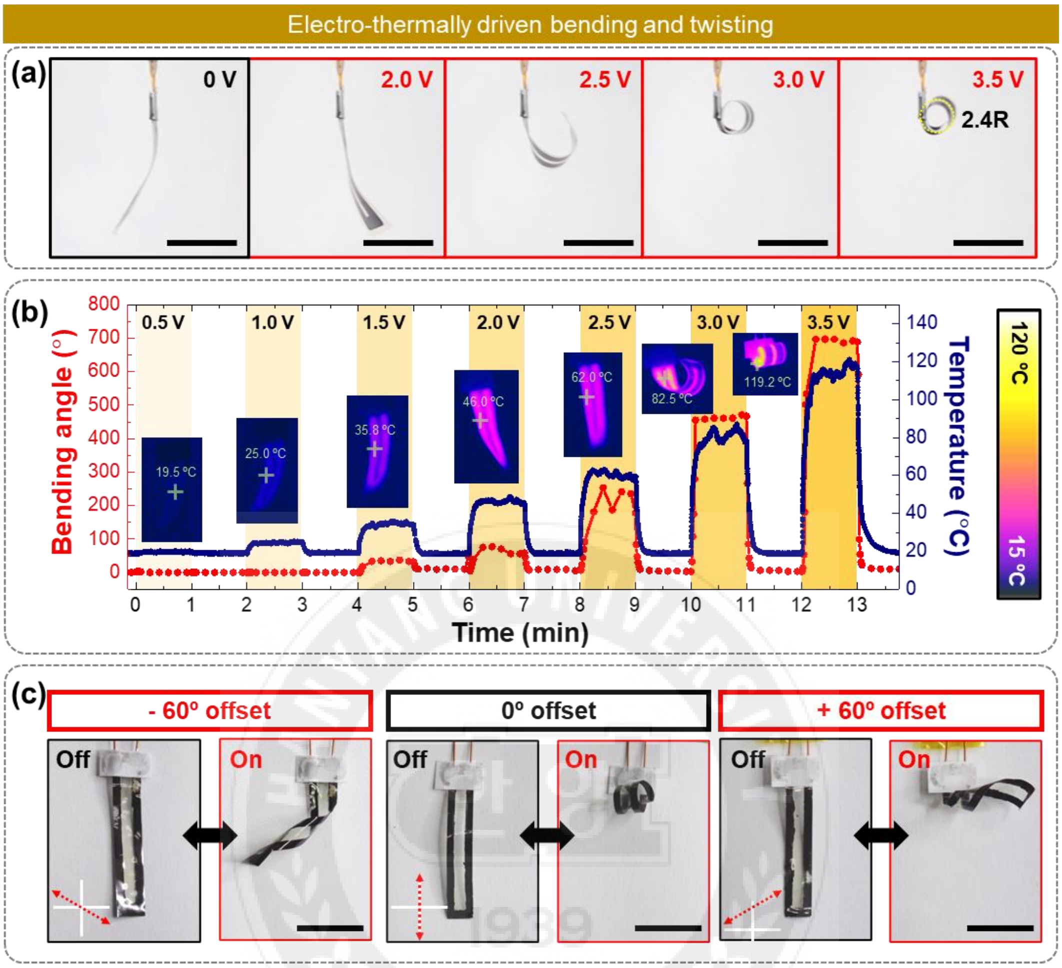

Figure 4.27 Electro-thermally driven actuation of the MLBs. (a) Bending actuation of the MLB with different magnitudes of applied voltage. Scale bars: 1 cm. (b) Variation in temperature and bending angle of the MLBs when different voltages were applied. Inset figures indicate thermal images of the MLBs with different magnitudes of applied voltage. (c) Front-view digital images of the MLB before and after electrothermal actuation at different offset angles of ±60° and 0°. Scale bars: 1 cm.

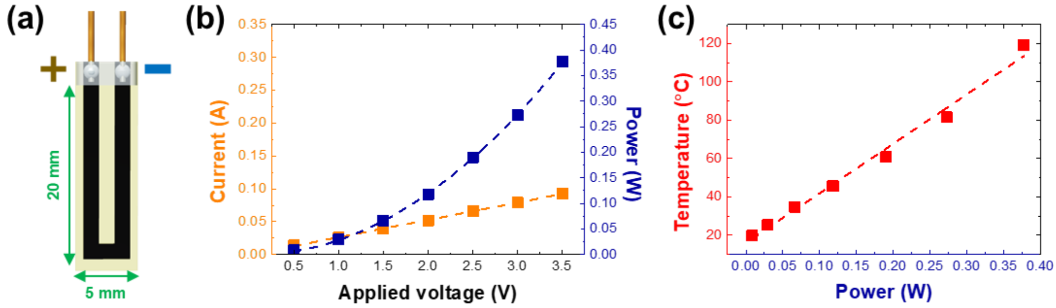

Figure 4.28 (a) Schematic illustration and corresponding dimensions of MLB utilized in electro-thermally driven bending actuation. Each end of the MXene pattern was connected to an electric wire attached by silver paste. The coated silver paste was encapsulated by using epoxy glue to prevent any undesirable detachment. (b) Current and power were demonstrated as a function of the applied voltage. (c) The temperature of MLB was demonstrated as a function of power as plotted in (b).

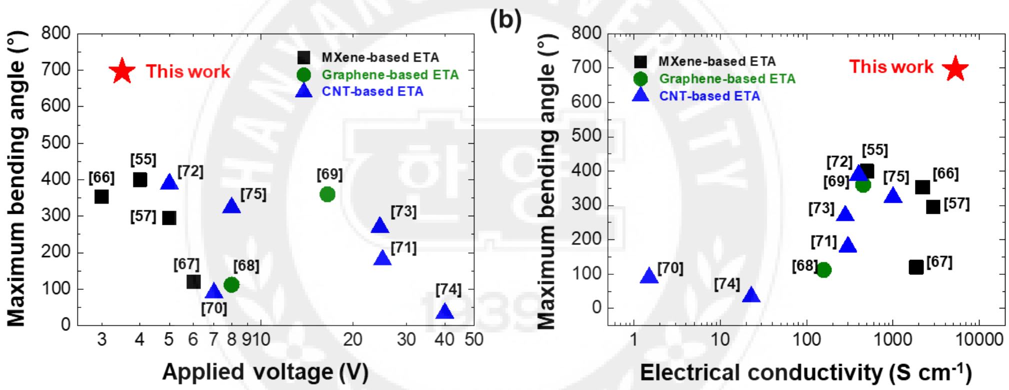

Figure 4.29 Maximum bending angles of previously reported electro-thermally driven actuators (ETAs) based on MXene, graphene, and carbon nanotubes (CNTs) were demonstrated as a function of (a) applied voltage and (b) electrical conductivity.[55, 57, 66-75] The detailed parameters are described in Table 4.3.

Ref	Maximum bending angle (°)	Time (s)	Maximum temperature (°C)	Applied voltage (V)	Electrical conductivity (S cm^{-1})	Materials
1 [57]	295	15	180	5.0	2,915	MXene-CNT composites/ PDMS bilayer
2 [66]	353	20	140	3.0	2,200	NdFeB@PDMS composite film /MXene/PTFE trilayer
3 [67]	120	40	110	6.0	1,860	Polytetrafluoroethylene (PTFE)/MXene/PI trilayer
4 [55]	400	n.a.	84	4.0	500	MXene/Bacterial cellulose (BC) bilayer
5 [68]	112	10.5	220	8.0	157	Graphene/Carbon fiber/PI trilayer
6 [69]	360	5	n.a.	16.5	451	Reduced GO (RGO) /PDMS bilayer
7 [70]	90	48.6	160	7.0	1.5	MWCNTs/Polyurethane (PU) composite and MWCNTs/silicone rubber composite bilayer
8 [71]	180	12	150	25.0	300	CNT/PDMS bilayer
9 [72]	389	10	48	5.0	400	Super aligned CNT (SACNT)/BOPP bilayer
10 [73]	270	1.2	170	24.5	280	SACNT/PDMS bilayer
11 [74]	34	5	98	40.0	23	CNT/PDMS composites
12 [75]	324	12	351	8.0	1,000	CNT film /PDMS bilayer
This work	**697**	**14**	**118**	**3.5**	**5,300**	**Collectively assembled MXene/LCE bilayers (Co-MLBs)**

Table 4.6 Summary of the bending performance of electro-thermally driven actuators (ETAs) based on MXene, graphene, and carbon nanotubes (CNT).

Owing to the high electrical conductivity of MXene, joule heating was induced by applying a single-digit level DC voltage, thereby activating electro-thermally driven actuation. The 0° MLB with 43.6 Ω was bent when a DC voltage was applied through a connected DC power supply (**Figure 4.27a**). **Figure 4.28a** provides the dimensions of electro-thermally driven bending. As the applied voltage was increased, the measured current passing through the MXene layer linearly increased according to Ohm's law ($V = IR$, V: Voltage, I: Current, and R: Resistance) and the power ($P = I^2R$) also increased by following the second order polynomial function (**Figure 4.28b**). The generated heat during voltage application has a proportional relationship with the power according to Joule's law (Equation (4.6)).

$$Q = I^2Rt = Pt \quad (4.6)$$

Where Q is the amount of heat, R is the resistance, t is the time that current passes through the MXene, and P is the power generated during voltage application. The temperature of MLB during voltage application showed a linear relationship with the generated power (**Figure 4.28c**). Hence, the bending angle of the MLB increased according to the enhanced electro-thermal effect when the applied voltage increased (**Figure 4.27b**). The radius of curvature, maximum bending angle, and temperature of the MLB were 2.4 R, 697°, and 118 °C, respectively, when a voltage of 3.5 V was applied for 14 s. Electro-thermally driven directional twisting was also demonstrated by using the ±60° MLB (**Figure 4.27c**). In particular, the MLB also demonstrated a high electro-thermally driven actuation performance owing to its higher electrical conductivity than other graphene, carbon nanotubes, and MXene-based electro-thermally driven actuators (ETA), as shown in **Figure 4.29a and 4.29b**.[55, 57, 66-75] The detailed parameters of previously reported ETAs are provided in **Table 4.6**.

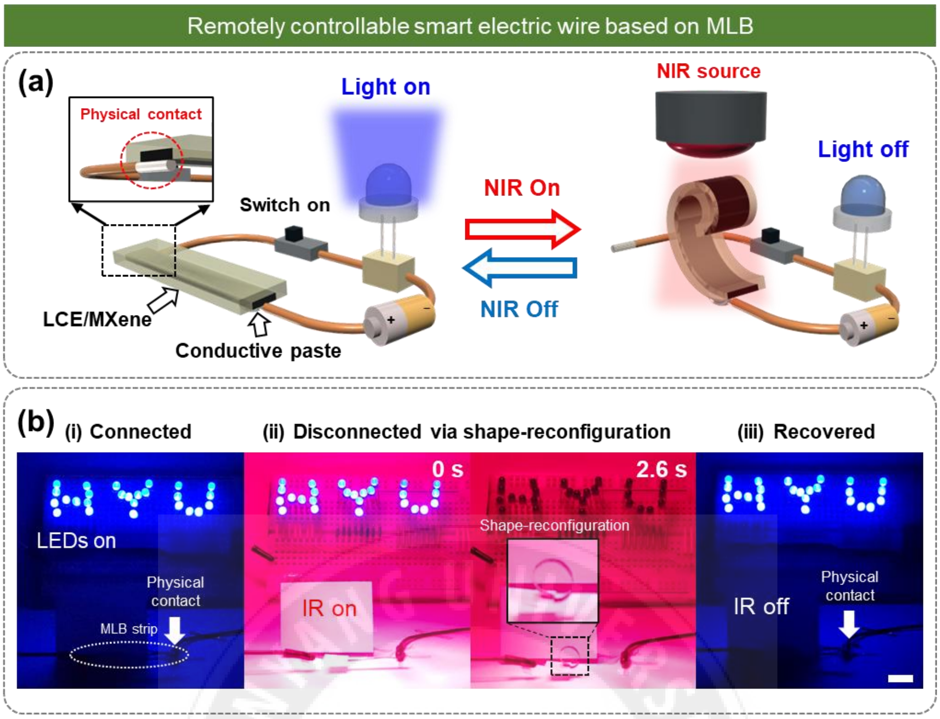

Figure 4.30 Untethered switching of the electrical circuit. (a) Schematic illustrations and (b) digital images of remotely turning on and off the LEDs via photo-thermally driven bending of the MLBs connected in the electrical circuit. Scale bar: 1 cm.

The MLB was used in a remotely controllable switch in an electrical circuit by untethered photo-thermally driven actuation. The schematic in **Figure 4.30a** illustrates the operating mechanism of controlling the electrical circuit by irradiating the NIR light to MLB. When one end of the MLB contacted a wire in an electrical circuit with LEDs, the LEDs were stably illuminated because the MLB possessed high conductivity and low contact resistance. The MLB was disconnected from the electrical circuit when the MLB was bent during NIR irradiation, and the LEDs were turned off. Finally, the MLB connected in the circuit demonstrated the reversible and untethered operation of the LEDs via repetitive bending actuation under NIR light irradiation (**Figure 4.30b**). Hence, the MLB has great potential for application as a shape-reconfigurable electronic wire in the automatic operating system of electrical devices.

4.2.4 Collectively Assembled MLBs for Multi-Functional Locomotion

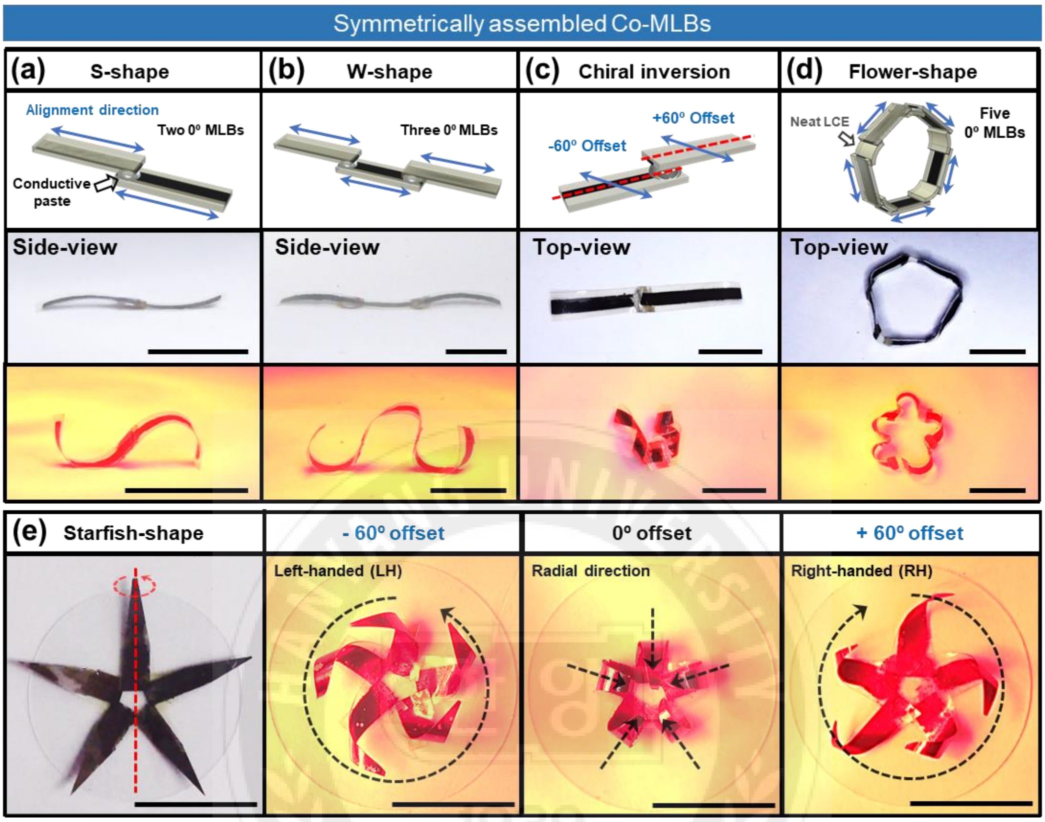

Figure 4.31 Symmetrically assembled MLBs. (a) S-shape-reconfiguration of cross-assembled structure of two 0° MLBs, (b) W-shape-reconfiguration of cross-assembled structure of three 0° MLBs, (c) chiral inversion of cross-assembled structure of +60° MLBs and −60° MLB, (d) Flower-shape-reconfiguration of a circularly assembled structure composed of five 0° MLBs and five neat LCEs. (e) Chiral structures of starfish-shaped Co-MLBs. Scale bars: 1 cm.

Collective assembly is an effective strategy for creating scaled-up 3D structures with enhanced geometrical diversity.[76,77] The collectively assembled MLB can show various shape-reconfigurations by adjusting the assembled direction, assembled angle, and geometrical parameters of unit MLB including length and alignment direction. Herein, several MLB strips served as building blocks and were assembled by using conductive silver paste. We introduced four types of symmetrically assembled structures, as well as the corresponding reversible and untethered shape-reconfigurations under irradiation with 0.4 W cm^{-2} NIR light (**Figure 4.31**). First, the S-shape and W-shape-reconfigurations were demonstrated by connecting the ends of two and three 0° MLBs (**Figure 4.31a and 4.31b**). The assembled −60° MLB and +60° MLB demonstrated a chiral inversion structure containing both right-handed and left-handed coils (**Figure 4.31c**).

To achieve the multi-curvilinear structure in the radial direction, five 0° MLBs and neat LCE pieces were assembled alternatively to form a closed-loop shape (**Figure 4.31d**). Under NIR light irradiation, the MLBs in the closed-loop structure were bent towards the LCE layer direction, and the closed-loop Co-MLBs demonstrated a flower-like shape-reconfiguration. Also, inspired by the axisymmetric structure in nature, a starfish-shaped Co-MLBs was assembled, and it demonstrated a radially bent structure and twisted structure showing opposite chirality with left- or right-handedness (**Figure 4.31e**).

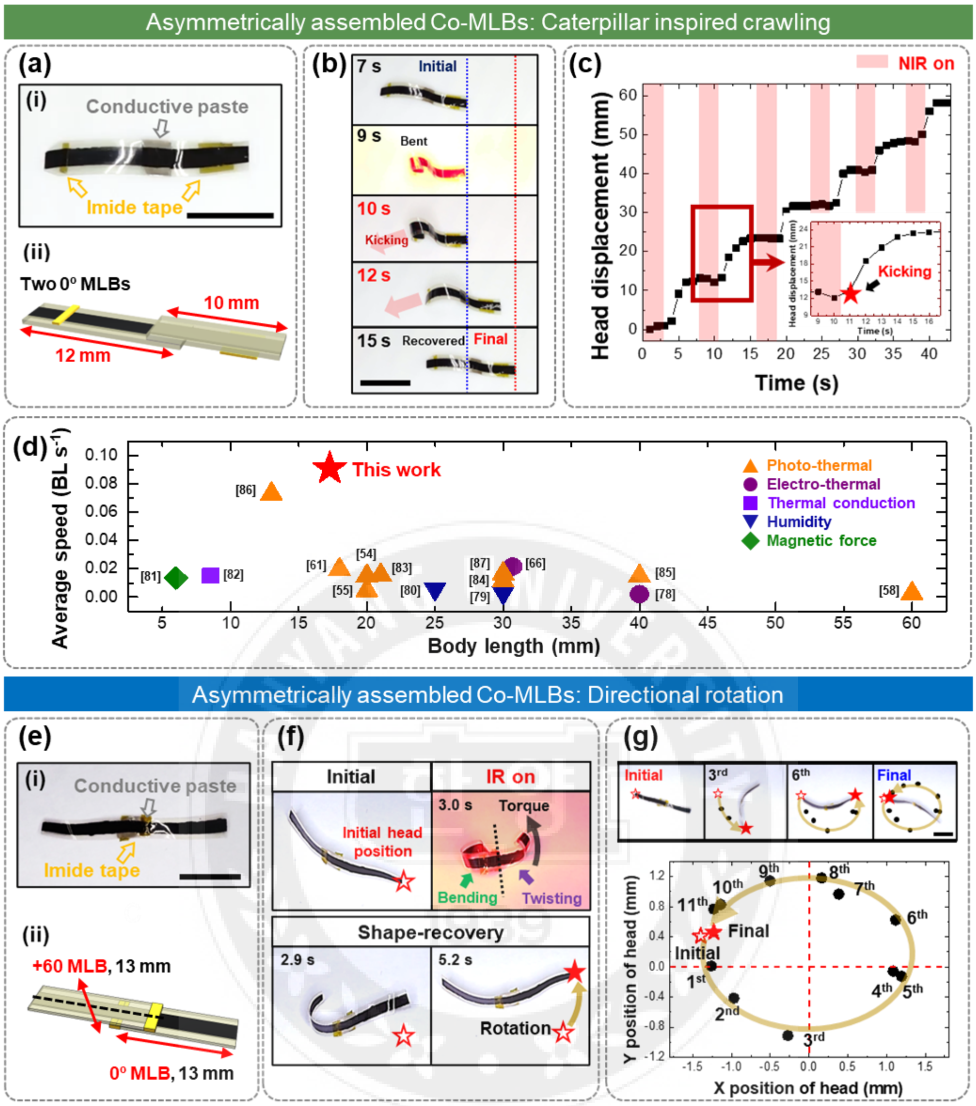

Figure 4.32 Asymmetrically assembled MLBs (Co-MLBs). (a-c) Caterpillar-inspired crawling. (a) (i) Digital image and (ii) schematic of Co-MLBs, cross-assembly of two 0° MLBs with different long-axis lengths of 10 mm and 12 mm, respectively, (b) straight locomotion of Co-MLBs under repeated irradiation with NIR (0.4 W cm^{-2}), c head displacement as a function of time. The red box indicates the onset of NIR irradiation. The inset graph is an enlarged graph of the 2nd cycle of NIR irradiation. (d) Summary of the crawling performance of previously reported MXene-based soft actuators.[54, 55, 58, 61, 66, 78-87] The detailed parameters are described in Table 4.4. (e-g) Directional rotation. (e) (i) Digital image and (ii) schematic of the Co-MLBs, cross-assembly of the 0° MLB and + 60° MLB with the same long-axis length of 12 mm. (f) Rotation mechanism of the Co-MLBs with hetero-offset angle under one NIR on/off cycle. (g) Digital images of Co-MLBs with hetero-offset angles after each NIR on/off cycle. The changes in the head position are marked according to the NIR on/off cycles. XY coordinates with the position change of the head and center of the Co-MLBs with hetero-offset angles. Scale bars: 1 cm.

Ref	Avg velocity (BL s^{-1})	Avg velocity (mm s^{-1})	Body length (BL, mm)	Stimuli	Materials
1 [66]	0.022	0.662	30.7	Electro-thermal	NdFeB@PDMS composites film/ MXene/PTFE trilayer
2 [78]	0.002	0.083	40	Electro-thermal	MXene/Graphene bilayer
3 [79]	0.003	0.098	30	Humidity	MXene/CNF/PDA trilayer
4 [80]	0.006	0.150	25	Humidity	Tannic acid modified MXene and BC composites
5 [81]	0.014	0.082	6	Magnetic force	SWNT-V_2CT_x MXene/ SWNT-$Ti_3C_2CT_x$ MXene and NdFeB Ecoflex composites hybrid
6 [82]	0.015	0.129	8.5	Thermal conduction	MXene/Paraffin wax multilayer
7 [54]	0.015	0.295	20	Photo-thermal	MXene/Graphene oxide (GO) bilayer
8 [83]	0.015	0.324	21.02	Photo-thermal	I-3-isocyanatopropyltriethoxysilane (IPTS) modified MXene/PE bilayer
9 [61]	0.019	0.349	18	Photo-thermal	MXene/LDPE bilayer
10 [84]	0.011	0.339	30	Photo-thermal	MXene-doped polydiallyldimethylammonium chloride (pDADMAC)
11 [55]	0.005	0.092	20	Photo-thermal	MXene/Bacterial cellulose (BC) bilayer
12 [85]	0.015	0.588	40	Photo-thermal	Cellulose and MXene composites/Polycarbonate (PC) bilayer
13 [86]	0.073	0.946	13	Photo-thermal	Silver nanowire and MXene composites /LLDPE bilayer
14 [87]	0.016	0.491	30	Photo-thermal	MXene/LDPE bilayer
15 [58]	0.003	0.150	60	Photo-thermal	MXene/PET bilayer
This work	**0.091**	**1.570**	**17.27**	**Photo-thermal**	**Collectively assembled MXene/ LCE bilayers (Co-MLBs)**

Table 4.7 Summary of the crawling performance of previously reported MXene-based locomotors under various stimuli.

During mass transportation of small creatures, directional actuation can be achieved by asymmetric geometries, which break the symmetry of actuation magnitude. To achieve a caterpillar-like crawling motion, two 0° MLBs of different lengths (12 mm and 10 mm) were used to break the symmetry of the mass center (**Figure 4.32a(i) and 4.32a(ii)**). Upon irradiation with NIR light, the longer MLB was bent to a greater extent than the MLB with a shorter length. The longer MLB, therefore, plays the role of a leg that generates the kicking action.

Imide tape was also used to reduce friction with the ground and to enhance the breaking symmetry. The assembled MLB demonstrated directional crawling via a programmed disparity in the degree of bending of each MLB component under repetitive NIR irradiation (**Figure 4.32b**). The head of the assembled MLB moved forward by 58.1 mm for 40 s over 10 cycles of NIR irradiation on (3 s) and off (5 s) (**Figure 4.32c**). The maximum locomotion velocity was measured as 5.2 mm s^{-1}, and the average locomotion velocity corresponded to 0.091 BL (Body Length) s^{-1}. During the crawling process, reversible bending of the shorter MLB also plays an important role in reducing friction with the ground, which improves the moving efficiency. As shown in **Figure 4.32d**, the co-MLBs showed a notably greater average locomotion velocity than other MXene-based soft locomotors. **Table 4.7** summarizes the detailed parameters of previously reported MXene-based locomotors. In addition to the linear locomotion, a self-rotational movement was achieved by assembling two MLBs with hetero-offset angles. A 0° MLB and a −60° MLB were assembled to achieve the rotational motion (**Figure 4.32e(i) and 4.32e(ii)**). Under NIR light, the −60° MLB induces a directional torque via a twisting motion, whereas the 0° MLB positioned on top of it bends, dragging the entire body (**Figure 4.32f**). When the NIR irradiation is halted, the twisted −60° MLB and bent 0° MLB both recover their original flat shapes. Here, the head position of the assembled MLB moves through an arc by the rotational movement. The combination of torsional twisting and dragging triggers the unidirectional rotational movement of the assembled MLB. Under twelve NIR on (3 s)/off (5 s) cycles, the head position was rotated by moving through a circular orbit, after which the head position returned to its initial position (**Figure 4.32g**).

4.2.5 Snap-Through-Inspired Actuation of the Collectively Assembled MLB

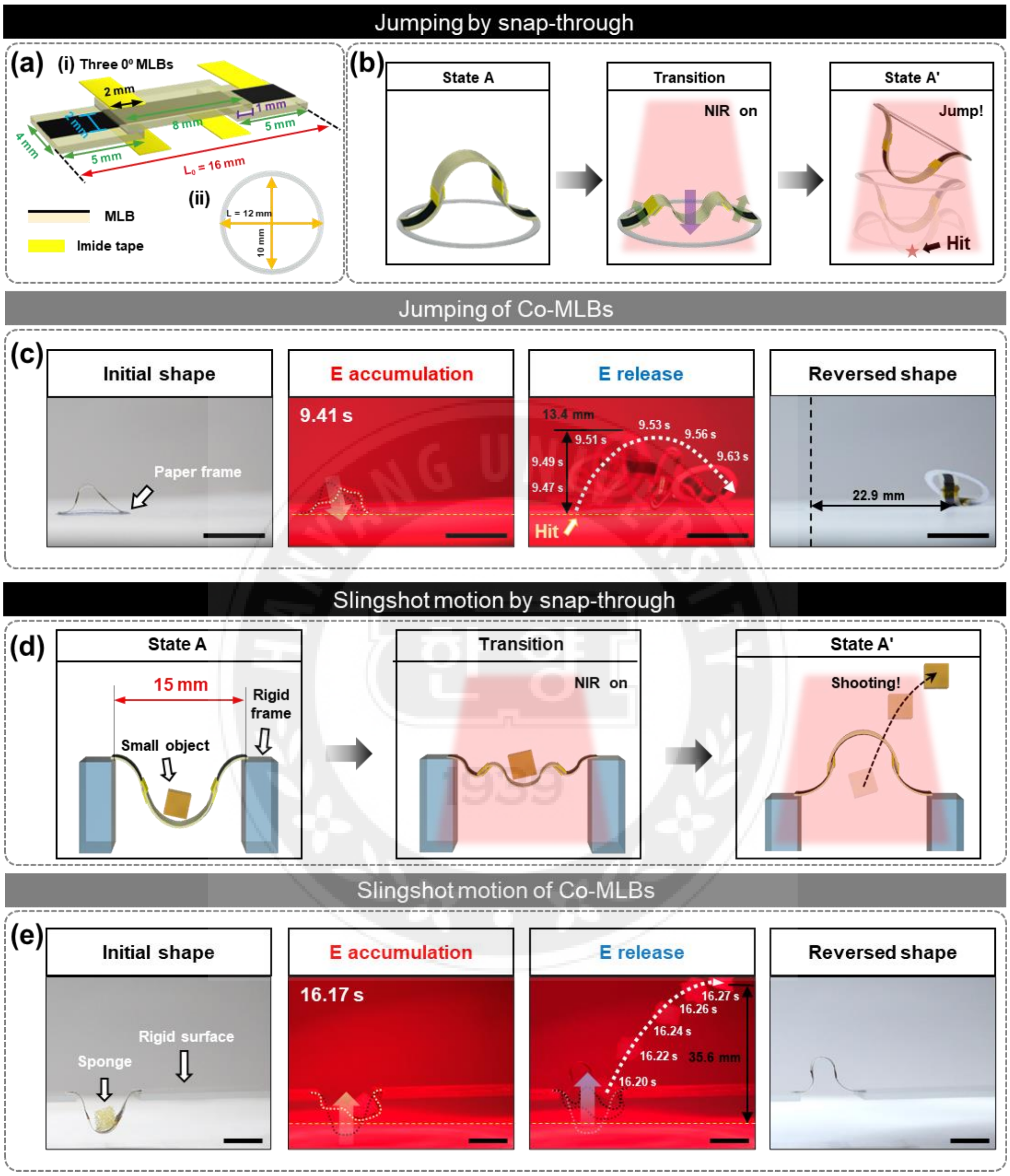

Figure 4.33 Snap-through inspired actions. (a) Dimensions of Co-MLBs composed of three 0° MLBs and a paper frame for jumping motion. (b) 3D schematic indicated the jumping mechanism of Co-MLBs attached to a paper frame. (c) Jumping motion of Co-MLBs connected to the paper frame with a circumcircle diameter of 12 mm and inscribed circle diameter of 10 mm. (d) 3D schematic indicated the mechanism of slingshot action. (e) Slingshot-inspired actuation of Co-MLBs. The Co-MLBs launch a piece of the sponge by rapidly releasing accumulated elastic energy as kinetic energy. Scale bars: 1 cm.

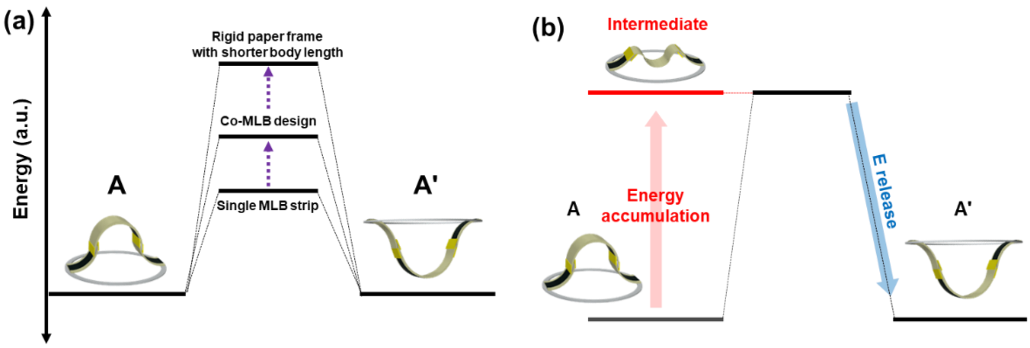

Figure 4.34 Mechanism of snap-through behavior. (a) Potential energy diagram of two bistable states A and A′. (b) Energy transition during energy accumulation and release process.

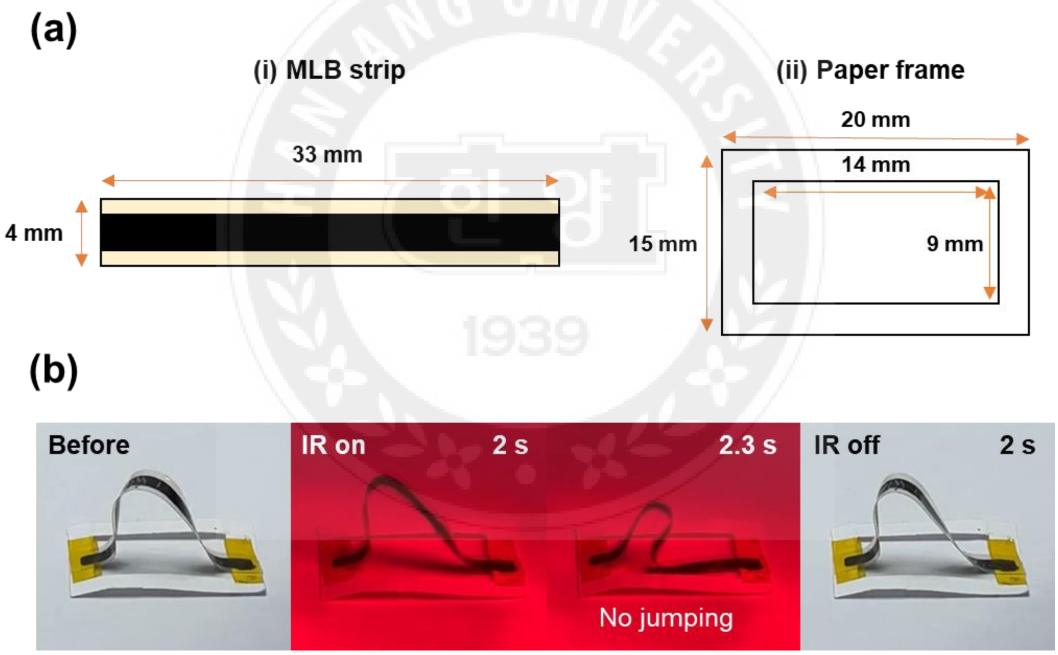

Figure 4.35 Jumping failure of a single MLB strip connected to a paper frame. **(a)** Dimensions of an applied MLB strip and a paper frame. **(b)** Shape change and recovery before and after infrared (IR) light irradiation.

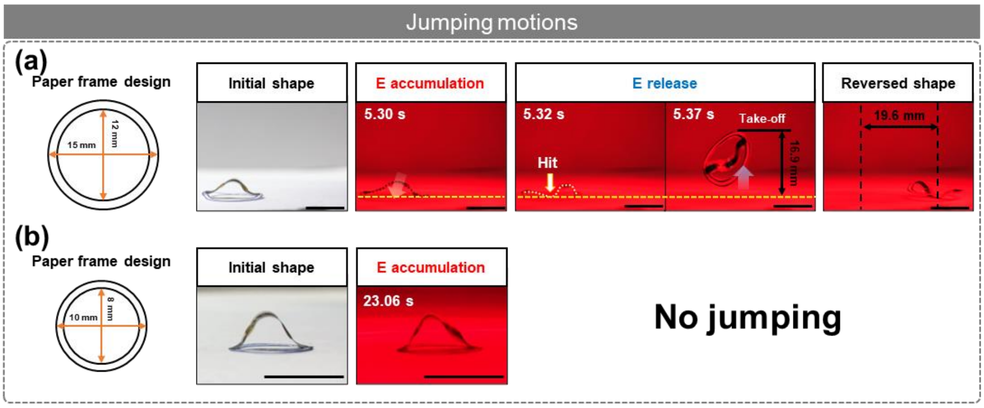

Figure 4.36 (a-b) Jumping results of Co-MLBs connected to the paper frame with (a) a circumcircle diameter of 15 mm and inscribed circle diameter of 12 mm, and (b) with circumcircle diameter of 10 mm and inscribed circle diameter of 8 mm. Scale bars: 1 cm.

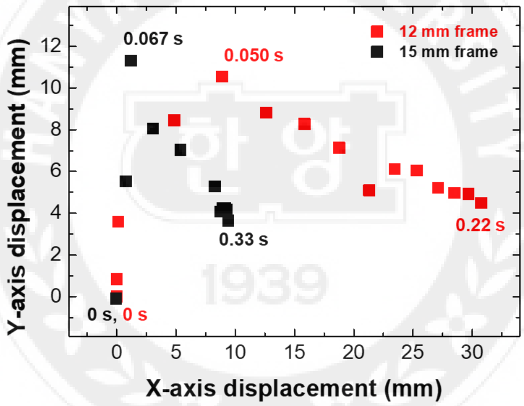

Figure 4.37 Positional change of the center position of the paper frame with Co-MLB attached during jump according to the diameter of the circumscribed circle of the paper frame.

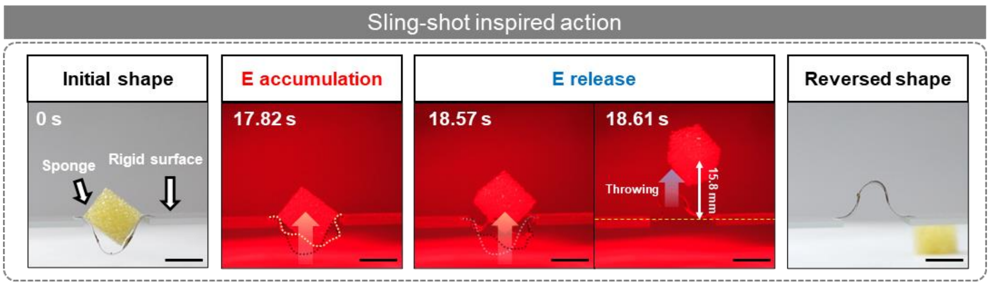

Figure 4.38 Slingshot-inspired actuation of Co-MLBs. The Co-MLBs launch a piece of the sponge with a weight of 11.5 mg heavier than the Co-MLB (9.4 mg) by rapidly releasing accumulated elastic energy as kinetic energy. Scale bars: 1 cm.

In small-scale soft robotic systems, the nature-inspired snap-through behavior is an effective strategy for achieving explosive and rapid actuation through the accumulation and instantaneous release of energy.[11,88,89] To achieve remotely controlled jumping, we introduced Co-MLBs and rigid paper frames. As flexible and soft materials easily release the stored external stress by thermal motion, designing a structure that can restrict deformation is an effective strategy for storing elastic energy.[90, 91] The Co-MLBs for jumping therefore adopt a structure that consists of two short MLB strips alternatively connected to both ends of a long MLB strip (**Figure 4.33a**).

For snap-through, both ends of the Co-MLBs were attached to a rigid paper frame, which has a shorter circumcircle diameter than the long axis length of the Co-MLBs. The Co-MLBs attached to the paper frame initially showed a stable curvilinear structure, named A state, which is one of the stable isometries (**Figure 4.33b**). When the NIR was irradiated, the Co-MLBs effectively stored elastic energy and deformed to the metastable intermediate state. The two short MLBs attached to both ends of a long MLB were bent in opposite directions relative to the long MLB. This restriction of shape-reconfiguration allows for an effective storage of the elastic energy (**Figure 4.34a**). Furthermore, the paper frame also hindered the shape-reconfiguration of Co-MLBs, increasing its capability to store elastic energy. When the accumulated elastic energy overcame the energy barrier, the Co-MLBs rapidly hit the ground by converting the elastic energy into mechanical energy (**Figure 4.34b**). Finally, the jumping behavior was achieved by the principle of action and reaction, and the structure of Co-MLBs transferred to another stable isometry, named A' state. In the case of a single MLB strip attached to a paper frame, it cannot jump due to insufficiently stored elastic energy (**Figure 4.35**). When we used the paper frame with a circumcircle diameter of 12 mm and width of 2 mm, the Co-MLBs with a body mass of 9.7 mg demonstrated a maximum jumping height of 18.4 mm and a total jumping distance of 35.51 mm (2.96 BLs) with a maximum jumping velocity of 97.2 cm s^{-1} (**Figure 4.33c**). When the ratio between the circumcircle diameter of the paper frame (L) and the long axis length of the Co-MLBs (L_0) gets closer to 1, the released energy by snap-through decreases due to the reduction in the threshold strain.[88] As L increased to 15 mm, the total jumping distance decreased to 2.40 BLs due to the reduction in the released energy (**Figure 4.36a and 4.37**). However, when L was decreased to 10 mm, no jumping was observed because the increased space constraints elevated the threshold strain required for snap-through (**Figure 4.36b**). The same dimension of Co-MLBs was also used to shoot an object into the air by snap-through. Inspired by the slingshot, we fixed the two ends of the Co-MLBs with a rigid substrate. The sponges were placed on the Co-MLBs and the Co-MLBs were irradiated by the NIR light, which caused a gradual energy accumulation (**Figure 4.33d**). Then, the sponges with a weight of 0.7 mg and 7.8 mg were launched in the air with a maximum height of 35.6 mm and 15.8 mm, respectively (**Figure 4.33e and 4.38**).

4.3 Conclusions

In Chapter 4, we successfully demonstrated diverse 3D shape-reconfiguration and locomotion by introducing the MXene patterned LCE bilayer (MLB) with a superior electrical conductivity of ~ 5,300 S cm^{-1} compared to the previously reported MXene-based actuators and electrically conducting LC polymer composites. During repetitive dynamic mechanical deformations, the stable bilayer structure remains secure without delamination due to the strong interfacial adhesion. Due to both the high photo-thermal conversion efficiency and electrical conductivity of the MXene, the MLB allows for performing photo-/electro-thermally driven direction-controlled actuation according to the morphogenesis of the LCE layer. During electro-thermally driven actuation, the MLB was bent with a maximum bending angle of 697° by applying a single-digit level voltage of 3.5 V. As a proof-of-concept, the MLB strip was connected to an electrical circuit as a reconfigurable wire that can stably turn on and off LEDs by photo-thermally driven actuation. Lastly, to enhance the diversity and functionality of the shape-reconfiguration, we introduced the concept of a collective assembly. The collectively assembled MLBs (Co-MLBs) demonstrated shape-reconfiguration from a flat 2D strip shape to multi-curvilinear (S-, W-, Flower-shape), and chiral structures based on the combination of the assembly structure and alignment offset in the MLBs. Beyond the static shape-reconfiguration, dynamic body mass transportation via crawling and directional rotation was demonstrated with the asymmetric Co-MLBs as one of the soft robotic applications. Furthermore, the introduction of a snap-through system enables the Co-MLBs to achieve rapid and explosive motions, represented by jumping and shooting an object. Synthetically, we demonstrated the rapid and reversible soft-robotic maneuver of the soft-electronics by considering the compatibility of hetero-materials and introducing a bilayer structure and assembled design.

4.4 Experimental Section

Materials: (1,4-Bis-[4-(6-acryloyloxyhexyloxy) benzoyloxy]-2-methylbenzene) (RM82; Synthon Chemicals GmbH & Co. KG), (1,4-bis-[4-(3-acryloyloxypropyloxy) benzoyloxy]-2-methylbenzene) (RM257; Synthon Chemicals GmbH & Co. KG), 1,4-benzenedimethanethiol (BDMT; Merck), butylated hydroxytoluene (BHT; Merck), Irgacure 369 (I-369; Ciba), the layered ternary carbide MAX-phase powder (Ti_3AlC_2; Carbon-Ukraine Ltd.), hydrochloric acid (HCl; Merck), and lithium fluoride (LiF; Merck) were used as received without further purification.

Preparation of $Ti_3C_2T_x$ MXene solution: Single few-layer $Ti_3C_2T_x$ (MXene) was obtained by chemically etching the Al layers (A of MAX) from the $Ti_3C_2T_x$ MAX phase by following the previously reported methods.[36,92] LiF (2 g) and 9 M HCl (40 mL) were stirred together for 30 min at 35 °C in a flask. Thereafter, $Ti_3C_2T_x$ (MAX phase, 2 g) was slowly added to the LiF/HCl solution until completely dissolved, and the mixture was stirred for 24 h under an argon atmosphere. After the etching reaction, the obtained mixture (40 mL) was diluted with DI water and centrifuged. The supernatant containing HCl and excess ions from the etching process was discarded. The obtained sediments were diluted and centrifuged repeatedly until the pH reached 6. Finally, the delaminated $Ti_3C_2T_x$ dispersion was centrifuged again to separate relatively heavy moieties (i.e. unreacted MAX phase, multi-layer MXenes), and the supernatant was obtained as a concentrated colloidal dispersion.

Preparation of MXene patterned LC glass cell: To prepare the MXene patterned LC cell, two glass slides, each with an MXene pattern and polyamide-based alignment layer, were utilized. First, to prepare the MXene patterned glass slide, a patterned mask was attached to the cleaned glass slide. The masked glass slide was subjected to plasma treatment for 5 s with a radiofrequency (RF) power of 6.8 W, by using a plasma cleaner (Harric Plasma, PDC-32G-2). The MXene solution was sprayed on the masked glass slide and dried by blowing with dry N_2 gas. Secondly, 0.125 wt% of polyamide (Elvamide, Dupont) dissolved in methanol was spin-coated on another glass side to prepare the alignment layer-coated glass slide. Subsequent mechanical rubbing induced a surface relief grating on the glass slide coated with the alignment layer. Finally, the MXene-patterned LC cell was obtained by attaching the MXene-patterned glass slide and the alignment layer-coated glass slide with a spacing of ~50 μm.

Synthesis of the MXene patterned LCE: The LCE mixture comprised 68.0 wt% RM82, 19.5 wt% RM257, 11.0 wt% BDMT, 0.5 wt% BHT, and 1.0 wt% of I-369, according to previously reported methods [5]. The LCE mixture was injected in the gap of the MXene patterned LC cell at 90 °C to generate an isotropic state by capillary action. The MXene patterned LC cell, filled with the molten LC mixture, was cooled to 25 °C, followed by UV-induced photopolymerization by irradiation with 365 nm light at an intensity of 30 mW cm^{-2} for 30 min (CoolLED, pE-4000). The light intensity was measured by an optical power meter (Thorlabs, PM100D).

Characterizations: The zeta potential of the MXene aqueous solution (0.05 mg mL^{-1}) was measured by dynamic light scattering (DLS; Zetasizer Nano ZS, Malvern Instruments); the pH was varied using HCl and NaOH solution. The topography and current image of the MXene nanosheet were obtained in tapping mode by atomic force microscopy (AFM; Park Systems, XE-70) and conductive-AFM (C-AFM) scanning, respectively. The measured data were processed with XEI software (Park Systems). The internal alignment structure of MLB was observed by scanning electron microscopy (SEM; Thermo Fisher Scientific, Verios G4 UC) and polarized optical microscopy (POM; Nikon, LV100NPOL). Elemental analysis was conducted by using energy dispersive X-ray spectroscopy (EDS; EDAX, Elect Plus) and X-ray photoelectron spectroscopy (XPS; Thermo Fisher Scientific, K-Alpha+). The infrared spectrograms were obtained by using a Fourier-transform infrared spectrometer (FT-IR; Thermo Fisher Scientific, Nicolet iS50). For attenuated total reflectance (ATR) mode, the diamond ATR accessory was equipped with the FT-IR. The electrical resistance of the MXene films on the glass slide and the MLB was measured with a digital multimeter (Keithley, DMM7510). The height profile of the MXene films on the glass slide was obtained by using a surface profiler (KLA Tencor, D-300). The average cross-sectional area was determined by integrating the measured heights for 5 different regions. The thermo-mechanical properties and the coefficient of thermal expansion were measured by using a temperature sweep test with conditions of 1 Hz and 5 °C min^{-1} and a controlled force test with conditions of 0.001 N and 3 °C min^{-1}, respectively, by dynamic mechanical analysis (DMA; TA Instruments, Q800).

Demonstration of various stimuli-responsive actuation: The photothermally driven actuation of the MLB was demonstrated by exposure to NIR light with an intensity of 0.4 W cm^{-2} (UNIX, UIM-250). The electro-thermally driven actuation was demonstrated by applying a controlled voltage (0.5–3.5 V) using a DC power supply (KEITHLEY, 2231A-30-3). The temperature variation during the photo- and electro-thermally driven actuation was measured by utilizing a thermal-imaging camera (FLIR, E40).

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Chapter 5 Conclusions and Future Directions

5.1 Conclusions

In this work, I presented novel approaches to creating a 3D programmable platform for shape-reconfigurable and locomotive electronics using the FPP technique and preparing electrically conductive LC polymer composites.

In Chapter 2, I demonstrated various 3D architectures based on photocurable resin via FPP and collective assembly. The photocured polymer was reconfigured right after the FPP from an initially flat shape to 3D free-standing structures by releasing accumulated internal stress resulting from a crosslinking density gradient. Detailed geometries of 3D structures were successfully controlled by adjusting spatiotemporal parameters including pattern design, and UV irradiation time of pre- or post-curing process. The collective assembly enables the creation of regular and repetitive structures, such as metastructures, while also offering diversity in geometry, including variations in size and complexity. Furthermore, the assembled design performed enhanced load resistance capabilities like a honeycomb structure and was utilized as a framework of electronics by metal coating. However, it became necessary to develop a material that could be programmed to reconfigure its shape at a desired time and location in response to external stimuli, overcoming the limitation of shape-reconfiguration occurring throughout the structure only at the moment when internal stress is released immediately after FPP.

For this, mechanically robust yet stretchable, shape-reconfigurable electronics were demonstrated by creating an azo-LCN/rGO bilayer in Chapter 3. Using a customized rGO-patterned LC cell, a rGO layer that is 10 times thinner than the azo-LCN layer was in-situ transferred onto the azo-LCN layer after photopolymerization. The azo-LCN/rGO bilayer exhibited a stable interface due to π-π interactions between the mesogens and rGO. This bilayer structure showed an electrical conductivity of 380 S cm^{-1} and a high storage modulus of approximately 6.4 GPa at room temperature. While the storage modulus of the azo-LCN/rGO was enhanced by 4.9 times compared to neat azo-LCN, the CTE mismatch between the azo-LCN/rGO layers and the softening effect under light stimuli created a synergistic effect, resulting in enhanced UV-induced photo-chemically driven actuation compared to neat azo-LCN. Additionally, the azo-LCN/rGO bilayer demonstrated multi-stimuli responsive actuation under UV and IR irradiation, as well as direct heat conduction. In particular, the Kirigami design enables the azo-LCN/rGO bilayer to perform hybrid shape-reconfiguration, combining mechanical stretching beyond its elongation at break with photochemical actuation, all without a reduction in electrical conductivity. Although the azo-LCN/rGO bilayer can stably maintain its reconfigured shape, the cis-to-trans back isomerization of azobenzene is relatively slow compared to the trans-to-cis photoisomerization, making it unsuitable for achieving reversible shape-reconfiguration on a timescale of a few seconds. Specifically, for the azo-LCN/rGO bilayer, recovery to its initial shape requires heating while the structure is mechanically constrained.

In Chapter 4, I demonstrated the LCE/MXene bilayer through the engineering of new materials and their integration, showcasing photothermally driven reversible shape-reconfiguration. LCE exhibited a lower glass transition temperature near room temperature compared to azo-LCN (~60 °C), enabling fast recovery from the reconfigured shape to the initial shape within seconds at room temperature. As a filler, MXene provided superior electrical conductivity of 5300 S cm^{-1} without the need for additional reduction processes, unlike rGO. The bilayer also exhibited a stable interface, attributed to hydrogen bonding between the mesogen and oxide and hydroxide terminal groups on the surface of MXene.

This robust interface allowed the bilayer to endure over 1,000 cycles of repeated bending actuation without delamination or reduction in electrical conductivity. In addition to its durability, LCE/MXene demonstrated exceptional actuation performance and superior electrical conductivity compared to other reported electrically conductive actuators, making it suitable for applications such as a smart switch capable of remotely turning LEDs on and off through photothermally driven bending. High electrical conductivity also enabled electro-thermally driven actuation with just 3.5 V. To achieve multi-functional actuation beyond shape-reconfiguration, an asymmetrically assembled design was incorporated into the LCE/MXene bilayer to make changing the position of the mass center during shape-reconfiguration. The assembled bilayer demonstrated both straight and rotational movements, depending on length differences and alignment offset angles between individual LCE/MXene units. Notably, the integration of this assembled design with snap-through instability enabled the soft bilayer to perform rapid jumping and sling-shooting behaviors. Overall, I believe that this work will provide broad insight to develop next-generation electronics.

5.2 Future Directions

The functionality and shape programmability of LC polymer composites can be further enhanced by incorporating new functional fillers and advanced processing. The following ideas expand on this: (1) In this dissertation, shape-reconfiguration of LC polymer composites was achieved mainly by photothermal effect and photoisomerization of azobenzene under IR or UV irradiation, respectively. Photonic stimuli can precisely deliver uniform intensity over long distances with minimal spatial constraints, but they require an open space due to the exponential decrease in penetration depth caused by light absorption according to the Beer-Lambert law. To address this limitation, introducing magneto-responsive fillers (e.g. NdFeB, iron particles) can be considered to achieve various remotely controlled shape-reconfigurations at blocked space due to the deep penetration depth of magnetic fields. Furthermore, the direction of the magnetic field can be easily controlled when using an electromagnet by adjusting the direction of the current. (2) In this dissertation, the molecular alignment of LC polymer composites was controlled by surface-induced alignment techniques. This technique has many advantages due to its simplicity, low cost, and scalability, but it faces limitations in achieving uniform programmed alignment in LC polymers when the thickness exceeds 50 μm. Furthermore, the resulting geometry of the LC polymer is limited to a 2D film shape. To expand structural diversity through advanced alignment programming, additive manufacturing can be considered, as it provides an initial 3D shape. Beyond the 2D film shape, the initial 3D structure can be reconfigured into another 3D form, modulating functionalities. Additionally, multi-material printing techniques allow for the introduction of functional fillers at specific locations within the LC polymer, facilitating the optimal use of these fillers and enabling the development of customized, shape-reconfigurable electronics.

Appendix

List of publications

1. M. Zhang, S. Jeong, **W. Cho**, J. Ryu, B. Zhang, P. Crovella, A. J. Ragauskas, J. J. Wie*, and C. G. Yoo*, "Green co-solvent-assisted one-pot synthesis of high-performance flexible lignin polyurethane foam", ***Chem. Eng. J.***, 499, 156142 (2024) **IF = 15.1**
2. **W. Cho**†, S. Kim†, H. Lee, N. Han, H. Kim, M. Lee*, T. H. Han*, and J. J. Wie*, "High-Performance Yet Sustainable Triboelectric Nanogenerator Based on Sulfur-Rich Polymer Composites with MXene-Segregated Structure", ***Adv. Mater.***, 2404163 (2024) **(Back Cover)** **IF = 29.4**
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4. H. Moon†, J. G. Lee†, **W. Cho**, J. Jeon, J. E. Park, and J. J. Wie*, "Structure-Property-Actuation Relationships of Shape-Fixable Magnetic Vitrimer Micropillar Arrays", ***Sens. Actuators B Chem.***, 402, 135092 (2024) **IF = 9.221**
5. J. E. Park, H. Je, C. R. Kim, S. P, Y. Yu, **W. Cho**, S. Won, D. J. Kang, T. H. Han, R. Kwak, S. G. Lee*, S. Kim*, and J. J. Wie*. "Programming Anisotropic Functionality of 3D Microdenticles by Staggered-Overlapped and Multi-Layered Microarchitectures", ***Adv. Mater.***, 2309518 (2023). **IF = 32.086**
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9. S. Jhang[†], **W. Cho**[†], S. K. Lee, A. R. Yu, J. G. Lee, K. Jung, Y. Pu, C. G. Yoo, and J. J. Wie*, "Functional Group Effect of Chemically Modified Microcrystalline Methyl Cellulose on Thermoplastic Polyurethane Composites", ***Cellulose***, 30, 6917-6931 (2023) **IF = 6.123**

10. J. H. Hwang[†], S. H. Kim[†], **W. Cho**[†], W. Lee, S. Park, Y.S. Kim, J.-C. Lee, K.J. Lee*, J. J. Wie*, D.-G. Kim*, "A Microphase Separation Strategy for the Infrared Transparency-Thermomechanical Property Conundrum in Sulfur-Rich Copolymers", ***Adv. Optical Mater.***, 2202432 (2023) **(Inside Front Cover article) IF = 10.05**

11. **W. Cho**[†], J. Hwang[†], S.Y. Lee[†], J. Park, N. Han, C.H. Lee, S.-W. Kang, A. Urbas, J.O. Kim, Z. Ku*, J. J. Wie*, "Highly Sensitive and Cost-Effective Polymeric-Sulfur-Based Mid-Wavelength Infrared Linear Polarizers with Tailored Fabry–Pérot Resonance", ***Adv. Mater.***, 35, 2209377 (2023) **IF = 32.086**

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18. **W. Cho**, and J.J. Wie*, Design of Polymeric Materials for Triboelectric Nanogenerators (TENGs), ***Physics and High Technology***, 30, 12 (2021)

19. **W. Cho**, D. Y. Hahm, J. H. Yim, J. H. Lee, Y. J. Lee, D.-G. Kim, Y. S. Kim, and J. J. Wie*, “Programmable Building blocks via Internal Stress Engineering for 3D Collective Assembly”, ***Adv. Mater. Technol.***, 5, 200758 (2020) **(Front cover article)** **<u>IF = 5.395</u>**

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Author:
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Multi-functional locomotion of collectively assembled shape-reconfigurable electronics

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Author: Woongbi Cho,Dong Jun Kang,Min Jeong Hahm,Jisoo Jeon,Dong-Gyun Kim,Yong Seok Kim,Tae Hee Han,Jeong Jae Wie
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국 문 초 록

조웅비

유기나노공학과

한양대학교 일반대학원

기존의 전자소재들의 회로는 전형적으로 단단한 물성을 갖는 기판위에 구성되어 있다. 따라서, 낮은 변형율을 갖기 때문에 다양한 곡률을 갖는 곡선형 구조에 맞춤형으로 도입하기 까다롭다는 한계가 있다. 따라서, 유연한 고분자 소재를 일렉트로닉스의 유연 플랫폼으로 제작하고자 하는 연구들이 활발히 진행 중이다. 예를 들어 하이드로겔 및 엘라스토머와 같이 유연한 고분자 소재가 주로 활용되며, 이들을 활용한 유연 디바이스 및 스트레처블 디바이스의 경우 높은 변형율로 인해 기계적 변형을 통해 특히 다양한 곡선형 구조를 갖는 신체와 같은 구조체에 손상 없이 적용될 수 있다. 하지만, 지금까지 활용되던 유연한 소재들의 경우 형상 변형을 위해서 물리적으로 디바이스들을 잡은 뒤 힘을 가해야 하기 때문에 소형 유연 디바이스의 경우 공간적인 제약으로 인해 형상 제어에 한계가 존재한다. 본 학위 논문에서는 수동적인 형태 변형에 국한된 유연 디바이스, 스트레처블 디바이스를 넘어 외부 자극의 선택적 감응을 통해 스스로 자신의 형상을 사전에 프로그래밍 된 형상으로 변화시킬 수 있는 스마트 고분자를 도입하여 차세대 전자장치의 플랫폼으로 활용할 수

있는 전략에 대해 설명한다. 비접촉식으로 고분자 구조체를 원하는 형태로 변형시키기 위해 구조 내에 인위적으로 응력 구배를 유도하고자 하였으며, 이는 광 흡수제를 포함하는 광 경화성 레진에 전면 광중합을 도입하는 방법, 분자 배향의 방향이 조절된 액정 고분자 복합소재를 제작하는 방법으로 구성된다. 먼저, 전면 광중합 방식 도입을 통해 광 경화성 레진에 광 흡수제를 도입하는 방식으로 경화된 고분자 구조체 내부에 가교밀도 구배를 달성하였고, 이에 따른 내부 응력의 구배를 발생시킬 수 있었다. 광 경화가 끝난 후, 구조체 내의 위치별 응력 해소 방향의 차이가 발생하여 3차원 형태로의 형상 재구성이 발생되고, 최종 형상은 도입된 광 패턴, 광 조사 시간 등의 시공간적 매개 변수 조절을 통해 제어된다. 제작된 각각의 구조는 단위 객체로 활용되어 더 크고 복잡한 형상을 갖는 군집 구조로 조립되었으며, 금속 코팅을 통해 전도성 3차원 프레임으로 응용되었다. 이어서, 분자 구조의 이방성을 갖는 액정 고분자 복합소재를 제작하여 형상 변형이 광 경화가 끝난 직후로 제한되지 않고 외부 자극을 원하는 위치 및 시점에 선택적으로 부여하여 형상 변형을 유도할 수 있는 전도성 스마트 소재를 개발하였다. 액정 고분자는 고유의 비등방성 구조 및 분자 배향으로 인해 열 수축/팽창의 이방성을 가지고 있어 외부 온도 변화를 감응하여 프로그래밍 된 형태로 형상 재구성이 가능하다. 이를 전기전도성이 뛰어난 2차원 소재인 환원 그래핀 및 맥신과 이중층 형태로 복합화 하여 액정 고분자의 배향성을 감소시키지 않는 동시에 각각의 2차원 물질들이 갖는 전기전도성을 손실 없이 액정 고분자에 도입하였다. 각각의 물질은 액정 고분자 물질과 2차 결합을 통해 안정적인 계면을 형성할 수 있어 반복적인 형상

재구성에도 형태를 안정적으로 유지한다. 또한, 환원 그래핀 및 맥신의 경우 우수한 광-열 전환 특성을 가져 적외선 조사를 통해 액정 고분자 복합소재의 온도를 원격으로 조절하여 형상 재구성을 달성하였다. 특히, 액정 고분자에 광 감응성 아조 벤젠 관능기를 도입하여 자외선 조사 하에서 발생하는 광이성질화 현상을 통해 원격으로 형상 제어가 가능하며, 액정 고분자의 가교 밀도를 낮춰 유리전이온도를 상온 근처에 위치하도록 설계하는 경우 광열 자극을 온/오프 하는 방식으로 가역적인 형상 재구성을 달성하였다. 액정 고분자 복합소재를 다양한 군집 구조로 조립하는 경우 다양한 형태의 형상 변형이 가능하며, 조립 시 비대칭성 및 스냅 스루 불안정성을 도입하는 경우 형상 변형시에 무게 중심 이동을 유발시켜 직진 운동, 회전 운동, 점핑 운동과 같은 다기능성 움직임을 구현할 수 있다. 형상을 프로그래밍할 수 있는 복합소재의 도입을 통해 차세대 전자장치로의 플랫폼 소재로써 활용될 잠재력이 높으며, 그 응용이 무궁무진 할 것으로 기대된다.

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